

W. M. KECK OBSERVATORY ADAPTIVE OPTICS NOTE
KAON 1542

The Keck AO Performance Estimator

*Predicting NGS, single-beacon LGS, and LTAO Strehl across a night from CFHT
MASS/DIMM data*

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1. Introduction

keck_ao_estimator is a standalone Python tool that models the delivered performance of the Keck adaptive-optics systems — natural guide star (NGS), single-beacon laser guide star (LGS), and laser tomography (LTAO) — from the public CFHT weather-tower MASS/DIMM data. It was built to answer, quantitatively, the question every observer asks: given the atmosphere that night, what Strehl should the AO system have delivered? For any night it produces a stacked timeline figure and a CSV table giving, at each atmospheric sample, the predicted NGS Strehl, the predicted single-beacon LGS and/or LTAO Strehl, the isoplanatic angle θ_0 and cone-effect parameter d_0 , and the seeing/ r_0 record itself.

Although it began as an after-the-fact Strehl estimator, the tool has grown into a general planning and analysis instrument. Ahead of a run it can fetch a night's data directly from the MKWC archive, resolve targets and tip-tilt stars, predict performance timelines for any target and guide-star configuration, map predicted Strehl or FWHM across the science field, compare tip-tilt sensors (STRAP against the infrared TRICK sensor), and explore hypothetical conditions — a chosen seeing, profile mix, or airmass — instead of a recorded night (§3.8). After a run it measures delivered Strehl and FWHM directly from NIRC2 and OSIRIS frames and compares them, star by star, against the prediction for the same conditions (§10). The same engine underlies a separate LGS sky-coverage analysis, reported in its own companion note.

The primary interface is the interactive GUI (§9), which exposes every capability above; its user's manual is KAON 1556. The command line (§4) drives the same engine for scripted and batch use. The tool covers both Keck telescopes: the telescope selection sets the fitting-error term, the NGS response model, the azimuth-dependent pointing limits, and the tomography default (K1: LTAO panel on; K2: single-beacon panel). All predictions are evaluated at a selectable science wavelength (K band by default); seeing and r_0 remain quoted at 500 nm by convention.

Package layout

The tool ships as the keck_ao_estimator package (pip-installable, with an optional [gui] extra for the Qt dependency). Installing it (pip install ".[gui]", or pip install . for the CLI/engine only) provides two console commands: keck-ao-estimator (the CLI) and keck-ao-estimator-gui (the GUI). Running from a source checkout without installing is also possible: python3 -m keck_ao_estimator.cli --help (CLI) or python3 -m keck_ao_estimator.gui.app (GUI), with PYTHONPATH=src:src_estimator set.

2. Inputs and data flow

Three whitespace-delimited files from the Mauna Kea Weather Center (MKWC) archive drive everything; they can be supplied locally (--dimm/--mass/--masspro) or downloaded directly by date with --fetch-date YYYYMMDD. MKWC names files by the morning/UT date: the night that begins the evening of 24 May 2026 lives in the 20260525.* files. Downloads are cached, and the cache is staleness-aware: if the UT night is not yet over the files are re-fetched automatically. A missing DIMM download is fatal (nothing can be computed without it), but missing MASS/masspro downloads — normal early on a live night, when MKWC posts DIMM first — only warn, and the run degrades gracefully to DIMM-only NGS estimates.

- <date>_dimm.dat — total seeing time series (DIMM), the input to the NGS model and to all total-seeing budget terms.
- <date>_mass.dat — free-atmosphere seeing time series (MASS), plotted in the seeing panel.
- <date>_masspro.dat — the MASS C_n^2 profile: six layer-integrated values J_i at 0.5, 1, 2, 4, 8, 16 km plus the free-atmosphere seeing. The profile drives the LGS/LTAO budget, θ_0/d_0 , and the LTAO layer-mismatch penalty. Duplicate timestamps (an MKWC quirk) are dropped at load, keeping the last occurrence.

Two timebases

Each estimate is computed on the timebase of the data it physically depends on. NGS needs only total seeing, so it is computed at every DIMM sample. LGS, LTAO, θ_0 and d_0 need the MASS profile, so they are computed at every MASS-profile sample, at the profile's own timestamp, taking the nearest DIMM sample within a tolerance (--match-tol, default 600 s) as the total-seeing input. Each MASS profile yields exactly one point: profiles are never re-used for several samples (so the LGS/ θ_0 curves stop honestly when MASS stops, instead of coasting on a stale reading) and never silently dropped. Connecting lines in every panel break across gaps wider than the tolerance, so sparse MASS coverage looks sparse in every panel, consistently.

3. Physics

3.1 From seeing to scaled wavefront error

Seeing ε and the Fried parameter r_0 encode the total integrated turbulence, with every altitude weighted equally. The error budget references every seeing-sensitive term to a nominal night ($\varepsilon_{\text{tot}} = 0.50''$, $\varepsilon_{\text{fa}} = 0.30''$) and scales its wavefront-error amplitude by the 5/6 power of the seeing ratio (turbulence power goes as $\varepsilon^{5/3}$; amplitude as its square root):

$$\varepsilon \approx 0.98 \lambda / r_0, \quad r_0^{-5/3} \propto \int C_n^2(h) dh, \quad S_{\text{tot}} = (\varepsilon / \varepsilon_{\text{ref}})^{5/6}$$

3.2 NGS model

The NGS Strehl uses empirical fits rather than a term-by-term budget: a Gaussian seeing-degradation factor times a Gompertz faint-end roll-off. In the expression below, S_0 is the bright-star, zero-seeing ceiling; s_K is the seeing expressed at K band (the 500 nm value scaled by the $V \rightarrow K$ wavelength factor); A sets how fast Strehl falls with seeing; and the double exponential handles guide-star faintness: m_0 is the magnitude at which the roll-off term has already cut the Strehl to $e^{-(e^0)} \approx 37\%$ of its bright-star value — the knee of the curve — and w sets how sharply the knee turns (brighter than m_0 the factor is near 1, fainter it collapses within roughly a magnitude or two).

$$S_K = S_0 e^{-A s_K^2} \exp[-e^{(m - m_0)/w}], \quad S_K = \varepsilon_{500} (V \rightarrow K)$$

$$\text{K2: } S_0 = 0.755, A = 0.738, m_0 = 13.76, w = 1.71$$

$$\text{K1: } S_0 = 0.61, A = 1.00, m_0 = 15.73, w = 1.53 \text{ (} -0.05 \text{ quadcell penalty)}$$

The K2 constants are the on-sky HAKA fit; K1, still on the pre-HAKA RTC and OCAM, uses its own reconstructed reference curve with two physical adjustments: a steeper seeing exponent ($A = 1.0$) because the lower-stroke K1 DM saturates in poor seeing, and a flat -0.05 Strehl penalty for the KAPA PRO quadcell-saturation effect. The Gompertz fits are empirical K-band fits; for other science wavelengths the K-band Strehl is backed out to an implied wavefront error and re-evaluated at the requested wavelength via the Maréchal relation. By default the Gaussian seeing term is evaluated with a Kolmogorov 5/3-power exponent re-anchored inside the fit's calibration range, which is identical within that range but rolls off at the physical rate beyond it — see §3.4 (6).

Off-axis NGS (--ngs-offset). When the guide star is offset from the science target by θ , the science direction suffers full angular anisoplanatism against the NGS-referenced correction and both NGS curves are multiplied by $\exp(-(\theta/\theta_0)^{5/3})$, with θ_0 evaluated per sample at the science wavelength. Where a MASS profile is available within the match tolerance its θ_0 is used (plotted as circles); elsewhere — including whole nights with no masspro data — an assumed θ_0 (--assumed-theta0, default $15''$ K-band at zenith, the Maunakea median, scaled to wavelength and line of sight) fills in, and those samples are drawn as triangles so the provenance of every point is visible. Setting --assumed-theta0 ≤ 0 disables the fallback and leaves honest gaps. The θ_0 actually used per sample is recorded in the CSV (ngs_theta0_arcsec).

Editable fit. The four Gompertz terms — ceiling S_0 , seeing exponent A , faint-end midpoint m_0 , roll-off width w — are exposed for the active telescope as --ngs-s0 / --ngs-a / --ngs-m0 / --ngs-w (default: the telescope's fitted value). The separate K1 post-fit quadcell-saturation penalty is --k1-quadcell-penalty (default 0.05, K1 only). These change the empirical fit — recalibration, not a budget what-if — and any override is recorded in the CSV provenance. In the GUI they are the “NGS Gompertz fit” block on the NGS tab, which repopulates to the active telescope's values when the telescope changes.

Provenance of the K2 constants. The K2 constants above are a fit to the full HAKA on-sky commissioning set. At $0.60''$ seeing, K band, they give a predicted K2 NGS Strehl of 0.641 at $R = 8$, 0.544 at $R = 11$, 0.350 at $R = 13$, 0.210 at $R = 14$, 0.084 at $R = 15$, and 0.016 at $R = 16$.

The K1 constants rest on a different footing. The K1 reference curve was reconstructed against an earlier K2 fit — its width $w = 1.53$ was taken from that fit, and its $R = 11.4$ crossover was derived against it — and neither anchor has been re-confirmed against the current K2 fit, so K1 keeps 0.61 / 1.00 / 15.73 / 1.53. The two widths are not shared, and K1's crossover should be read as approximate; re-deriving K1 against the current K2 fit is an open item.

Calibration state of the fit. The Gompertz fit is calibrated to optimized on-sky data, typically taken after a Fast & Furious (F&F) run — which is generally worth about 10 K-band Strehl points. The fitted ceiling S_0 therefore represents the system at its best; to estimate NGS performance without an F&F optimization, lower S_0 by 0.1.

3.3 LGS / LTAO error budget

The laser modes use an analytic nm-RMS error budget. High-order terms add in quadrature; the high-order and tip-tilt budgets are converted to Strehl with the Maréchal approximation and multiplied:

$$\sigma_{HO}^2 = \sigma_{alt}^2 + \sigma_{ang}^2 + \sigma_{fit}^2 + \sigma_{bw}^2 + \sigma_{scint}^2 + \sigma_{meas}^2 + \sigma_{NaFoc}^2 + \sigma_{static}^2 + \sigma_{margin}^2$$

$$S(\lambda) = \exp[-(2\pi\sigma/\lambda)^2], \quad S_{total} = S_{HO}(\lambda) \times S_{TT}(\lambda)$$

Table 1 lists the terms at the reference night, their seeing scalings, and how the two laser modes differ. LTAO replaces the single-beacon cone term (focal anisoplanatism) with a tomography error carrying only a weak 1/3-power residual free-atmosphere dependence, and pays a slower-loop bandwidth penalty: the LTAO high-order loop runs at a lower frame rate than single-beacon (~600 vs ~1500 Hz on K1), so the bandwidth term grows by a latency-floor-aware factor (§3.4 (5); ~1.44 by default). EM gain preserves the measurement SNR, so the measurement term is unchanged. An optional LTAO tilt- θ_0 gain (§3.4 (3b), off by default) can additionally reduce the tilt-anisoplanatism charge in the tip-tilt budget for a future multi-star tomography mode; at the shipped default the LTAO tip-tilt charge is identical to single-conjugate.

Term	Ref (nm RMS)	Scales with	Notes
Fitting	97.0 (K1) / 48.1 (K2)	total seeing $^{(5/6)}$	DM actuator density; the K1–K2 difference
Focal anisoplanatism	122.7	free-atm $^{(5/6)}$	single-beacon only; replaced by tomography in LTAO
Tomography	85.1	free-atm $^{(1/3)}$	LTAO only; plus the layer-mismatch penalty (§3.4)
Angular anisoplanatism	23.3 at 2" offset	free-atm $^{(5/6)}$	scales as (offset/2") $^{(5/6)}$; see §3.4 (4)
Bandwidth	42.5 ($\times \sim 1.44$ LTAO)	wind-weighted (§3.4)	servo lag; LTAO latency-floor penalty (§3.4 (5))
Scintillation	23.7	total seeing $^{(5/6)}$	
HO measurement	63.3	fixed	preserved in LTAO (OCAM2K EM gain)
Na focus	42.0	fixed	sodium-layer altitude variations
Static / calibration	122.5 (K1) / 109.2 (K2)	fixed	
Margin	130	fixed	budget reserve
Tip-tilt	164 total at ref	see §3.4 (3), (3b)	recomputed per TT-star configuration; an optional LTAO tilt- θ_0 gain (§3.4 (3b), off by default) can reduce the tilt-anisoplanatism rows

Where these numbers come from (v3_1_3). The reference column is the AO performance error budget spreadsheet v3_1_3, re-referenced to zenith. The sheet quotes its atmosphere-dependent terms at $ZA = 50^\circ$, while the pipeline separately projects each night's measured seeing onto the target's line of sight — using the sheet's rows as-is would apply the zenith penalty twice. Atmosphere-scaled terms are therefore divided by $(\sec 50^\circ)^{(1/2)} = 1.247$; the tomography residual by $(\sec 50^\circ)^{(1/5)} = 1.092$, fixed by its own $\epsilon_{fa}^{(1/3)}$ scaling law so that a reference night observed at $ZA 50^\circ$ reproduces the sheet; seeing-independent terms are carried as-is. The static group follows the sheet's own rows, including a DM finite-stroke term (31 nm) and an instrument term (60 nm); the K1 static subtotal is 122.5 nm.

How this reference set was checked. On sky (M79, 2026-01-31 UT, OSIRIS Kp) the older v3_1_1 values under-predicted the single-LGS on-axis Strehl badly — 0.238 against 0.327 measured — while v3_1_3 predicts 0.315, inside the night's own scatter. A residual LTAO under-prediction of ~0.06 remains and points at the LTAO bandwidth floor fraction, deliberately left untouched pending a 600 Hz loop-telemetry audit. The 22-scenario regression harness passes byte-identical against the frozen v3_1_3 references. The older set remains selectable as 3_1_1 (CLI --budget-version, or the WFE tab's Budget values picker); under 3_1_1 every changed parameter is reported through the modified-budget provenance, so a legacy-values run cannot masquerade as the reference budget.

K2 caveats. No K2 v3_1_3 sheet was available. K2's fitting reference was de-scaled on the ZA-50 provenance alone (60 → 48.1 nm) and K2's telescope-aberration group keeps the older sheet's value. The de-scaling is known to be incomplete rather than merely provisional: K1's fitting reference also carried the 0.6" → 0.5" seeing reference ($\div 1.1641$) and K2's did not, which breaks the actuator-pitch invariant the two references would otherwise satisfy (§3.1). Applying the same factor to K2 would give 41.3 nm. That has not been done, because it raises every K2 Strehl prediction and so needs a harness re-freeze and an explicit decision rather than a documentation edit. K2 numbers should therefore be read as provisional, and specifically as conservative in the fitting term.

Table 1 — LGS/LTAO error-budget terms. "Ref" is the allocation at the nominal night; seeing-scaled terms multiply by $(\varepsilon/\varepsilon_{\text{ref}})^{5/6}$ on amplitude.

3.3a How each term scales with the atmosphere, and why

Every night, the same fixed hardware sees a different atmosphere, and the budget rescales each error term from its reference-profile value (Table 1) to that night. Three scaling rules cover every term, and which rule a term takes is dictated by its physics, not chosen for convenience. This section sets out the justification term by term, because a scaling that is wrong in form cannot be repaired by any reference value.

The common amplitude law: seeing^{5/6}. Most wavefront-error terms are a residual fraction of the turbulence, and turbulence strength enters through the Fried parameter r_0 . A wavefront variance built from a (D/r_0) or (d/r_0) ratio scales as that ratio to the 5/3 power (the Kolmogorov structure-function exponent), so the error amplitude — the RMS — scales as $r_0^{(-5/6)}$. Since seeing $\varepsilon \propto \lambda/r_0$ at fixed wavelength, $r_0^{(-5/6)}$ is exactly $\varepsilon^{(5/6)}$. This is the origin of the seeing factors used throughout the budget:

$$S_{\text{tot}} = (\varepsilon/0.5'')^{5/6}, \quad S_{\text{fa}} = (\varepsilon_{\text{fa}}/0.30'')^{5/6}$$

The 5/6 exponent is not an approximation laid on top of the physics, it is the physics. What differs between terms is which seeing drives them — the total column, or the free atmosphere alone — and that is set by where in the system the error is created.

Total-seeing terms (scale with s_{tot}). These are created at the pupil by the whole turbulent column, so the integrated C_n^2 — total r_0 from the DIMM — is the right driver.

- **Fitting error** — the turbulence at spatial frequencies above the DM's correction cutoff $f_c = 1/(2d)$ for actuator pitch d :

$$\sigma_{\text{fit}}^2 = a_F (d/r_0)^{5/3}$$

Hence $\sigma_{\text{fit}} \propto r_0^{(-5/6)} \propto \varepsilon^{(5/6)}$. It is a pupil-plane spatial-frequency argument — every layer's high-frequency content arrives at the pupil regardless of altitude — so total seeing, not free-atmosphere, is correct. Internal check: the K1/K2 fitting ratio should follow the actuator-pitch law $(57/20 \text{ actuators across the pupil})^{5/6} = 2.394$, and the sheet's uncorrected rows do — $141 / 60 \text{ nm}$ gives 2.350, within 2 % — good evidence that the sheets encode the same $(d/r_0)^{5/3}$ law read along the pitch axis. The shipped v3_1_3 references, however, give $97.0 / 48.1 = 2.017$, which is 16 % off the law — a bookkeeping asymmetry rather than a physical result. v3_1_3 applies two independent corrections — $\div 1.1641$ to move the seeing reference from 0.6" to 0.5", and $\div 1.2473$ to move from ZA 50° to zenith — and K1 received both ($141.0 \rightarrow 121.1 \rightarrow 97.1$, matching the shipped 97.0) while K2 received only the second ($60.0 \rightarrow 48.1$). The arithmetic is unambiguous: $2.350 / 2.017 = 1.165$ against the seeing factor 1.1641, so the ratio did not drift, it moved by precisely the factor applied to one telescope and not the other. Aliasing, often lumped with fitting, scales identically.

Why that settles it, and what has not been changed. The invariance argument stands on its own, independent of the arithmetic: the fitting-error ratio between the two telescopes is set by actuator pitch, so a change of seeing reference multiplies both telescopes' fitting errors by the same factor and must leave the ratio unchanged. Any re-referencing that moves the ratio has therefore been applied asymmetrically. Carrying the 1.164 on K2 as well would give FITTING_ERR_K2 = 41.3 nm and restore the ratio to 2.347, back within 2 % of the law. That change has deliberately not been made: removing $\sqrt{(48.1^2 - 41.3^2)} \approx 24.6 \text{ nm}$ in quadrature from every K2 budget raises K2 Strehl predictions across the board, so it is a physics change requiring a harness re-freeze and a decision on whether to move a shipped default, apply it as a separately selectable version, or hold until a K2 v3_1_3 sheet is in hand. Until then the shipped K2 fitting reference remains 48.1 nm and K2 predictions should be read as carrying a known, quantified conservative bias in this term.

- **Bandwidth (servo lag)** — the wavefront changes during the loop delay. The Greenwood frequency $f_G \propto v_{\text{wind}}/r_o$, and the residual $\propto (f_G/f_{\text{loop}})^{5/6}$, so at fixed loop rate the amplitude again follows $r_o^{-5/6}$. This term additionally carries a wind-weighting (§3.4 (1)) and, for LTAO, a latency-floor rate penalty (§3.4 (5)); the seeing dependence itself is the standard 5/6.
- **Scintillation** — amplitude fluctuation from propagation; the standard weak-turbulence result places it on the same seeing^{5/6} amplitude law.
- **High-order measurement (partly), tip-tilt atmospheric term** — the WFS measurement noise floor and the atmospheric component of tip-tilt both track the turbulence the loop must measure and correct, and carry s_{tot} . (The photon-limited part of measurement is separately magnitude-dependent, §3.4 (3)/(3a).)

Free-atmosphere terms (scale with s_{fa}). These are created specifically by turbulence at altitude, so they are driven by the free-atmosphere seeing from the MASS profile, not the total column. Using total seeing here would wrongly charge them for the ground layer, which these effects do not see.

- **Focal anisoplanatism (cone effect)** — the single laser beacon samples a cone, not a cylinder, missing high-altitude turbulence outside the cone. It is an altitude effect by construction: no free atmosphere, no cone error. Scales as $s_{\text{fa}}^{5/6}$ (and worsens toward the horizon through the geometry, §3.6).
- **LTAO tomography residual** — the reconstruction error from sampling the volume with a finite laser asterism lives entirely in the free atmosphere the tomography must reconstruct. It scales more gently than the others, as $s_{\text{fa}}^{1/3}$, because tomographic averaging over the asterism suppresses part of the altitude dependence; the layer-mismatch surcharge (§3.4 (2)) rides on top of it.
- **Angular anisoplanatism** — for an off-axis reference (the laser at its offset, or an off-axis TT star), the correction decorrelates with angle at a rate set by the isoplanatic angle θ_o , which is a free-atmosphere quantity (high layers decorrelate fastest). Scales as $s_{\text{fa}}^{5/6}$ times the offset dependence (§3.4 (4)); vanishes on-axis.

Fixed terms (do not scale with seeing at all). These errors do not come from the observed atmosphere, so scaling them with seeing would be physically wrong — they are held at their nm-RMS reference values on every night.

Budget values picker. A Budget values selector sits above the slider groups: 3_1_3 (the default, zenith-referenced v3_1_3 set described in §3.4) or 3_1_1 (the older v3_1_1 sheet's values). Selecting 3_1_1 moves every affected slider through the ordinary what-if machinery rather than through a hidden path, so the summary line reads “Budget values v3_1_1 (legacy) selected — 12 parameter(s) off the v3_1_3 defaults (MODIFIED BUDGET)” and every export carries the moved parameters in its provenance: a legacy-values run can never be mistaken for the reference budget. Each slider's tooltip states where its reference value comes from (“AO performance error budget v3_1_3, zenith-referenced”). The CLI equivalent is `--budget-version {3_1_3,3_1_1}`.

- **Static / calibration** — telescope-optics figure, WFS zero-point, DM hysteresis and registration (§3.4 (7)). These are instrumental, present in a dark dome, and independent of tonight's turbulence.
- **Sodium-layer focus** — set by the sodium layer's altitude and thickness, an ionospheric property unrelated to ground/free-atmosphere seeing.
- **High-order measurement (photon-limited part)** — driven by guide-star flux and WFS frame rate, i.e. magnitude, not seeing (§3.4 (3)/(3a)); the fixed detector-noise floor likewise does not scale.
- **Margin / reserve** — a fixed budget allocation by definition.

Caveats on the amplitude law. Two second-order effects are deliberately not modelled. (i) Outer scale: real turbulence is von Kármán (finite L_o , tens of metres at Maunakea), which suppresses low-spatial-frequency power — but fitting and the high-order terms live well above that scale, so the pure-Kolmogorov 5/6 holds where it is applied. The one term where a finite L_o enters directly is the tip-tilt open-loop ceiling (§3.4 (3c)). (ii) DM stroke saturation: in poor seeing the DM runs out of stroke, adding error beyond $(d/r_o)^{5/3}$; this is a missing separate term (captured empirically only in the K1 NGS fit's steepened seeing exponent, §3.2), so the LGS budget may read slightly optimistic in very poor seeing. Neither affects the correctness of the scaling law itself — the term most worth suspicion is never the amplitude exponent but the temporal term (bandwidth, where frame-rate vs latency was the real issue, §3.4 (5)) and the empirical NGS fit (where extrapolation, not scaling, was the failure mode, §3.4 (6)).

3.4 Term-level budget models

Beyond the per-term seeing scalings of §3.3a, the default budget models the terms below in more physical detail. None of the models contains a parameter fitted to the validation nights; the first four are validated on the 2025-12-04 (ground-layer-dominated) and 2025-12-06 (free-atmosphere-dominated) K1 nights on HD 18770. A simplified legacy budget (`--legacy-budget`) replaces them with the fixed-allocation sheet behaviour; §7 tabulates the differences term by term.

(1) Wind-weighted bandwidth. Servo-lag error is set by how fast the turbulence evolves over the loop's response time, and the natural timescale is the Greenwood frequency f_G . Each layer contributes to f_G in proportion to its turbulence strength times its wind speed to the 5/3 power — faster-moving turbulence crosses the aperture and decorrelates sooner, so a fast high layer costs far more bandwidth than an equally strong slow ground layer. Scaling σ_{bw} with total seeing alone — as the legacy budget does — implicitly freezes the wind profile at its median mix. The default budget makes the wind weighting explicit: the measured turbulence is split into a ground part J_g (DIMM minus MASS, in $\varepsilon^{5/3}$ turbulence-power units) and a free-atmosphere part J_f (MASS), each assigned a representative wind — 8 m/s for the slow boundary layer, 25 m/s for the jet-influenced free atmosphere. Because $\sigma_{bw} \propto f_G^{5/6}$, the wavefront-error amplitude scales as the square root of the wind-weighted power, and the bandwidth term is rescaled by the ratio of that quantity between the night and the reference night (first equation below; second line defines the symbols). At the reference profile the factor reduces to s_{tot} , so a median night matches the plain seeing scaling; ground-dominated nights fall below it (bandwidth relief — the Dec 4 case) and free-atmosphere-dominated nights rise above it (a surcharge — Dec 6):

The two representative wind speeds (boundary layer 8 m/s, free atmosphere 25 m/s) are inputs, --wind-ground / --wind-free, since the true profile is not available from the seeing data; supply a night estimate (e.g. from a GFS forecast). At the 8/25 defaults the term is unchanged.

$$f_G^{5/3} \propto \int C_n^2(h) v(h)^{5/3} dh \Rightarrow \text{bw scale} = \sqrt{\frac{J_g v_g^{5/3} + J_f v_f^{5/3}}{J_{g,ref} v_g^{5/3} + J_{f,ref} v_f^{5/3}}}$$

$$J_g = \varepsilon_{tot}^{5/3} - \varepsilon_{fa}^{5/3}, \quad J_f = \varepsilon_{fa}^{5/3}, \quad v_g = 8 \text{ m/s}, \quad v_f = 25 \text{ m/s}$$

(2) Tomography layer-mismatch penalty (LTAO only). The K1 tomographic reconstructor uses a static layer prior — altitudes 0, 0.5, 1, 2, 4, 8, 16 km with fixed turbulence fractions [0.4557, 0.1295, 0.0442, 0.0506, 0.1167, 0.0926, 0.1107]. The aloft altitudes coincide with the six MASS bins, so each profile sample can be compared directly against the prior. The mismatch m (total-variation distance between the night's normalized bin profile and the prior's normalized aloft fractions) degrades the correction toward the single-beacon cone term with a quadratic ramp. Both endpoints are existing budget terms: $m = 0$ reproduces the stock tomography term, $m = 1$ removes the tomographic benefit entirely. The ramp is quadratic because an MMSE reconstructor's loss is second-order in prior mismatch — the beacons still measure the actual turbulence, and a moderately wrong prior mainly misassigns altitude, at small cost across the asterism's angles. A linear ramp was tried first and erased the LTAO benefit over single-beacon at the $m \approx 0.5$ typical of ordinary nights, contradicting on-sky experience; the quadratic law preserves that benefit while still closing the extreme-mismatch 2025-12-06 night.

$$m = \frac{1}{2} \sum_i |f_i - p_i| \in [0, 1], \quad f_i = J_i / \sum_j J_j, \quad \sigma_{alt}^2 = \sigma_{tomo}^2 + m^2 (\sigma_{FA}^2 - \sigma_{tomo}^2)$$

(3) Parameterized tip-tilt star. The sheet's TT budget (164 nm at the reference night) is built for one configuration: an $R = 15.2$ TT star 19.3" off axis. The tool instead recomputes the TT error from the underlying one-axis mas rows for any TT star (--tt-mag, --tt-offset): measurement error scales with photon noise ($\times 10^{(-0.2 \Delta m)}$), tilt and centroid anisoplanatism scale linearly with offset (zero on-axis), and dispersion, non-common-path, wind shake and the TT margin stay fixed. At the sheet's own configuration the recomputation reproduces the fixed TT term to better than 1%. For a bright on-axis TT reference (e.g. the science star itself at $R \approx 10$) the TT error drops to ≈ 72 nm, a $\times 1.2$ –1.26 Strehl factor common to both laser modes at K band.

(3a) STRAP measurement row, recalibrated on sky. STRAP centroids a seeing-limited R-band spot, and the sheet's photon-only measurement row (7.25 mas at $R = 15.2$, slope 0.2) is far too optimistic against on-sky data. The TRICK commissioning paper (Rampy et al., Near-infrared tip-tilt sensing at Keck, Table 1) provides three same-star, on-axis STRAP/TRICK H-Strehl pairs; dividing out the common high-order Strehl implies STRAP total tip-tilt residuals of $\sim 8.4 / 12.0 / 17.8$ mas one-axis at $R = 12.0 / 13.0 / 15.5$ — versus the sheet's ~ 6 –7 mas. The measurement row the tool carries is the least-squares fit through those points (fixed rows subtracted); it holds through its last measured anchor at $R = 15.5$, in mas one-axis:

$$\sigma_{meas} = 6.9 \times 10^{0.116(R-12)} \times s_{tot} \text{ mas}$$

The sub-photon slope ($0.116 < 0.2$) reflects the closed loop's gain optimization at the faint end. Model-vs-measured Table-1 Strehl ratios: 0.910/0.852/0.561 vs 0.916/0.783/0.543. The refined row is the only STRAP row the tool carries. The NGS path is unaffected — it senses tip-tilt on the NGS WFS rather than on STRAP — so the NGS fit and the NGS FWHM anchor do not depend on it.

Faint of the anchor, a constant slope has nothing to rest on and the physics changes: STRAP leaves the photon-noise regime for the background- and read-noise regime, where the error steepens. The row therefore continues along the performance sheet's quadcell slopes, grafted anchor-continuously at $R = 15.5$:

$$\sigma_{\text{meas}}(R) = \sigma_{\text{meas}}(15.5) \times 10^{0.204(R - 15.5)} \quad (15.5 < R \leq 17.5)$$

$$\sigma_{\text{meas}}(R) = \sigma_{\text{meas}}(17.5) \times 10^{0.230(R - 17.5)} \quad (R > 17.5)$$

giving 22.2 / 35.6 / 58.6 / 99.5 mas one-axis at $R = 16 / 17 / 18 / 19$, the faintest of these clipped by the open-loop ceiling of (3c). The slopes come from the performance-sheet totals for a 10"-off-axis STRAP star, which agree with the fitted row to within $\pm 13\%$ for $R \leq 14$ and steepen beyond — the photon-to-background transition a constant slope lacks. The sheet's own knee sits about 1–1.5 magnitudes too bright against the measured $R = 15.5$ anchor (28.5 vs 17.8 mas), which is why the hybrid is anchored on sky rather than adopted wholesale. The steepening applies to both selectable STRAP laws, and the default budget star ($R = 15.2$) sits below the knee, so the frozen reference values are unchanged. The grafted knee is unconfirmed on sky beyond $R = 15.5$ — an $R > 16$ ladder point would pin it (the same measurement §3.9 recommends).

(3b) Tomographic tilt-anisoplanatism reduction (LTAO only). The tip-tilt budget of (3) is, by itself, mode-blind: its two tilt-anisoplanatism rows — 9.17 mas and 1.28 mas one-axis at the budget configuration, both linear in TT-star offset and both inherited from the 2004 single-conjugate sheet — would be charged identically to single-beacon LGS and to LTAO. In LTAO the tool can reduce them by a settable effective tilt- θ_0 gain, applied as $(1/\text{gain})^{(5/6)}$ to those two rows and to nothing else; the default gain is 1.0 — no reduction (see below). The physical argument is that the differential tilt between the TT star's direction and the science direction is generated by turbulence at altitude — focus and astigmatism aloft appear as field-varying tilt at the pupil. The LGS asterism cannot sense global tilt, which is why a tip-tilt star is needed at all, but it does sense exactly those altitude modes, so the tomographic reconstructor can correct the field dependence of tilt even though it cannot correct tilt itself. Under LTAO the TT star's tilt therefore decorrelates over a larger effective θ_0 than the single-conjugate sheet assumes; under single-beacon LGS nothing changes.

Where it would matter, and by how much. With the gain enabled, the reduction is negligible for a bright on-axis reference and decisive for the far, faint stars that set sky coverage: at a 30" TT-star offset the two anisoplanatism rows are about 85 % of the total tip-tilt variance, rising to about 93 % at the 51" patrol edge, while on-axis they vanish entirely. At a gain of 1.1 and the budget TT star ($R = 15.2$ at 19.3", reference profile) the tip-tilt term would fall 224.2 → 213.3 nm and the LTAO K-band Strehl rise by about +0.006; at 30" offset with a magnitude-13 TRICK star the LTAO Strehl would rise by about +0.018. Single-beacon LGS numbers are bit-identical before and after, at any gain.

Why the default is 1.0. KAON 1303 §5.5 settles the calibration this knob was flagged as waiting on, and it argues against a default reduction: the LGS-tomography null modes are a field-varying tilt concentrated in the quadratic modes (focus and astigmatism), and the low-order tomographic algorithm estimates those quadratic terms only with three tip-tilt stars — with two, and especially with one, they land in the final budget. The step that buys the reduction is one star → two, and multi-star tip-tilt was descoped. Three independent lines agree with no reduction at the single-star default: the KAON 1303 §5.5 analysis; the M79 single-beacon-versus-LTAO comparison, null at 1.1σ ; and the 2026-01-31 measured tilt decay, whose $\theta_0\text{_{TT}} = 69''$ matches the no-gain sheet rows' equivalent 60–83" across the patrol field. The knob remains for a future multi-star mode — a gain above 1.0 enables the reduction — and --legacy-budget is exempt at any setting, remaining byte-faithful to the frozen 2004 sheet. Relative to a 1.1 gain, the no-reduction default costs about 0.005 LTAO K-band Strehl (0.007 at most) at the default tip-tilt star; single-beacon predictions are identical either way.

Where the parameter lives. The engine constant is `DEF_LTAO_TT_THETA0_GAIN = 1.0` (tiptilt.py, with the KAON 1303 §5.5 rationale in its comment block), applied in `lgs_budget_terms()`. The command-line flag is `--ltao-tt-theta0-gain G`, and the CSV provenance line records `ltao_tt_theta0_gain` whenever it is set away from the default, so a run at a moved gain cannot be mistaken for a default one. In the GUI it is an "LTAO TT θ_0 gain" spinbox on the LGS tab (range 1.0–3.0, step 0.05) directly under "LTAO bw floor frac"; it recomputes live, marks the field map stale, and is saved and restored in configs, with configs saved without the field loading at the default (KAON 1556 §3).

(3c) Tip-tilt open-loop ceiling. A loop cannot do worse than no correction. The parameterized TT budget's anisoplanatism rows grow without bound with the TT-star offset (and, in §3.8's scenarios, with the anisoplanatism re-weighting), but physically the worst case is no tilt correction at all: the science image becomes a seeing-limited spot whose motion is the uncorrected atmospheric tilt — itself limited by the outer scale L_0 of the turbulence, which removes the largest tilt-producing scales. The TT term is therefore capped at the uncorrected one-axis tilt for the night's seeing and the configured outer scale. The ceiling is anchored to KAON 1318 Table 1 — uncorrected two-axis tip-tilt versus L_0 , atmosphere and telescope jitter merged — interpolated in $\log L_0$ and scaled with seeing as s_{tot} . The default $L_0 = 50$ m is the Mauna Kea median given by both KAON 1318 (Fig. 5) and KAON 1303 (§5.5), consistent with the Keck heritage 75 m of van Dam et al. (2004); it gives 74.4 mas one-axis at the reference profile. --outer-scale sets a per-night value, and inf recovers the infinite-outer-scale Kolmogorov G-tilt over the aperture:

$$\sigma_{\text{tilt}}^2 = 0.170 (D/r_0)^{5/3} (\lambda/D)^2$$

In that limit the ceiling evaluates to ≈ 110 mas one-axis at the reference seeing; the finite- L_0 anchor lowers it to 74.4 mas at the 50 m default. The cap is achromatic ($r_0 \propto \lambda^{6/5}$) cancels λ/D and scales with seeing exactly as the budget's s_{tot} (both $\propto r_0^{(-5/6)}$). The same ceiling bounds the NGS tilt used by the FWHM path. Nothing here is fitted to the validation data — the anchor is the published KAON 1318 table. The nominal budget sits at $\sim 26\%$ of the ceiling at the default L_0 , so it binds only for extreme offsets, very faint stars, or aloft-heavy scenarios — never on the reference scenarios (regression-verified byte-identical).

(3d) TRICK IR tip-tilt sensor (K1). K1's tip-tilt can be sensed by the R-band STRAP quadcell (the default budget) or the IR TRICK sensor in H or K (--tt-sensor trick-h/trick-k; --tt-mag is then the guide star's H/K magnitude). TRICK centroids the AO-corrected, near-diffraction-limited guide-star image, so it holds tip-tilt to a much fainter guide magnitude than the seeing-limited quadcell. Only the measurement row changes — the tilt-anisoplanatism, bandwidth, wind-shake and margin rows are shared (tip-tilt is achromatic image motion). The measurement row, in mas one-axis:

$$\sigma_{\text{meas}} = 5.34 \times 10^{0.60(m_{\text{wfs}} - 14.0)} \times s_{\text{tot}} \text{ mas}$$

with m_{wfs} the guide magnitude in the sensing band (engine constants TRICK_MEAS_REF = 5.34, TRICK_MAG0 = 14.0, TRICK_BETA = 0.60). These are calibrated to the on-sky 2026-05-28 (M3, LTAO) Strehl-vs-guide-magnitude data: performance is floor-limited (flat Strehl) to a sensing-band magnitude ≈ 14 , then rolls off steeply. The measured ladder gives a plateau of 0.480 ± 0.007 and 0.433 ± 0.034 at $H = 14$. Run at each rung's own star and the night's own conditions, the shipped engine reproduces the measured ladder to a weighted $+0.038$, or $+0.019$ after the crowding correction (§3.9; the offsets used there carry the TRICK-to-imager registration caveat stated with the ladder). On K1 the TRICK/OSIRIS dichroic ties the science band to the other of H/K, so trick-k forces science H and trick-h forces science K (auto). K1 only.

Off-axis spot degradation. TRICK centroids the AO-corrected core at the star, whose quality falls with the star's angular separation from the laser — so the TRICK measurement row inflates off-axis by the factor

$$\sqrt{1 + (\theta/25'')^2}$$

with θ the star–laser separation (the field map uses the true separation, the timeline uses the TT offset as a proxy). For bright TRICK stars the row is so small the inflation is invisible out to 40–50" (as the commissioning data show); a faint ($m \approx 14$) TRICK star converges toward and then loses to STRAP tens of arcsec off-axis, reproducing the paper's prediction that beyond a certain off-axis distance STRAP will outperform TRICK. With the §3.4 (3a) recalibration, the model reproduces the paper's paired on-axis Table-1 ratios (see there) and the 2026-05-30 M15 off-axis convergence trend (TRICK/STRAP H-Strehl ratio falling with offset); M15's absolute on-axis gap ($\sim 1.5\times$ vs modeled $\sim 1.2\times$) remains partly unexplained — sequential (non-simultaneous) STRAP/TRICK runs under evolving conditions.

(4) Laser-pointing offset. The sheet's 44 nm angular-anisoplanatism allocation assumes a beacon only 2" from the science direction. Operationally the K1 laser is offset 4.97" from the science target — 4.8" West and 1.3" South of the pointing origin in the standard north-up/east-left configuration, i.e. radius 4.97" at position angle 254.8° N → E — and in LTAO the 4-beacon asterism center sits at the same separation from science, the relevant separation for the tomographic solution, while the K2 laser is not offset at all. The term scales with the actual offset, going to zero on-axis:

$$\sigma_{\text{ang}} = 44 \text{ nm} \times s_{\text{fa}} \times (\theta/2'')^{5/6}$$

The variance is $\propto \theta^{5/3}$. Defaults are the per-telescope operational offsets (--lgs-offset overrides). At 4.97" the charge is $\times 2.14$ the allocation — still large on free-atmosphere-dominated nights, and common to both laser modes. Charging LTAO at the asterism-center offset is a first-order treatment: tomography interpolates within the asterism, so some of this error may overlap the tomography term.

Where the 4.97" comes from. The value is a campaign measurement: a bench stage-alignment property rather than a nightly pointing error, so it lives in the detector frame and is stable night to night, and it agrees with the rest of the campaign: the 2026-08-01 direct boresight measurement gave 4.7" W, 1.0" S, and the LTAO on-axis field-map peaks cluster at 5.8–6.1", further out because the asterism-weighted optimum sits off the single-beacon boresight. Only the magnitude enters the budget, which is radial; the position angle is used solely to place the laser marker on the field map (§9). Because the term scales as $(\theta/2'')^{5/6}$, the 4.97" offset charges $\times 2.14$ on the 2" allocation; on the 20260525 K1 reference night the single-LGS median is 0.2524 and the LTAO median 0.2734. K1 NGS does not use this term and K2's offset is zero, so it contributes exactly 0.0000 there.

(5) Latency-floor LTAO bandwidth penalty. The LTAO high-order loop runs slower than single-beacon — on K1, ~600 Hz for LTAO against ~1500 Hz single-beacon — which raises the bandwidth (servo-lag) term. A pure frame-rate law at an assumed half rate would multiply the whole bandwidth term by $2^{5/6}$. Two corrections apply, established by the K2 NGS budget analysis of §11: the true rate ratio is $1500/600 = 2.5$, not 2; and servo-lag error is not purely frame-rate-limited but dominated by a fixed loop-latency floor (real-time-controller compute plus DM response) that does not change with frame rate. The penalty is therefore applied only to the rate-dependent part of the bandwidth variance:

$$\sigma_{\text{bw}}^{\text{LTAO}} = \sigma_{\text{bw}}^{\text{single}} \times \sqrt{f_{\text{floor}} + (1 - f_{\text{floor}}) (f_{\text{single}}/f_{\text{tomo}})^{5/3}}$$

where f_{floor} is the fraction of the single-beacon bandwidth variance that is fixed latency (--ltao-bw-floor-frac, default 0.70, consistent with the NGS budget). The two corrections nearly cancel: the true $2.5\times$ rate ratio alone would give $2.5^{5/6} = 2.15$, but the latency floor pulls the effective factor down to 1.44 at the default floor fraction. Setting the floor fraction to 0 recovers the pure rate law (2.15); 1 removes the penalty entirely. The measurement error is still not penalized (OCAM2K EM gain preserves SNR at the lower rate). The active factor is printed to the console and recorded in the CSV provenance line.

Effect. Relative to the pure rate law, the latency-floor factor raises LTAO Strehl by ~0.01–0.04 depending on seeing (more in poor seeing, where bandwidth dominates), widening the LTAO-over-single-beacon margin the tool predicts — the same margin the layer-mismatch model (2) preserves. On the two December validation nights the in-window LTAO median moves from 0.462 to 0.472 (Dec 4, ground-dominated) and 0.243 to 0.263 (Dec 6, free-atmosphere-dominated); the change is small and its effect on the delivered match is mixed, so the model rests on its physical basis rather than on a two-night fit.

(6) Kolmogorov NGS seeing law (high-airmass fix). The NGS Gompertz fits were calibrated on open-loop K-band FWHM of ~0.19–0.38", all at ZA 0–30°. Their Gaussian seeing term $\exp(-A s_K^2)$ is empirically fine inside that range, but a Gaussian free-falls when extrapolated — and at high airmass the line-of-sight projection pushes the fit's input 2–3 $\times$ beyond the calibration maximum. The quadratic exponent turns airmass into an $X^{6/5}$ hit on $-\ln(S)$, where the physical closed-loop residual variance grows only linearly with airmass (the turbulence path integral). On 2026-07-06 (HIP 88553 at airmass 2.34, zenith seeing 0.72–0.92") the delivered NGS Strehl was ~0.50–0.56 while the Gaussian form predicted 0.29–0.42. The seeing term therefore defaults to the Kolmogorov variance law, re-anchored at the calibration mid-range:

$$\exp(-A s_K^2) \rightarrow \exp(-A' s_K^{5/3}), \quad A' = A s_{\text{anchor}}^{1/3}, \quad s_{\text{anchor}} = 0.30''$$

Inside the calibrated range the two laws agree to better than 0.005 Strehl, so the §6 validation is unaffected; beyond it the $5/3$ law rolls off at the physical rate. On the HIP 88553 window the prediction moves from 0.29–0.42 to 0.41–0.49 against the delivered 0.50–0.56 (the remaining gap is consistent with the DIMM sampling a different sightline at ZA 65° and one seeing sample being a transient spike). The law applies to both telescopes — K1's DM-stroke steepening is preserved relatively — and airmass is only the vehicle: a genuinely bad $>1''$ zenith night would have been under-predicted identically. --ngs-seeing-law gaussian selects the Gaussian fit form instead; the active law and anchor are recorded in the console summary and CSV provenance. Note the LGS/LTAO budget needed no

analogous change: its terms scale as seeing^{5/6} in amplitude, so variance $\propto$ airmass was already built in — the NGS fit's quadratic was the odd one out.

(7) Static/calibration error, broken out per telescope. The high-order static/calibration allocation is broken out into the five physical sub-groups of the error-budget sheet, each the quadrature sum of its rows (nm RMS): telescope aberration (uncorrectable static + dynamic), WFS zero-point calibration (static + dynamic + leaky-integrator), DM (hysteresis + drive digitization), uncorrectable AO-system + instrument, and DM-to-lenslet registration (misregistration + pupil-scale). They enter the Maréchal sum in quadrature. The telescope-aberration group is per-telescope: K1's primary + segment figure is worse than K2's — uncorrectable static telescope aberration 66 nm on K1 vs 47 nm on K2 — so the group is 68.2 nm (K1) / 50.0 nm (K2). The static subtotal is 91.3 nm (K2) / 102.4 nm (K1); adding Na focus 57 nm reproduces the sheet's 107 nm static-block subtotal. The per-telescope values apply to the legacy budget too (a fixed allocation, not part of the legacy methodology). All five groups are individual WFE sliders (the telescope pair greys with the unselected telescope, like the DM-fitting rows); the CSV/PNG provenance lists the per-telescope decomposition.

3.5 θ_0 and d_0 from the profile

The isoplanatic angle and the cone-effect parameter are computed per sample from the six C_n^2 bins, calibrated to reproduce the HAKA report's reference case ($\theta_0 = 3.13''$, $d_0 = 1.36$ m at 0.30'' free-atmosphere seeing, 500 nm). θ_0 is altitude-weighted ($h^{5/3}$) and scales with wavelength to the 6/5; the commonly quoted ~15'' for Maunakea is a K-band figure — the 500 nm value is ~2–3''. d_0 is wavelength-independent and is carried in the CSV.

$$\theta_0^{-5/3} \propto \sec \zeta \int C_n^2(h) h^{5/3} dh, \quad \theta_0(\lambda) \propto \lambda^{6/5}$$

3.6 Line-of-sight projection

MKWC seeing is reported corrected to zenith. With --zenith-angle, or per sample from the target's own airmass when --target is given, the tool projects the atmosphere onto the actual line of sight: seeing grows as airmass^{3/5}, which propagates into every seeing-scaled term, worsens the cone effect, and shrinks θ_0 . Inside the observing window(s) the target's airmass is used; outside, the fixed baseline.

3.7 PSF size (FWHM)

Strehl fixes the PSF peak but not its width, so --report fwhm (and both) adds a per-sample FWHM at the science wavelength. The estimate is built from a physical PSF model, and — importantly — is reported under two conventions, because two on-sky measurement tools disagree by ~20 % in poor conditions and the model reconciles both (§6.1).

Three-component PSF. Each sample's PSF is the sum of, in energy:

- **(1) Diffraction core** — the Airy pattern for a $D = 9.96$ m circular aperture (the circle inscribed in the Keck hexagon), convolved with the residual tip-tilt jitter. Tip-tilt is image motion, not a wavefront-phase error: it smears the core rather than only lowering its peak. The jitter σ is the budget's tip-tilt term expressed back as one-axis milliarcseconds. The core carries the high-order Strehl $S_{ho} = S / M(\sigma_{tt})$, where M is the Maréchal factor of the tip-tilt term — dividing the tilt out here prevents charging it twice. The convolution is done exactly in the OTF domain.
- **(2) Corrected-band shoulder** — a Moffat ($\beta = 4.765$) of FWHM equal to the AO control radius $\theta_c = (N/2) \cdot \lambda/D$, where N is the DM actuator count across the pupil (K1: 20; K2/HAKA: 57). It carries the corrected-band residual energy (servo lag, measurement, aliasing ...).
- **(3) Seeing wings** — the uncorrected halo, a Moffat matched to the line-of-sight seeing disk (Kolmogorov $\lambda^{(-1/5)}$ scaling), carrying the fitting-error share.

The non-core energy ($1 - S_{ho}$) splits between shoulder and wings in proportion to variance: $E_{wings} = (1 - S_{ho}) \cdot (\sigma_{fit}/\sigma_{ho})^2$, shoulder the remainder, where σ_{fit} is the per-sample fitting term and σ_{ho} the total high-order residual. There are no tuned constants in the PSF — θ_c comes from the DM and the split from the budget.

Three FWHM conventions. The same model PSF is reduced three ways. A note on mechanism first: none of the three is a confirmed mechanism match for the summit IDL tools behind §6.1's validation (fwhmastro.pro, OSIRISSTREHL_QL2.pro, mpfitpeak.pro) — see the notes below.

- **half-max (psf_fwhm_mas):** the first radius where the profile falls to half its peak. Because the Airy core out-peaks the halo ~100:1 at moderate Strehl, the half-max stays on the core and this convention is largely

insensitive to the shoulder. No real image tool is confirmed to measure exactly this; it is a clean, tool-independent reference.

- **Gaussian-fit, no background (fwhm_gaussfit_mas):** an isotropic Gaussian fit with free amplitude but no free background, over a box of radius `--fwhm-box-mas` (default 300 mas). This is the mechanism match for OSIRISSTREHL_QL2.pro, the OSIRIS quicklook tool's own (rarely-used) Strehl button — confirmed from source, though that tool's own box is a fixed ≈ 30.7 mas at K band, not the 300 mas default kept here to preserve the published §6.1 numbers. A fit does feel the shoulder, so it reads higher in poor conditions than the half-max convention.

Gaussian-fit with background (fwhm_gaussfit_sky_mas): the same isotropic Gaussian fit but with a free background term, over the same `--fwhm-box-mas` box. This is the mechanism match for the OSIRIS quicklook tool's separate, more commonly used hand-drawn-box feature (an informed inference — MPFITPEAK's default Gaussian-plus-free-background, not confirmed from source, but the closest analog to a user dragging a box on real sky). A free background absorbs flat halo/sky flux instead of forcing it into a wider fitted σ , so this convention reads narrower than the no-background fit on the same PSF. Note that the standalone AO Strehl tool itself (fwhmastro.pro) is a full 2-D elliptical fit; the engine's PSF is azimuthally symmetric, so no convention here is a complete mechanism match for it — these three conventions match the two OSIRIS quicklook mechanisms and provide a clean tool-independent reference, not a stand-in for the elliptical fit.

A fourth convention: as the SR tool reads it. `fwhm_srtool_mas` is the default convention and the first entry in every FWHM selector. The other three ask what a tool of a given mechanism would read off the analytic profile; this one instead runs the estimator's own Measured-SR tab — its `find_fwhm.pro` port (§10) — on the model PSF rendered as a NIRC2 detector frame with the 1.2–1.4" annulus sky removed. It is therefore the only convention comparable to a measured FWHM without a convention caveat, and it is what the Measured-SR tab's Predicted FWHM box reports. Reports and figures quote it unless another convention is explicitly selected (`--fwhm-curves`); the other three remain available and every CSV still carries all four column sets.

Validation, and two results that contradict the natural expectation. Real NIRC2 frames were measured with the tool, and each convention then evaluated at that frame's own measured Strehl — so only the convention is under test, not the Strehl model. Median (convention – measured), in mas:

Frame set	n	srtool	half-max	gaussfit	gaussfit-sky
20260727 o Her (isolated standard)	60	–0.37	–1.41	–3.84	–4.09
20260730 Galactic Centre (crowded)	45	–8.86	–9.99	–11.62	–12.58
20260728 M92 (crowded)	18	–16.62	–17.71	–18.61	–20.26

Sign. At a given Strehl the model reads slightly low, not high. The several-mas overestimates in the predicted-vs-delivered FWHM tables are therefore attributable to the Strehl prediction, not to the FWHM convention, and no change of convention will close them. What this entry fixes is the ≈ 1 mas that was the convention's.

Mechanism. The annulus sky — the obvious candidate, since the tool measures on image – sky — is $\leq 2 \times 10^{-4}$ of the peak here and moves the FWHM by under 0.01 mas. The whole ≈ 1.1 mas is the 1-px annulus binning: the innermost bin ($r < 0.707$ px) is a mean, so the apparent peak sits below the true one, the half-max level falls with it, and the crossing moves outward. The tool reads wider than a continuum half-max, not narrower. Read the o Her row as the convention test — the crowded rows are dominated by blended neighbours inflating the measured FWHM, and only their ordering is informative.

All three agree near $S \approx 1$ and diverge as S falls; the spread across them is a diagnostic of where the light sits (core-dominated vs shoulder-dominated), and the free-background convention sits narrower than the no-background one at fixed box radius. At $S \leq 0.25$ a lone FWHM is ill-posed — quote it with the Strehl.

NGS tilt-servo residual. A bright on-axis NGS has, structurally, no seeing-scaling tip-tilt in the budget (its jitter is a fixed wind-shake/margin floor), so without a correction the modelled NGS FWHM would be flat. The missing physics is servo-lag residual on atmospheric tilt, amplitude $\propto \varepsilon^{5/6}$. This is supplied by `--ngs-tilt-servo` (default 6.189 mas one-axis at the reference profile), calibrated to a single on-sky anchor (≈ 52 mas at K-band seeing 0.6") and applied only in the FWHM path — it never enters the reported Strehl.

AO PSF FWHM (K-band (2200 nm)) vs. time — 2026-06-30 (K1, LTAO)

seeing disk (no AO, not plotted): median 610 mas range 364-1221 mas

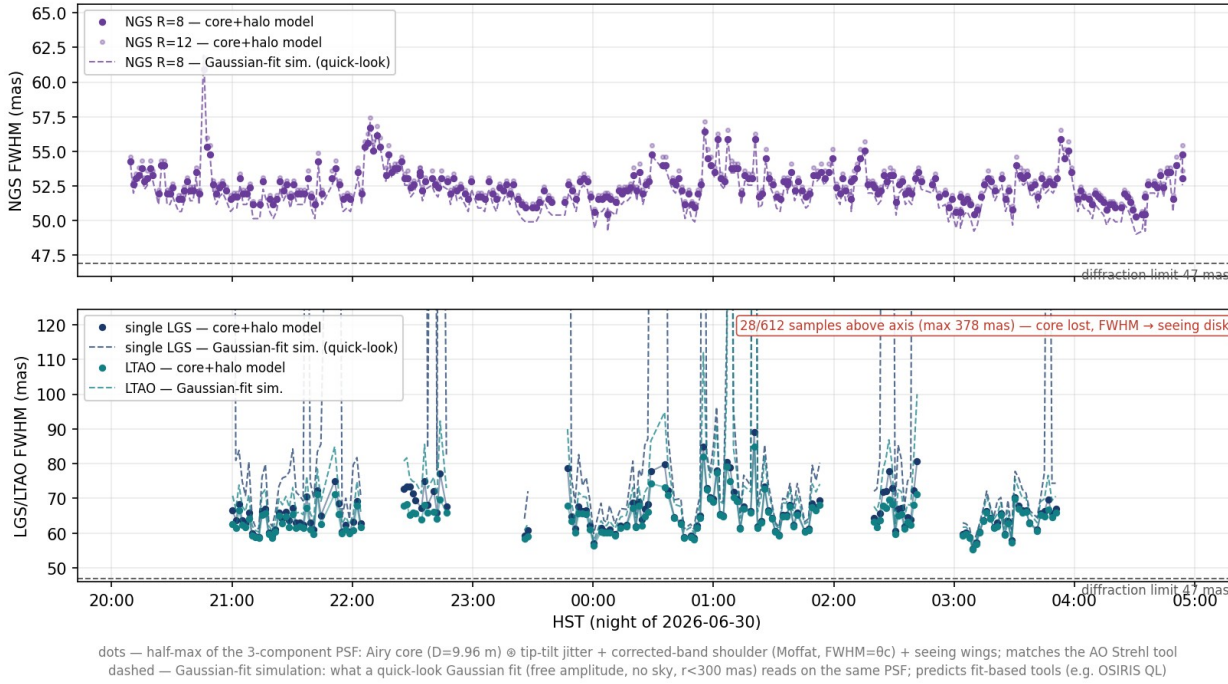


Figure 1 — FWHM vs time for the night of 2026-06-30 (K1, high airmass): dots = half-max convention, dashed = Gaussian-fit simulation. NGS (top) tracks seeing slowly; LGS/LTAO (bottom) shows the two conventions splitting as conditions worsen.

3.8 Hypothetical-condition scenarios (field-map prediction)

The field map and the error-terms view (§9) can be driven by a synthetic atmospheric snapshot instead of a night's data, so performance can be predicted for conditions the user dials in: zenith DIMM and MASS seeing, θ_0 (K band, zenith) and a zenith angle (synthetic_field_snapshot()) in the engine). No Run — and no loaded night — is required: with nothing loaded, the GUI computes the map and the term breakdown directly from the current control settings and the scenario. Ingredients, none fitted:

- **Line-of-sight projection.** Inputs are zenith values (the MKWC convention); seeing is projected by the §3.1 factor $X^{(3/5)}$ and θ_0 by $X^{(-8/5)}$.
- **Synthesized Cn^2 profile.** The LTAO layer-mismatch penalty (§3.4 (2)) needs a 6-bin profile. Its total is pinned by the free-atm seeing through the Kolmogorov inversion $\varepsilon = 0.98 \lambda/r_0$, $r_0 = [0.423 \text{ k}^2 \Sigma J]^{(-3/5)}$ — applied to a masspro file's own six bins this reproduces its seeing column to ~1 % (median, 244 profiles). Its shape is the measured median Mauna Kea free-atmosphere profile at the six MASS altitudes (0.5/1/2/4/8/16 km above the summit), tilted in altitude, $J_i \propto \text{shape}_i \cdot h_i^\alpha$, with α solved (bisection) so the profile's θ_0 matches the requested one: a small θ_0 pushes turbulence aloft, a large one pulls it down, and the layer mismatch m (vs the reconstructor prior, §3.4 (2)) follows from the same profile. The MK shape is tropopause-dominated — the strongest free-atm layer is the 8 km MASS bin (whose triangular response integrates ~6–12 km above the summit, i.e. the jet/tropopause), with the 4 km layer comparatively weak. Aloft values are the RAVEN Maunakea median (Ono et al. 2016; consistent with the MASS/SCIDAR campaign of Tokovinin et al. 2005, where the 8 and 16 km layers dominate the free atmosphere). This is distinct from the reconstructor prior of §3.4 (2), which is a fixed RTC design assumption left untouched — so a median MK night carries a real, nonzero layer mismatch ($m \approx 0.19$) against that prior.
- **Anisoplanatism re-weighting.** The budget's free-atm scalings ($s_{fa} = (\varepsilon_{fa}/0.30'')^{(5/6)}$) assume the median MK profile's altitude distribution — equivalent to assuming $\theta_0 \propto 1/\varepsilon_{fa}$. When a scenario decouples them, the altitude moment $\Sigma J h^{(5/3)}$ is re-weighted by $g^{(5/3)}$ with $g = \theta_0 \text{MK}(\varepsilon_{fa})/\theta_0 \text{requested}$ (exact, by construction of the α solve), so the laser angular-anisoplanatism term and the TT-star anisoplanatism rows carry the factor $g^{(5/6)}$. Every night-data path uses factor 1, preserving the invariant that the map's centre reproduces the timeline's science estimate.
- **θ_0 self-consistency by default.** In the GUI, θ_0 auto-tracks the median-MK value at the current seeing ($g = 1$) unless explicitly overridden, so a profile-shape deviation is always a deliberate assertion.

The reference scenario (0.5"/0.3", median MK shape) independently lands $\theta_0^K = 17.7''$ at zenith, beside the tool's long-standing 15" assumed- θ_0 default. The GUI Prediction tab plots the synthesized profile two ways sharing the altitude axis: layer-integrated $Cn^2 \cdot dh$ per bin (where the 8 km tropopause peak is visible — this is what MASS measures and the model uses) and Cn^2 density (textbook units, but MASS's thick bins smear the tropopause so it reads ground-heavy). The ground layer (total – free-atm, below MASS sensing) is shown separately in both.

Layer mode. The Prediction tab's Layers page (KAON 1556 §8) sets the scenario's turbulence directly on the reconstructor's altitude grid — ground plus the six MASS bins at 0.5, 1, 2, 4, 8 and 16 km — as turbulence fractions $f_i = J_i / \sum J$, the reconstructor's own units (§3.4 (2)), summing to 1. The scale is fixed by the total (DIMM) seeing through the same Kolmogorov inversion as above, so $J_i = f_i \times J(\epsilon_{\text{tot}})$; the free-atm seeing follows as the seeing of the six aloft layers, $\epsilon_{\text{fa}} = \epsilon_{\text{tot}} \times (\sum_{i \geq 1} f_i)^{(3/5)}$, and the ground layer's seeing-equivalent likewise from f_0 — seeing combines as a 5/3-power sum, never a plain one (0.4" and 0.3" of turbulence make 0.53", not 0.7"). The six aloft layers are then the scenario's Cn^2 profile verbatim: the layer-mismatch m of §3.4 (2) is that profile's total-variation distance from the prior's aloft fractions (exactly 0 when the layers are the prior), and θ_0 (auto) is its own isoplanatic angle from the altitude moment. No altitude tilt α is solved — the synthesized-profile machinery above is bypassed — but the anisoplanatism re-weighting still applies unchanged, with the reference θ_0 the median-MK-shape value at the same free-atm seeing, so a ground-heavy layer set relieves the aniso terms and an aloft-heavy one penalizes them exactly as an equivalent θ_0 request would; an explicit θ_0 still overrides. The two-number DIMM/MASS path is untouched by any of this. In the engine, layers.py carries RECON_HEIGHTS_M, fractions_to_layers(), recon_prior_layers(), layers_seeing() and its inverses, and synthetic_field_snapshot(..., cn2_layers=J7) takes the seven layers verbatim (alpha reported as NaN, the layers returned under cn2_layers); there is no CLI flag for layer fractions.

3.9 Tip-tilt sensor laws and their limits

Two findings from the 2026 KAPA characterization campaign and the companion sky-coverage analysis bear on the tip-tilt measurement rows. Both concern the faint end, and both are recorded here because the estimator's behaviour there is explicitly an assumption rather than a validated prediction.

STRAP: an optional shape correction, and why it is off by default. The campaign's predicted-vs-delivered ladder showed the residual (measured – predicted Strehl) tracking TT magnitude — about –0.115 at $R = 10$ and +0.084 at $R = 15.6$, crossing zero near $R = 13.2$ — suggesting the model's magnitude dependence is too steep. Matching the measured shape, with the row re-anchored so the crossing magnitude is unchanged, gives STRAP_MEAS_REF 6.90 → 7.83 mas and STRAP_BETA 0.116 → 0.070. This is available as --strap-law onsky2026 and is OFF by default.

The trend is attributed to an instrumental artefact. STRAP inserts an ND filter on stars brighter than $R \approx 10$ –10.5, and the filter introduces defocus and hence centroid error. Every negative point in the ladder sits at $R \leq 10$; that one bin supplies 8 of the 21 points and the entire downward slope. Excluding it, the slope is consistent with zero and the crossing magnitude is unconstrained across the whole sampled range:

Basis	n	Slope / mag	t	Crossing R
All 21 ladder points (what the correction fitted)	21	+0.0345 ± 0.0050	6.86	12.99 ± 0.36
ND-affected bin ($R \leq 10$) excluded	13	+0.0189 ± 0.0110	1.72	11.4 ± 2.0

The correction is therefore withdrawn as a campaign finding. It remains selectable, but the constant STRAP_PIVOT_ONSKY = 13.2 is precisely the crossing magnitude this audit withdraws, so it no longer has a measured basis, and the monotone ladder should be read as an artefact demonstration rather than a measurement of the law. Two loose ends are flagged rather than reconciled: the ND insertion rule itself is assumed ($R < 11$), not sourced — no header keyword records the ND state, with OBSNNAME reading "open" on all 4165 campaign science frames, so a Keck AO citation is still needed — and one rung at $R = 9.8$ shows a residual less negative than the $R = 10$ rows, which the ND hypothesis alone does not explain.

Independent of the artefact, the correction was never adoptable as a fit. Attributing the whole residual to this row is arithmetically impossible: it would require removing ~63 600 nm² of TT variance at $R = 15.6$ from a term totalling ~69 500 nm² whose magnitude-independent floor (bandwidth, wind shake, margin) is ~18 500 nm² — the unconstrained fit demands a negative magnitude slope, i.e. tip-tilt error improving as the star gets fainter. $\beta = 0.070$ is also flatter than photon statistics (a photon-limited quad-cell gives 0.2; the existing row is already sub-photon at 0.116, justified by faint-end gain optimisation). And the faint-end consequences are large and unmeasured: the two laws diverge strongly faint of the data, where the default row steepens along the sheet slopes into the open-loop

ceiling (§3.4 (3a), (3c)) — 59–100 mas one-axis at $R = 18$ –19, essentially zero delivered K Strehl — while the campaign’s faintest STRAP rung is $R = 15.6$. The recommended next measurement is a STRAP magnitude ladder on isolated field stars in good seeing reaching $R \approx 17$ –18; every ladder in the 2026 campaign sits in a crowded cluster field, mostly on mediocre nights — the one configuration that cannot separate a sensor law from field and conditions systematics.

TRICK: the faint-end row collapsed, and is now capped. TRICK_BETA = 0.60 is an empirical faint-end roll-off calibrated to the 2026-05-28 M3 ladder, which spans $H \approx 9.5$ –14.5. Extrapolated beyond that range it grew the measurement row by roughly $4\times$ per magnitude, saturating against the open-loop tilt ceiling (74.4 mas one-axis at the default $L_0 = 50$ m, §3.4 (3c)) — the capped row saturates fully by $H \approx 17.8$ — and driving the uncapped predicted Strehl to zero within rounding (≤ 0.001) by $H = 16$ — while TRICK is stated to close the loop to $H = 17$, and while a high-order LGS loop with no tip-tilt correction still delivers something. Driving Strehl to zero is the Maréchal mapping used far outside its validity. The fix applies the same physical argument already used for the off-axis spot cap: once the corrected core is gone the spot is seeing-limited, TRICK is centroiding the same kind of image STRAP always centroids, and extra magnitude can only cost signal-to-noise — it cannot degrade the spot further. The measurement row is therefore capped by the seeing-limited (STRAP-form) law at TRICK’s own sensing magnitude, with the spot inflation not applied on top of the capped branch. The two branches cross at $H \approx 14.7$, within noise of the $H \approx 14.5$ drop-off the 0528 ladder measured.

What the cap does, in numbers. On-axis, median scenario, LTAO on K1, K band (2200 nm), at the current defaults ($L_0 = 50$ m ceiling, 74.4 mas one-axis). Single-beacon LGS values sit just below the LTAO ones: 0.293 / 0.131 / 0.018 capped and 0.198 / 0.000 / 0.000 uncapped at $H = 15$ / 16 / 17. Only rows beyond the $H = 14.7$ knee appear, because that is the only place the cap acts: at every magnitude the M3 ladder actually reaches, the capped and uncapped branches are identical to the last digit, so there is no “before the cap” number to quote for any measured rung.

H (on-axis, median scenario, K band)	SR, cap off	SR, cap on
15.0	0.208	0.307
16.0	0.001	0.138
17.0	0.001	0.018

What that table is really saying. Everything the cap changes is unmeasured by construction. The branches cross at $H \approx 14.7$ and the deepest rung any campaign ladder reaches is $H = 14.5$, so no row in the table has a measurement beside it and none should be supplied: the nearest thing, 0528’s M13 K rung at $H = 15.0$, is a different field in a different band whose H rungs are flagged suspect, and it is deliberately not tabulated as a substitute. Note which direction the cap runs. The uncapped column is the non-physical branch, not the conservative one — it is the runaway the cap exists to stop, collapsing to 0.006 by $H = 16$ as the measurement row saturates against the open-loop tilt ceiling — so the cap raises the faint end rather than trimming it. The capped values at $H = 16$ –17 are therefore a physically bounded assumption rather than a validated prediction, and the sky-coverage faint end that rests on them (companion sky-coverage note) inherits that status. The harness passes byte-identical, since the default TT magnitude sits on the uncapped branch.

The M3 ladder against the model, at matched conditions (2026-08-07). The measured column removed above was mismatched against its prediction in three separate ways: the prediction was on-axis while every rung of the ladder is a real cluster star at a real offset, the prediction was the median scenario while the ladder was observed on one night, and the measurement is BrGamma (2166 nm) while the prediction was K (2200 nm). Each rung is evaluated at its own tabulated offset (subject to the registration caveat below), at the night’s own MKWC conditions at that rung’s own frame time, and at 2166 nm, with LTAO on K1 and the §3.4 (3b) tilt- θ_0 gain enabled at 1.1. The measurements are the 2026-08-05 re-measure of the ladder, drift-corrected. One property of the ladder’s construction matters for reading it: every rung is the same single hand-picked science star, observed under a different guide star, so star-to-star field variation differences out and what varies across rungs is guide-star dependence. It is also why the three brightest rungs return an identical Strehl — the loop is floor-limited there, and it is one measurement, not three that happen to agree.

H	offset "	measured SR	predicted SR	predicted – measured
9.4 †	5.79	0.4728 ± 0.0228	0.5578	+0.085
9.5 ‡	12.23	0.4728 ± 0.0346	0.4193	–0.054
10.0	5.99	0.4728 ± 0.0106	0.5304	+0.058

11.5	9.40	0.4887 ± 0.0106	0.5160	+0.027
12.0	9.24	0.5402 ± 0.0312	0.4932	-0.047
12.2	9.25	0.3951 ± 0.0398	0.4985	+0.103
13.2	9.45	0.5160 ± 0.0681	0.4990	-0.017
14.0	9.40	0.4326 ± 0.0342	0.4381	+0.006
14.5 †	15.29	0.2983 ± 0.0109	0.2530	-0.045

Reading that table. Six of the nine rungs are clean and three are flagged. ‡ H = 9.5 sits 907 s from the nearest MASS sample, beyond the tool's 600 s match tolerance, so its free-atmosphere input is borrowed from a quarter of an hour later. † H = 9.4 and H = 14.5 are single-frame blocks that the re-measure itself dropped, with times and ROI positions recovered from the first-pass frame table and the night log. Over the six clean rungs the weighted mean residual is $+0.038 \pm 0.007$, falling to $+0.019$ once the crowding shift below is applied to the measurements, and it carries no significant magnitude trend (-0.017 ± 0.007 SR/mag on six points). Offsets are TRICK's own ROI positions converted at 50 mas/px, referenced to a TRICK field centre assumed coaxial with the OSIRIS imager centre. That registration assumption is wrong: the rungs tabulated near 9.2–9.5" are stars sitting at the centre of the OSIRIS imager field of view, which implies the two centres are separated by roughly that amount rather than coincident. The tabulated offsets are therefore ROI positions under an incorrect registration, not true guide-star offsets, and the matched-conditions evaluation — which consumed them as tip-tilt offsets — inherits that error. The true registration has not yet been measured; the residuals here carry this caveat until the ladder is re-reduced against it. Conditions are quoted at zenith; M3 sat near 80° elevation, worth under 0.003 Strehl, and the frames' own EFFWAVE of 2169 nm against the 2166 nm used here is worth about 0.001.

The model against this ladder. At the tabulated offsets and matched conditions the H = 14.0 rung reads 0.4381 predicted against 0.4326 measured, and across the ladder the model runs about two hundredths high — the size of the crowding correction's own uncertainty. The cap acts only beyond H = 14.7, so every measured rung sits on the uncapped branch.

Crowding audit of the ladder (PSF-fit cleaning, 2026-08-05). M3 is a globular cluster, and the standing objection to every ladder in the 2026 campaign is that a crowded field cannot separate a sensor law from field systematics. The PSF-fit neighbour subtraction (§10) was run against this ladder to test that directly — the first real science field on which the empirical PSF could be built at all (§8). It separates the two cleanly: cleaning moves the level of the ladder and not its shape. The shift is flat with guide magnitude, at -0.0028 SR/mag across H 9.4–14.5, and the knee stays between H 13.2 and 14.5. TRICK_BETA = 0.60 and TRICK_MAG0 = 14.0 are shape parameters, and they survive the audit. What the audit cannot settle is the level.

Two shift values, and which one applies. The paired median over all matched field stars is $+0.038$, but that population is dominated by stars at $SR \approx 0.28$ while the ladder's own star sits near $SR \approx 0.5$, where the measured shift is smaller: restricted to the $SR \geq 0.40$ regime it is $+0.019 \pm 0.007$, and that is the value used above. Applying the all-star median to a peak star would overstate the correction by about a factor of two. The two populations differ in another way worth stating plainly — the shift is measured on several field stars per block, while the ladder it is applied to is the one science star. The level is in any case not adoptable as a correction: at $SR \approx 0.5$ the ladder sits above the $|\Delta SR| \leq 0.02$ range over which neighbour subtraction is validated (§10), where the residual error changes sign and the tool flags the result as an upper bound. $+0.019$ is comparable to that documented envelope bias, so it is an upper limit on real recovery rather than a measured level correction.

4. Running the tool from the command line

The interactive GUI (§9; user's manual in KAON 1556) is the primary interface, and nearly all users should start there — every control in it maps onto the engine described in §3. The command line documented in this section drives the same engine for scripted, batch, and reproducible use; its flag tables also serve as the reference for what each GUI control does.

Invocation. Once the package is installed (§1), the tool is run as keck-ao-estimator (CLI) and keck-ao-estimator-gui (GUI) — e.g. `keck-ao-estimator --telescope K1 --target ....`

Basic run on local files (K2 defaults, single-beacon panel):

```
keck-ao-estimator --dimm 20260525_dimm.dat \
  --mass 20260525_mass.dat --masspro 20260525_masspro.dat
```

Fetch a night straight from the MKWC archive by its morning/UT date stamp:

```
keck-ao-estimator --fetch-date 20260525
```

K1 (LTAO panel on by default), with a science target, observing window, and the science star as its own on-axis R = 9.9 tip-tilt reference:

```
keck-ao-estimator --fetch-date 20251204 --telescope K1 \
  --target --target-name 'HD 18770' --ra <RA> --dec=<DEC> \
  --window 23:00-00:15 --tt-mag 9.9 --tt-offset 0
```

Notes: negative declinations need the equals form (--dec=-29d00m28s). On firewalled summit machines astropy's IERS auto-download is already disabled in the script. --legacy-budget selects the simplified legacy budget (§7); --band/--wavelength selects the science wavelength; --lgs-offset overrides the per-telescope laser offset (K1: 4.97", K2: 0"); --ngs-offset/--assumed-theta0 configure an off-axis NGS (§3.2); --ltao-bw-floor-frac tunes the LTAO bandwidth penalty (§3.4 (5)); --ltao-tt-theta0-gain sets the optional LTAO tilt-anisoplanatism reduction (§3.4 (3b); default 1.0 = off); --outer-scale sets the outer scale used by the tip-tilt open-loop ceiling (§3.4 (3c); default 50 m, inf = Kolmogorov); --ngs-seeing-law selects the NGS seeing law (§3.4 (6)); --no-terms-plot suppresses the companion figure and --no-combined the single-file quick-look image (§5); a missing image extension on --out is normalized to .png; --force overwrites existing outputs.

FWHM and physics inputs. The following optional flags expose the FWHM reporting and the physics inputs; their defaults leave the standard behaviour unchanged:

Flag	Default	Effect
--report {strehl,fwhm,both}	strehl	add / substitute the FWHM figure & CSV columns (§3.7, §5)
--wind-ground M_S	8	boundary-layer wind for the wind-weighted bandwidth term
--wind-free M_S	25	free-atmosphere wind for the same term
--ngs-s0 / --ngs-a / --ngs-m0 / --ngs-w	fitted	override the NGS Gompertz fit terms (active telescope; recalibration)
--k1-quadcell-penalty S	0.05	K1 flat quadcell-saturation Strehl penalty
--ngs-tilt-servo MAS	6.189	NGS atmospheric tilt-servo residual (FWHM path only)
--tt-sensor {strap,trick-h,trick-k}	strap	tip-tilt sensor: refined STRAP (§3.4 (3a)) / K1 TRICK (§3.4 (3d), swaps the science band)
--fwhm-box-mas MAS	300	fit-domain radius shared by both Gaussian-fit FWHM conventions (§3.7, 2026-07-19 (b)); the OSIRIS quicklook tool's own box is hand-drawn, so 300 is kept to preserve the published §6.1 numbers, not a measured value
--outer-scale L0	50	fit-domain radius shared by both Gaussian-fit FWHM conventions (§3.7, 2026-07-19 (b)); the OSIRIS quicklook tool's own box is hand-drawn, so 50 is kept to preserve the published §6.1 numbers, not a measured value

NGS, and LTAO Strehl (K-band (2200 nm)) vs. seeing — 2026-05-24 (K1)

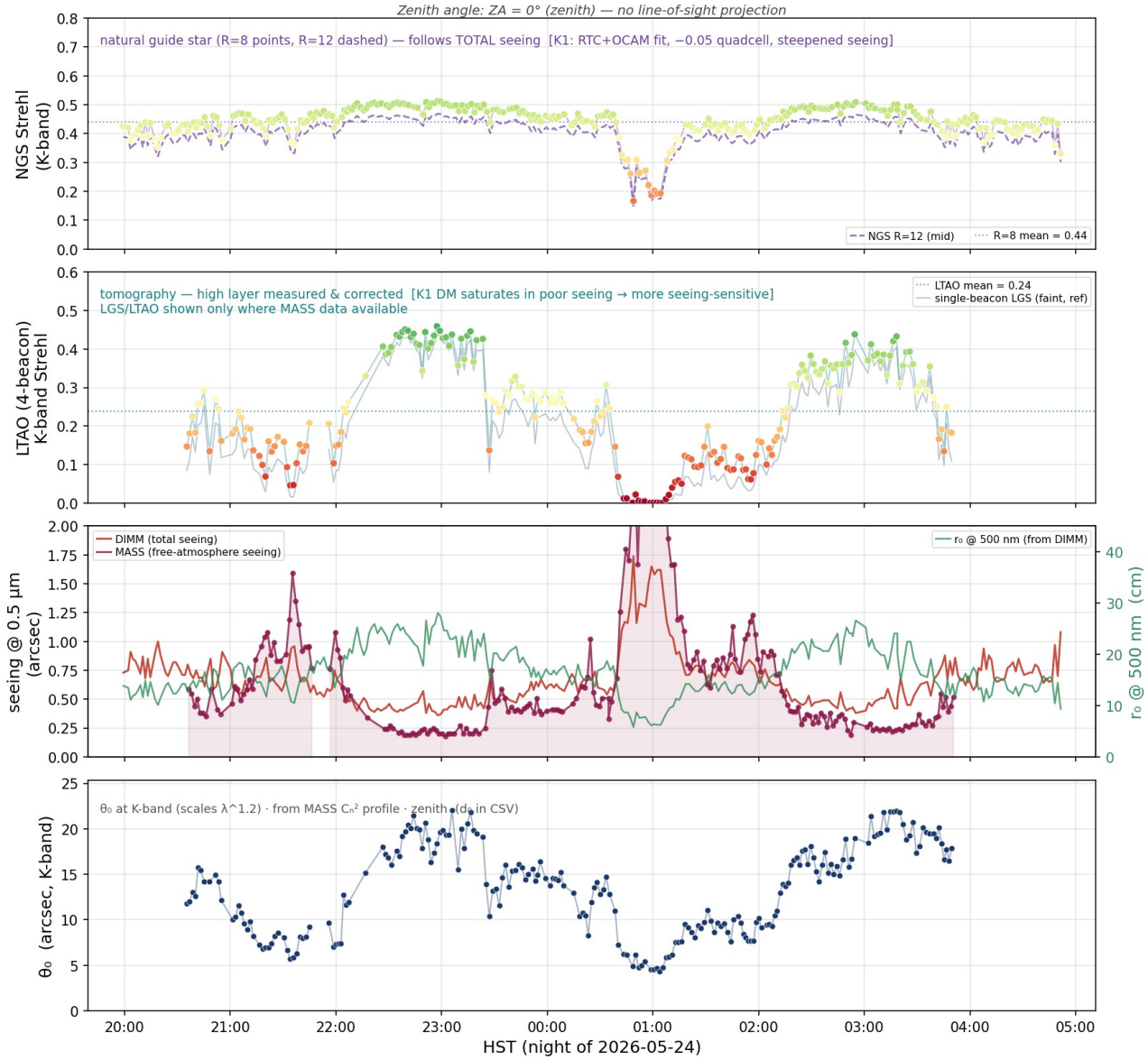


Figure 2 — K1 run on the night of 2026-05-24: NGS Strehl (top), LTAO with faint single-beacon reference (second), DIMM/MASS seeing with r_0 (third), θ_0 at K band (bottom). Line breaks mark real gaps in MASS coverage. With --target, the science target's airmass is overlaid (solid where unvignetted, dashed inside the Keck deck-shadow wedge), the observing window is boxed, and all seeing-scaled terms are projected onto the target's line of sight inside the window.

5. Outputs

Each run writes a PNG (auto-named from the data's UT date and telescope unless --out is given) and a CSV with the same base name. The CSV is a merged timeline of both timebases, one row per sample, distinguished by the leading source column: dimm rows carry the DIMM seeing, r_0 , the NGS Strehl estimates and (for off-axis NGS) the θ_0 actually used; mass rows carry the free-atmosphere seeing, LGS/LTAO Strehl, θ_0 , d_0 , the layer-mismatch value m , the matched DIMM total seeing used as input, and the full set of per-sample error-budget terms in nm RMS (err_*_nm columns: fitting, scintillation, angular anisoplanatism, bandwidth and the focal-aniso/tomography altitude term for both laser modes, and tip-tilt; the four fixed terms are stated in a comment line). The Strehl columns are exactly the Maréchal quadrature of these terms, so any prediction can be audited term by term. Comment lines at the top record the night, telescope, budget version, TT-star and laser-offset configuration, so every file is self-documenting.

The companion error-terms figure

Whenever MASS profiles exist, the script also writes `ao_strehl_<UT>_<telescope>_terms.png` (suppress with `--no-terms-plot`): the non-static budget terms vs time. Rows 1–3 show the nm-RMS terms — fitting, scintillation, angular anisoplanatism (labelled with the offset actually charged, with a dashed 1" imperfect-centering reference), tip-tilt (labelled with the configured star, with a dashed $R = 10$ on-axis reference), and the mode-dependent pairs sharing panels: bandwidth (single vs LTAO half-rate) and the altitude term (focal aniso vs tomography+mismatch — whose gap is the tomographic gain). The bottom row shows each term's share of the total wavefront-error variance as 100%-stacked areas for both laser modes, fixed terms included; variances are what add in the Maréchal product, so percent-of-variance is the physically meaningful notion of contribution. All panels break their lines across MASS coverage gaps, like the main figure. In the GUI, the Error-terms tab shows the same per-sample figure for a loaded night; under an active Prediction scenario it switches to a snapshot term breakdown at the scenario's conditions (§9).

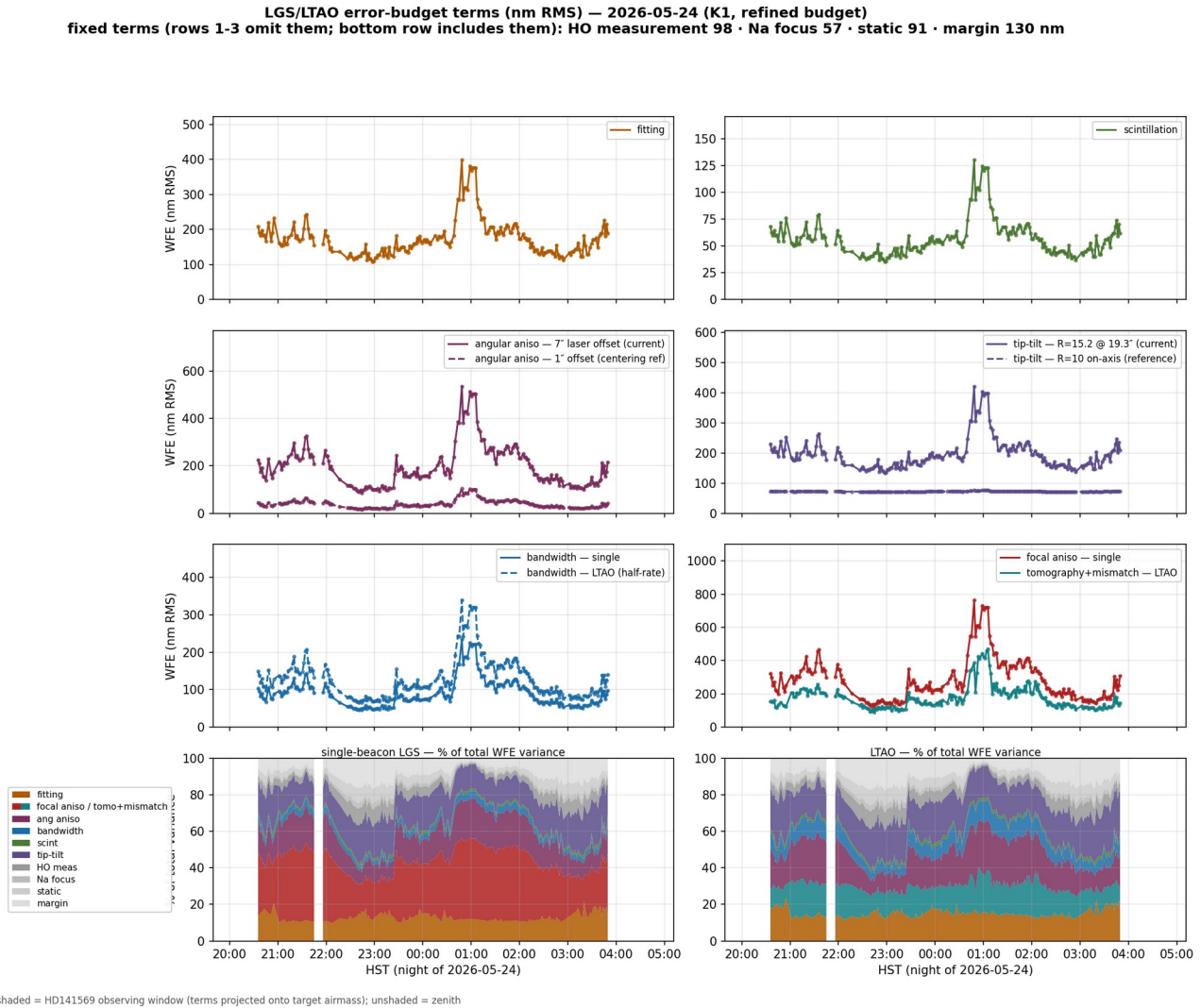


Figure 3 — the companion error-terms figure for the 2026-05-24 K1 night (UT stamp 20260525). Bottom row: percent-of-variance breakdown for single-beacon LGS (left) and LTAO (right).

The combined quick-look image

For one-file night review the script also writes `ao_strehl_<UT>_<telescope>_all.png`: the main timeline and the error-terms figure composited side by side (timeline left, terms right) at full native resolution — the two PNGs are stitched, not re-rendered, so the combined image can never drift from the separates. It is produced whenever the terms figure is (i.e. whenever MASS profiles exist); `--no-combined` skips it. All three images plus the CSV follow the UT-date naming convention.

FWHM figure and columns (`--report fwhm/both`). `fwhm` replaces the main figure with a two-panel FWHM timeline (linear mas, diffraction limit annotated); both overlays FWHM on right-hand axes of the Strehl figure. By default the

figure draws only the half-max convention; --fwhm-curves {srtool, halfmax, gaussfit, gaussfit-sky, both, all} (GUI: a “FWHM” combo by Report) selects which — a plot-only choice, as all three at once is busy. The CSV always gains sixteen columns — *_fwhm_srtool_mas (as the SR tool reads it, the default), *_fwhm_mas (half-max), *_fwhm_gaussfit_mas (Gaussian-fit, no background), and *_fwhm_gaussfit_sky_mas (Gaussian-fit with background) for each of NGS-bright, NGS-faint, single-LGS, LTAO — plus # fwhm_model=... / # fwhm_gaussfit=... / # fwhm_gaussfit_sky=... provenance lines. Absent under the default strehl report.

6. Validation against delivered performance

The refined budget was tested against delivered OSIRIS Strehl on two K1 HD 18770 nights of nearly identical total seeing ($\sim 0.6''$) but opposite profile shapes. The science star served as its own on-axis TT reference ($R = 9.9$; --tt-mag 9.9 --tt-offset 0). The LTAO-refined column uses the latency-floor bandwidth factor of §3.4 (5); medians within the observing windows:

Night (profile)	LTAO stock	LTAO refined	LTAO delivered	sLGS refined / delivered
2025-12-04 (ground-dominated, $\theta_0 \approx 28''$ K)	0.36	0.47	≈ 0.50	0.47 / ≈ 0.40
2025-12-06 (free-atm-dominated, $\theta_0 \approx 11''$ K)	0.34	0.26	≈ 0.23	0.17 / ≈ 0.17

Table 2 — in-window median K-band Strehl, stock vs refined budget vs delivered frames. The refined budget closes most of the ground-dominated night’s underprediction (driven by the TT-star and bandwidth terms) and essentially all of the free-atmosphere night’s overprediction (driven by the layer-mismatch penalty and the laser-offset correction). LTAO-refined values reflect the latency-floor bandwidth factor (1.44, §3.4 (5)); the naive $2^{5/6}=1.78$ factor would give 0.46 and 0.24 respectively — a shift of ~ 0.01 – 0.02 that is small next to the profile-driven differences.

Because seeing and airmass barely change between the two nights while the profile shape inverts, the two nights isolate the free-atmosphere dependence that dominates laser-mode performance. Figure 4 shows the predicted-vs-delivered Strehl and the θ_0 profile diagnostic: on the ground-dominated night (left) the predicted LTAO and single-beacon curves nearly coincide and both track the delivered frames; on the free-atmosphere night (right) the predicted LTAO sits well above single-beacon and both match their delivered populations, with the delivered scatter following the per-frame free-atmosphere variability rather than the nearly-flat airmass.

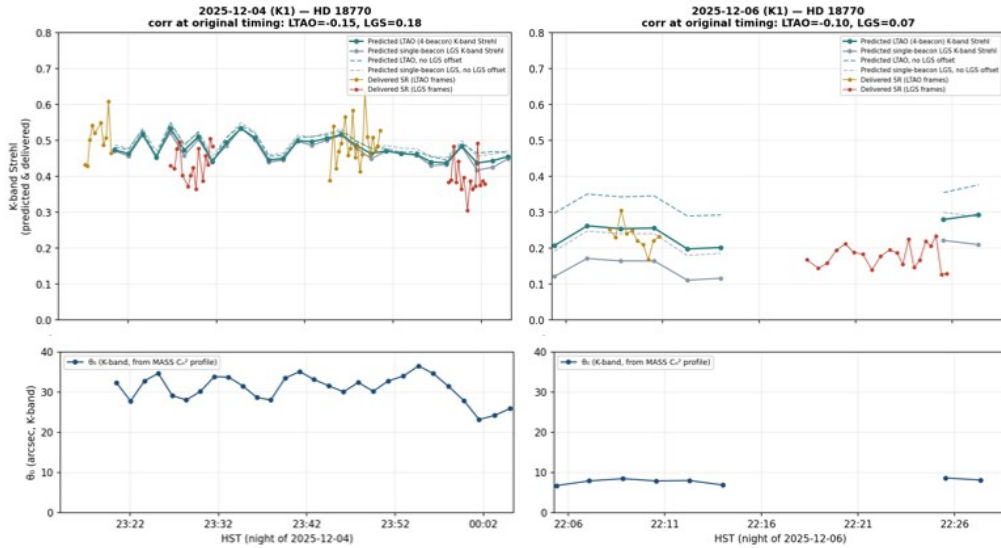


Figure 4 — predicted vs delivered K-band Strehl (top) and θ_0 at K band from the MASS C_n^2 profile (bottom) for the two validation nights (no time shift). The dashed predicted curves show the no-LGS-offset case for reference; delivered LTAO and LGS frames are plotted separately. The low θ_0 on 2025-12-06 ($\sim 8''$ vs $\sim 30''$) is the free-atmosphere-dominated profile that drives the two nights apart.

Re-check under the refined STRAP row (§3.4 (3a)). Both HD 18770 nights used an $R = 9.9$ on-axis tip-tilt reference — bright enough that the STRAP recalibration barely moves the measurement row (≈ 3.9 vs ≈ 0.6 mas against a ≈ 5.7 mas fixed floor). Re-running the two nights (data re-fetched from MKWC) under the refined row shifts

the predicted in-window LTAO K medians by -0.013 (2025-12-04) and -0.006 (2025-12-06) — small next to the profile-driven differences the section discusses, so Table 2 and Figure 4 stand as published.

6.1 FWHM validation

On 2026-06-30 00:35–01:36 HST (K1, airmass ≈ 1.8) two image tools measured the same 32 OSIRIS frames: the AO Strehl tool (fwhmast.pro, a single number) read a median FWHM of 62.8 mas; the OSIRIS quicklook tool's hand-drawn-box feature (reported $x \times y$) read 76.7 mas (ratio 1.20, $r = 0.89$) — the more plausible source of that historical number than the quicklook tool's separate, rarely-used Strehl button (§3.7, 2026-07-19 (b)). The three-component model, with θ_c from the DM and the energy split from the budget (no tuned constants), reproduces both: half-max 66.0 mas ($+5\%$ vs the AO Strehl tool) and the Gaussian-fit, no-background convention 71.8 mas (-6% vs OSIRIS QL). This numeric agreement is empirical for this one frame set, not a matched methodology: tracing the real tools' IDL source (§3.7, 2026-07-19 (b)) shows the AO Strehl tool is a 2-D elliptical fit and the quicklook tool's hand-drawn-box feature likely carries a free background, neither of which the half-max or no-background conventions actually reproduce mechanistically — they simply happen to land close on this PSF. The sign of (fit – half-max) flips between NGS (core-dominated, fit reads slightly narrow) and poor-condition LGS (shoulder-dominated, fit reads $\sim +9\%$), which is the physical behavior of a partially corrected PSF regardless of which real tool, if any, a given convention is read as standing in for.

7. Legacy vs refined budget at a glance

The tool implements two selectable budgets: the refined budget of §3.4 (the default) and a simplified legacy budget (--legacy-budget), retained to reproduce older CSVs still in circulation. Table 3 summarizes the term-level differences. On a median night the two budgets agree closely by construction — every refined term reduces to the legacy behaviour at the reference profile and default TT star — so differences appear at the profile extremes: legacy underpredicts LTAO on ground-dominated nights and overpredicts both laser modes on free-atmosphere-dominated nights. The NGS seeing law (last row) is governed by its own --ngs-seeing-law flag rather than by --legacy-budget, since it corrects the fit's extrapolation independently of the laser budget.

Term	Legacy (--legacy-budget)	Refined (default)
Bandwidth	scales with total seeing s_{tot} (fixed median wind mix)	scales with wind-weighted C_n^2 : ground at 8 m/s, free atmosphere at 25 m/s; equals s_{tot} at the reference profile (§3.4 (1))
LTAO altitude term	tomography term alone, $93 \text{ nm} \times s_{\text{fa}}^{1/3}$	tomography term plus quadratic layer-mismatch ramp toward the single-beacon cone term when the profile deviates from the reconstructor prior (§3.4 (2))
Angular anisoplanatism	flat $44 \text{ nm} \times s_{\text{fa}}$ (the $2''$ allocation, both telescopes)	scales as $(\text{offset}/2'')^{5/6}$ with the actual laser offset: K1 $4.97''$ ($\times 2.14$, the campaign-measured value), K2 $0''$ (term vanishes); --lgs-offset overrides (§3.4 (4))
Tip-tilt	$\sqrt{(151 s_{\text{tot}})^2 + 63^2} \text{ nm}$, fixed to the budgeted $R=15.2$ star at $19.3''$	recomputed from the one-axis mas rows for the configured TT star (--tt-mag/--tt-offset); defaults match legacy to $<1\%$ (§3.4 (3)); a tilt- θ_0 gain knob exists for a future multi-star mode, default 1.0 = no reduction (§3.4 (3b), KAON 1303 §5.5)
LTAO bandwidth penalty	whole bandwidth term $\times 2^{5/6} = 1.78$ (pure half-rate frame-rate law)	latency-floor-aware factor on the rate-dependent part only: 600/1500 Hz ratio with a fixed loop-latency floor, giving $\times 1.44$ by default; --ltao-bw-floor-frac tunes it (§3.4 (5))
NGS seeing law	Gaussian $\exp(-A s_K^2)$ (free-falls when extrapolated beyond the fit's $\sim 0.19\text{--}0.38''$ calibration range)	Kolmogorov $\exp(-A' s_K^{5/3})$ re-anchored inside the calibration range: identical there (<0.005 Strehl), physically-paced roll-off at high airmass; --ngs-seeing-law gaussian reverts (§3.4 (6))
Everything else	identical	identical (fitting, scintillation, fixed HO/TT terms, θ_0/d_0 , airmass projection)

Table 3 — the six term-level differences between the legacy and refined budgets. --legacy-budget reproduces older runs exactly (the LTAO bandwidth reverts with it; the NGS seeing law has its own --ngs-seeing-law flag). CSV headers record which budget and options produced each file.

8. Assumptions and limitations

- The DIMM/MASS tower samples its own line of sight, not the telescope's; ground-layer turbulence is spatially patchy and the two can disagree, especially on ground-dominated nights.
- Representative winds (8 / 25 m/s) stand in for the true wind profile in the bandwidth term; per-night wind data would sharpen it.
- The layer-mismatch penalty's $m = 1$ endpoint (tomography degrades to the single-beacon cone term) and its quadratic ramp are a first-order model; the reconstructor's true loss law is not measured. The normalized bin fractions are noise-dominated when the free atmosphere is very weak, where the penalty is correspondingly small but not exactly zero.
- Charging LTAO angular anisoplanatism at the asterism-center offset is first-order; tomography interpolates within the asterism, so this may mildly double-count the tomography term.
- The LTAO tilt-anisoplanatism reduction (§3.4 (3b)) defaults to off (gain 1.0) — the no-reduction case KAON 1303 §5.5 supports; enabling the knob is the assumption, and its delivered value would also depend on whether the RTC's tip-tilt path uses the tomographic altitude information at all. It changes nothing on-axis and the most at the patrol edge.
- Fixed reserves (static, margin) are always charged; on benign nights the real system does not consume them, so predictions saturate below the best delivered frames.
- The Strehl conversions use the Maréchal approximation, which is optimistic below $S \approx 0.1$.
- MASS is known to have reduced sensitivity to very weak free-atmosphere turbulence; θ_0 provides a cross-check.
- **PSF-fit neighbour subtraction succeeds on OSIRIS cluster frames, not on the NIRC2 fields measured to date.** The empirical PSF requires a minimum of four isolated, well-exposed stars. Across the NIRC2 fields measured — a sparse 2MASS standard field, the globular cluster M92, and the Galactic Centre — none supplies them: sparse fields contain too few stars, and those dense fields contain few that are sufficiently isolated. On such frames the feature detects the condition, declines, and leaves the default measurement unchanged, recording the reason in the log. It is therefore inert on them, never incorrect. The condition is not a property of dense fields as such: on the 2026-05-28 M3 OSIRIS frames the model builds and the path runs, and that run is the crowding audit of §3.9. Whether a field qualifies is a question about its isolated-star population, not about its instrument or its density alone. The Galactic Centre in particular is treated as a documented boundary rather than a target: its most isolated bright star lies at $0.80''$, a model built there predicts several times the neighbour light a target's aperture can contain, and selecting the PSF stars by hand does not remedy this. The default measurement path also fails independently on those frames, returning unphysical Strehl values that the tool rejects (§10).
- **FWHM model.** The core neglects Keck segment gaps and central obscuration (pure circular Airy). The K2 shoulder width (θ_c with $N = 57 \approx 1.3''$) is not yet tested against HAKA imaging — the K1 validation above should be repeated on K2. The NGS tilt-servo coefficient (6.189 mas) is fitted to a single on-sky anchor. The OSIRIS $x \neq y$ elongation (up to 1.3) is real directional jitter that the isotropic model does not reproduce; the simulation predicts the geometric mean only. Relatedly, none of the three FWHM conventions (§3.7) is a mechanism match for the standalone AO Strehl tool itself, which fits a 2-D elliptical Gaussian; the engine's PSF model is azimuthally symmetric by construction (2026-07-19 (b)).

9. The interactive GUI

A PyQt GUI wraps the same engine — the command-line tool is unchanged and remains the reference entry point. The GUI imports the engine and drives it through its public API, so the two can never disagree numerically. This section is a tour of what each control does; for the physics behind any number, follow the section references back into the body of this document.

Layout. A left control dock with tabs mapping the CLI flags (Data, Target, NGS, LGS, WFE sliders, Prediction), a central matplotlib canvas with four tabs (Timeline, Field map, Error terms, Measured SR), and a bottom bar (Run, Report mode, status with budget provenance, Export PNG/CSV).

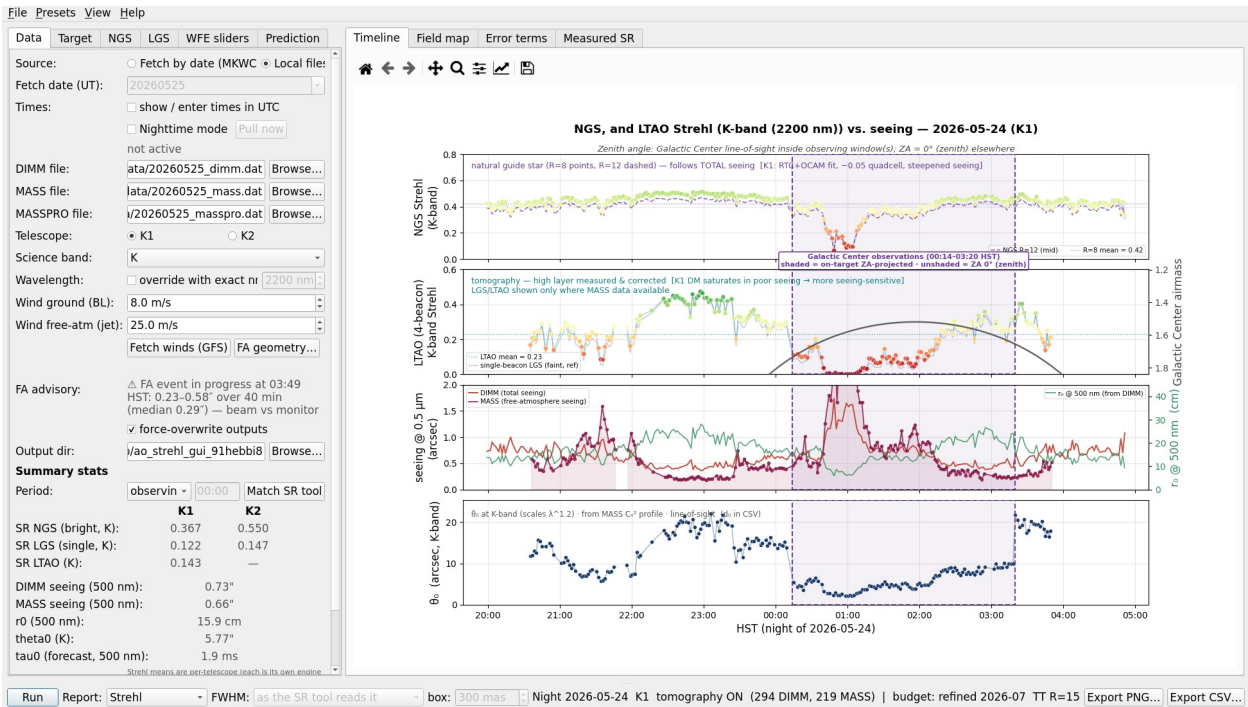


Figure 5 — main window (Timeline tab), K1 night loaded from local MKWC files.

Live what-if. After an initial Run, most controls update the plot live (~0.3 s): compute-only edits (NGS, TT star, LGS offset, WFE sliders, winds) recompute on the cached night; prep-affecting edits (telescope, wavelength, zenith, tomography, budget) re-prepare off-thread. The WFE sliders tab exposes every adjustable budget term, grouped by role — fixed terms, common terms (which project onto the NGS estimate too), and LGS-only terms — with inapplicable rows greyed out live: the unselected telescope's per-telescope terms (DM fitting error and the static telescope-aberration term), and the LTAO tomography residual whenever tomography is off (the K2 default). Any off-default value raises a red MODIFIED BUDGET indicator on the plot and records `budget_overrides=NAME=value` in exported CSV and a PNG footer, so a what-if figure can never be mistaken for the reference budget. A Report combo switches Strehl / FWHM / both live.

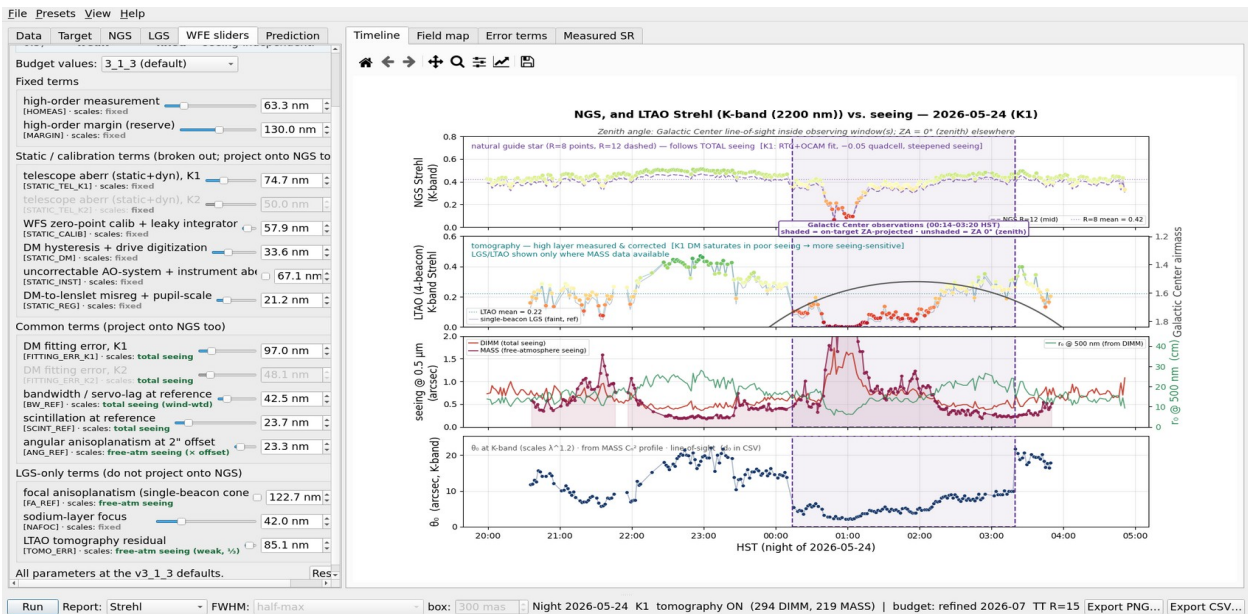


Figure 6 — WFE sliders tab, showing the Budget values picker at 3_1_3. Each slider sets a budget term's reference-profile nm value; the per-night seeing scaling is applied on top, so a slider means the same thing on every night. Inapplicable rows are greyed out for the active telescope and AO mode.

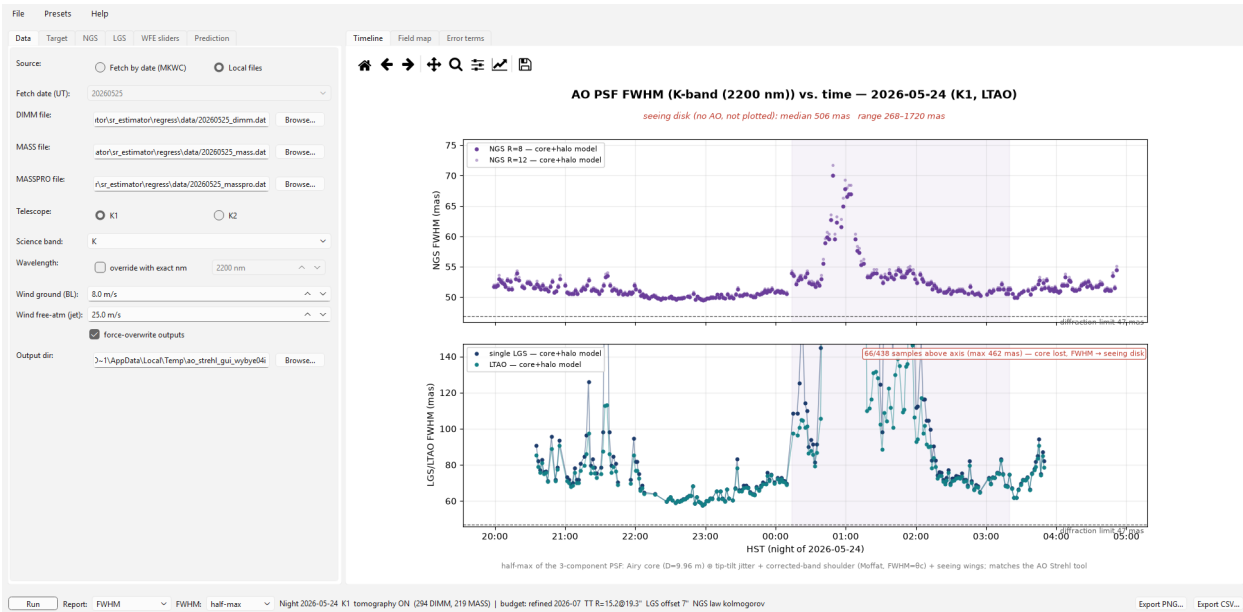


Figure 7 — Report = FWHM: the same night rendered as the two-panel FWHM timeline.

Budget what-ifs on the NGS estimate. The NGS model is an empirical on-sky fit — the budget rows are already inside it and cannot be turned individually — so a slider change reaches the NGS estimate as a Maréchal variance swap, applied per changed common-path term:

$$\sigma'^2 = \sigma^2 + (b^2 - a^2)$$

(The LGS-only rows never project; the inactive telescope's rows and, for an on-axis NGS, angular anisoplanatism are excluded). On the timeline this draws the dotted “projected NGS” overlay; on the field map in NGS mode the same swap is applied to the whole map, so upgrade studies read directly off the contours — e.g. a denser DM (K1 fitting 97 → 48 nm, Θ 84 nm in quadrature) visibly lifts the NGS Strehl across the field. The map is flagged “MODIFIED BUDGET — projected NGS (Θ/Θ N nm quad)”, or, when only LGS-only terms were moved, “no NGS-projecting change” — and LGS/LTAO maps continue to take the overridden budget directly. The Assumed θ_0 control greys out on nights with MASS profiles (θ_0 is then measured per sample and the fallback is ignored; the field-map title shows the θ_0 in use).

Target list. The Target tab keeps a list of the night's targets (name + RA/Dec): Save adds/updates one from the fields, Delete removes it, and picking one from the dropdown loads its coordinates. Loading a science frame on the field map adds that frame's target automatically and makes it the active target (its name/coordinates fill the fields). The whole window is sized to fit the display and never forced wider than the screen by long status text. The observing-window label on the timeline is hung in the margin above the panels in the GUI so it never covers data (the CLI keeps its frozen in-data position).

Coordinate formats. Every RA/Dec field (the Target tab, and the “star RA/Dec” mode of the NGS/TT-star offset entries) accepts three formats, detected independently per field so RA and Dec need not match: the original self-describing hms/dms (15h49m57.7s / −03d55m16s), colon-separated sexagesimal (15:49:57.7 / −03:55:16, RA in hours — the DS9/pipeline convention), or plain decimal degrees (237.49 / −3.92).

Target offset. A new Target-tab control shifts the typed RA/Dec by an arcsec amount — total+PA, Δ RA/ Δ Dec, or an absolute star RA/Dec, the same three-mode entry used for the NGS/TT-star offsets — before it reaches the engine. The typed RA/Dec stay independently editable and saveable as a named target; the offset is a persistent layer on top, defaulting to zero. A live readout shows the resolved effective position, which is what the airmass calculation, the field map, and the offset separation measurements all use — so applying a target offset moves the target, not the field map's sky backdrop: a loaded survey/FITS image stays visually fixed and the target marker moves within it, exactly as a real guide-star nudge would look. Useful for an offset guide star or a small dither without hand-computing a new RA/Dec.

Fix to base position. The TT-star offset, NGS offset, and laser offset controls each have a “fix to base position” checkbox. By default an offset is relative to whatever the target currently is, so it silently follows the Target offset — right for a target-relative dither, wrong for a real guide star or laser point, whose sky position does not move when the

target offset is nudged. Checking the box freezes the control's current entry as an absolute sky position; thereafter, as the target offset changes, the displayed $\Delta\text{RA}/\Delta\text{Dec}$ (and, for the laser, the engine-facing LGS-offset magnitude) auto-recomputes to keep that real point fixed — the hand-recomputation done for you. No effect unchecked, or in the star-RA/Dec mode (already absolute).

Multiple science targets. The field-map grid origin is an explicit field centre — where the LGS/laser is pointed, and where the on-axis value is reported — relabelled from “science target” now that a map can carry several. Any number of science-target markers can be dropped at offsets from that centre; each is annotated with its own predicted Strehl or FWHM at its field position and listed, with its own RA/Dec and offset from the field centre, in a Targets panel below the map. Right-clicking the map opens a menu to drop a science target at the clicked point, or to place the laser, TT star or NGS star there (the corresponding control jumps to it). The typical use is two or more targets a few arcsec apart in one science field with the laser on the field centre — read the delivered performance on each at once. Each target's value is evaluated with the same per-point anisoplanatism model that builds the map (engine helper `field_metric_at()`), so a target's number and the heat map beneath it always agree; it is not a pixel interpolation. Dropped targets are saved with the config.

Guide-star catalogue lookups. A Guide-star catalog combo (GSC 2.4, 2MASS, UCAC4, PanSTARRS DR2, Gaia DR2) with a Load button queries that VizieR catalogue around the field centre — radius the 60" field of regard — and plots the candidates on the map, drawn unfilled — a ring, not a dot, so a star loaded under a sky backdrop is never hidden underneath its own marker — sized by brightness in the current tip-tilt sensing band so the better guide stars are visibly bigger (size rather than colour, which would clash with the viridis heat map). Stars fainter than the sensor's practical limit (~ 17.5 STRAP, ~ 15.5 TRICK H/K) or with no derivable sensing magnitude are drawn small and hollow; switching the TT sensor re-sizes the field to the new band. Left-clicking a star inspects it — it is ringed and its magnitudes, including the sensing-band value, print in the status line, changing nothing. Right-clicking offers “Set as TT star” or “Set as NGS star”, which records the star's absolute RA/Dec (not a delta offset, so it stays correct if the target later moves) and its magnitude in the sensor's working band (R for STRAP/NGS, H or K for TRICK), so a real candidate can be adopted and its predicted delivered performance read immediately. Switching the TT sensor between STRAP and TRICK re-derives the tracked star's magnitude in the new band automatically (R, H and K are never numerically equal); editing the magnitude by hand stops this tracking, and the status line warns if no catalogue star or no usable magnitude is available in the new band. Right-clicking empty sky with no catalogue star underneath (“Put TT/NGS star here”) still places a delta offset, since there is no real coordinate to record. Magnitudes come from the catalogue's own band where available, else from a published colour transform (Gaia G, BP-RP $\rightarrow$ Cousins R, Evans et al. 2018; PanSTARRS $r, i \rightarrow R$, Lupton; GSC photographic $F \approx R$) and are flagged as estimated; near-IR H/K is only taken from real near-IR photometry (2MASS/UCAC4), never synthesised from optical colours. The query is the one networked feature besides the sky backdrop and runs off the UI thread.

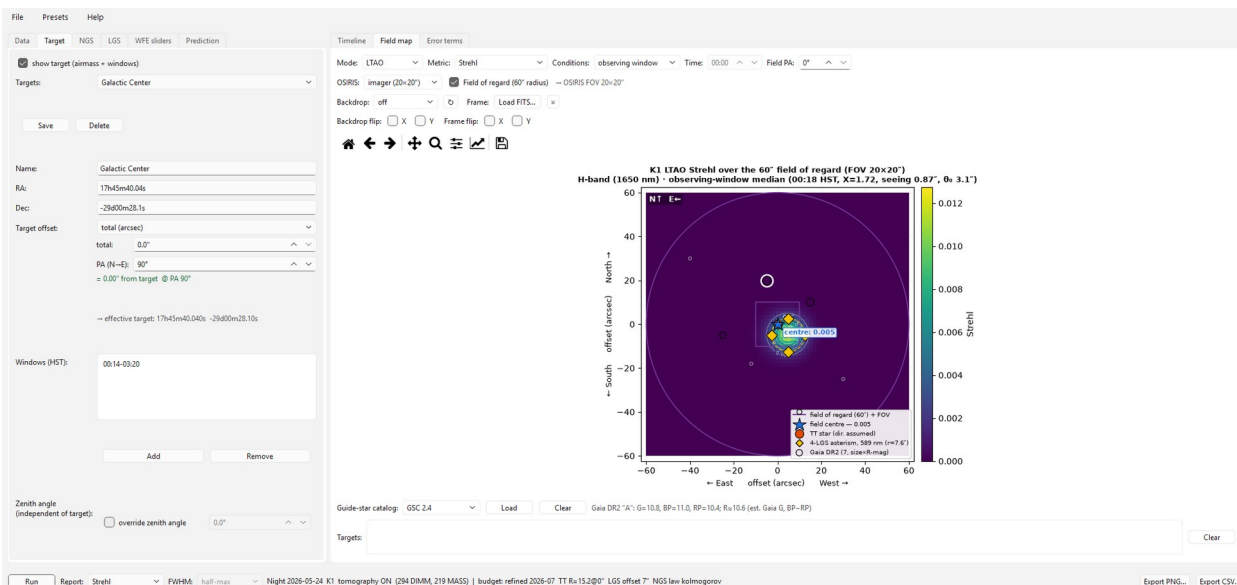


Figure 8 — the catalogue overlay: candidate stars are unfilled rings sized by sensing-band brightness (GSC 2.4, field of regard); the status line reports a TT-sensor band switch with no catalogue star currently tracked.

Offset entry. The NGS offset (NGS tab) and TT-star offset (LGS tab) can each be entered three ways, all resolving to the arcsec the engine takes: a total arcsec, sky-projected $\Delta\text{RA}/\Delta\text{Dec}$, or the offset star's RA/Dec (angular

separation from the science target) — for observers who have the star's coordinates but not the computed separation. A live readout shows the resolved value, and an unparseable coordinate disables Run.

NGS fit editor. The NGS tab also shows the empirical Gompertz fit for the active telescope with all four terms (S0, A, m0, w) editable plus the K1 quadcell penalty, and a live preview plot of Strehl vs guide-star magnitude at the selected science wavelength (extrapolated from the K-band fit, as the estimate does). Editing is recalibration, flagged as such; the reported Strehl budget is untouched.

LGS-tab Cn² profile. The LGS tab plots the night's real (as-measured, zenith) Cn² profile — the same twin-panel integrated/density view as the prediction tab — for the conditions selected on the field map: the mean over the whole night, the mean over the observing window, or the exact profile at a set time. It is the measured counterpart to the prediction tab's synthesized profile, and it tracks the field-map Conditions selector live.

Housekeeping. Config save/load (JSON of all controls), presets (reference budget; last-night-by-UT fetch; and, since 2026-07-19, tonight's-data-by-UT for a night still in progress), keyboard shortcuts, and export that renders a fresh full-resolution figure under the active overrides.

Toolkit. Written to the PyQt6 API via a one-line-swappable shim (qtcompat.py); runs on PyQt5 where PyQt6 is unavailable. Launched via the keck-ao-estimator-gui console command once installed (pip install ".[gui]"), or python3 -m keck_ao_estimator.gui.app from a source checkout; the GUI is internally organized as one module per control tab composed into the main window (no effect on behaviour or layout).

Field map. A second plot tab (before Error terms) maps performance (Strehl, or FWHM in any of its three conventions — half-max, Gaussian-fit, or Gaussian-fit-with-background, §3.7) across the science field, with North = +Y, East = -X. The field size is the selected science-camera FOV: NIRC2 10/20/40" on K2, and on K1 OSIRIS the 20" imager or a rectangular spectrograph field = (lenslet × plate-scale) × (64 × plate-scale) from a scale (0.02/0.035/0.05/0.10") + lenslet-width (16/32/36/42/45/48) pair. A field-of-regard toggle overlays the 60"-radius LGS tip-tilt guide-star patrol circle and zooms the view out to it; the performance map is then computed over the whole patrol field (a denser grid), with the science FOV drawn as a box, and a guide star tens of arcsec off axis is shown at its true position. The sky backdrop is composited from two independent layers. A Backdrop control fills the whole field with a wide-field sky image centred on the science field: either an online survey — DSS2 red/colour or 2MASS J/K fetched off-thread from the CDS hips2fits service, sized to the current view (the 60" patrol field when field-of-regard is on) — or, via Local FITS..., a wide local image (e.g. a GSAOI mosaic) resampled by its own WCS onto the same field grid, centred on the science field and re-centring when a frame is loaded or cleared. A separate Load FITS... control inscribes a local science frame (OSIRIS/NIRC2) on top of that backdrop at its own native angular size — an OSIRIS imager frame occupies only the central 20"×20" of the 60" field, with the survey filling the surrounding patrol area — with a dotted outline marking its footprint. The frame is centred and oriented on its own header pointing (a real celestial WCS if present, else one synthesized from the pointing + plate-scale + PA keywords, e.g. OSIRIS PSCALE/PA_IMAG) and thereby defines the field centre: the survey re-fetches to centre on the frame, the entered target coordinates are not needed, and the field map's star-coordinate offsets are measured from the frame centre. Loading a frame also sets the field-map snapshot to the frame's own observation time (its MJD-OBS/UTC → HST) and adds the frame's target (TARGNAME) to the night's target list. Either layer works alone or together. Raw counts get a percentile stretch and spectral cubes collapse to white light. A multi-extension frame (MEF — e.g. GSAOI's four detectors) is mosaicked: every science image extension is resampled through its own WCS onto the shared grid and combined (the inter-detector gaps stay transparent), with pointing/target/time read from the primary HDU. A loaded frame's WCS is trusted as-is, with no per-instrument parity guess — a genuinely wrong WCS is corrected by eye with the flip checkboxes below, or the Image PA override further down. Over whichever layers are shown the performance is drawn as coloured contour lines only (a filled heatmap over the image is unreadable), so candidate guide stars stay clearly visible.

At each field point the mode's anisoplanatism is re-evaluated relative to its correction references (NGS: the guide star; single/LTAO: the laser for high-order + the TT star for tip-tilt), at a single atmospheric snapshot (observing-window median by default; whole-night or a specific time selectable). It reproduces the science-direction value at the target — reported on the target marker (SR or the FWHM in mas). The FWHM map offers both the half-max and Gaussian-fit conventions. The science target (blue) and offset star (orange) are marked; a reference outside the field is drawn at the edge with its offset. The laser sits radially at the LGS-offset magnitude, position angle from a "Laser PA" control, which defaults to the campaign-measured 254.8° N → E for K1 (§3.4 (4)). In single-LGS mode the one 589 nm beacon is drawn yellow; in LTAO the four-beacon sodium asterism is a dashed yellow ring of radius 7.6" about that centre with the four LGS on it (anisoplanatism is still evaluated at the asterism centre, per the engine's LTAO model). NGS/TT positions come from the offset entry's $\Delta\text{RA}/\Delta\text{Dec}$ or star coordinates; a total-only offset carries its own position-angle field (default North), so a one-value offset can still be placed directionally, flagged on the plot as an assumed direction. The map is computed under the active WFE-slider overrides, so a modified budget carries into it like every other output.

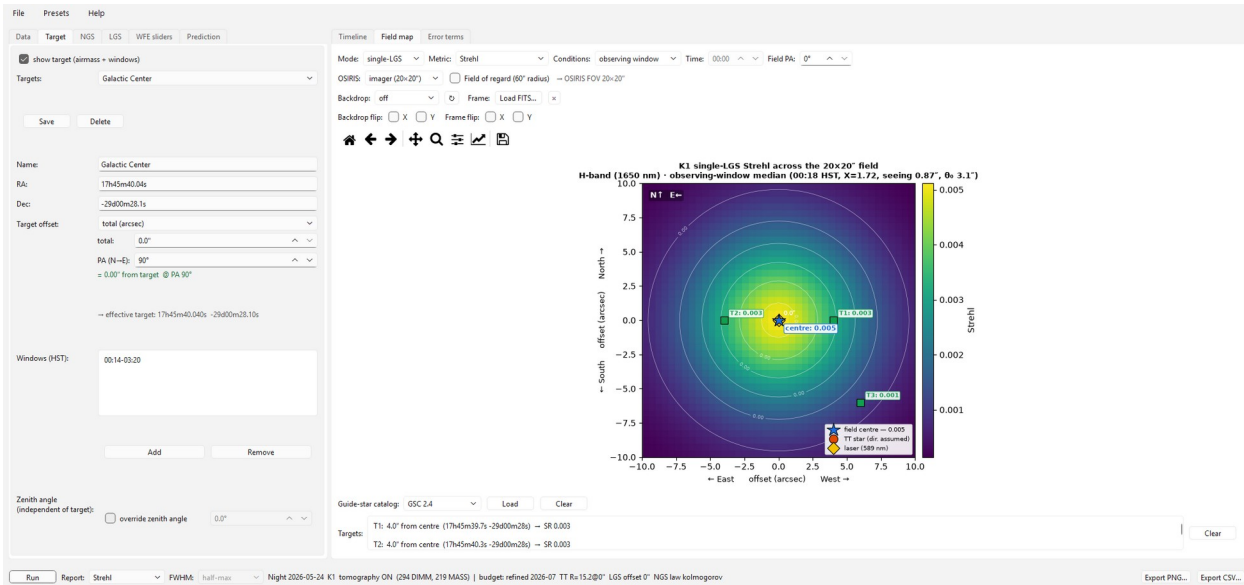


Figure 9 — the field-map tab (K1, single-LGS, Strehl across the 20×20" science field) with three science targets dropped (T1–T3), each now reporting its own RA/Dec alongside its predicted Strehl and offset from the field centre in the Targets panel below the map.

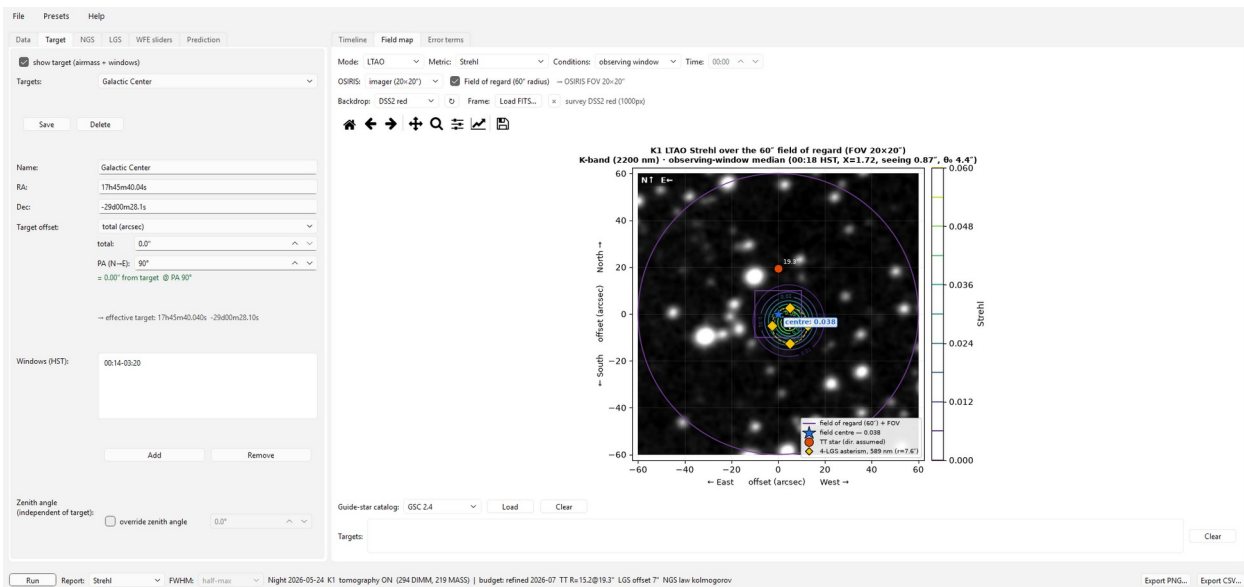


Figure 10 — the field of regard with a DSS2 sky backdrop: the 60" patrol circle, the science-FOV box, and contour-only Strehl drawn over the image so candidate guide stars remain visible; the Guide-star catalog combo sits in the control row.

Predicted scenarios (Prediction tab). The field map and the Error-terms tab can be switched from the night's data to a hypothetical scenario (§3.8): sliders for zenith DIMM / MASS seeing, θ_0 (K band) and zenith angle, with three presets that all fix the same total seeing (0.5") and vary only the free-atm split — Reference (0.5"/0.3"), Free-atm dominated (0.5"/0.45"), Ground-layer dominated (0.5"/0.15") — so they isolate how the turbulence location (not its strength) drives θ_0 and off-axis performance (θ_0^K ranges 11.8" → 35.3" across them at fixed 0.5" seeing). Each preset's θ_0 derives from the synthesized profile rather than being invented. The tab has two pages — Scenario, the DIMM/MASS pair, and Layers, which sets the same scenario as turbulence fractions per layer on the reconstructor's grid (§3.8 "Layer mode"); the seeing sliders run to 2.00", since Mauna Kea seeing is never worse than that. θ_0 auto-tracks the profile as the seeing moves unless its "auto" box is unchecked; an explicit override re-weights the laser/TT anisoplanatism terms (§3.8) and the readout reports the factor, the profile tilt and the LTAO layer mismatch. While enabled, the map's title is flagged PREDICTED SCENARIO (orange) with the full scenario spelled out, and the Conditions selector is gated off. The Error-terms tab meanwhile switches from the night's term time series to a snapshot breakdown at the scenario's conditions — grouped bars of every budget term in nm RMS, single-beacon LGS beside LTAO, computed by the same `lgs_budget_terms()` call the timeline and field map use

(`render_predicted_terms_figure()`, public engine API), with the scenario and the HO/tip-tilt totals and resulting per-mode Strehl in the title; WFE-slider overrides apply, with the MODIFIED BUDGET annotation, and toggling the scenario off restores the night's figure. No Run — and no loaded night — is required: with nothing loaded, the map and the term breakdown compute directly from the current control settings and the scenario, identically to what the same scenario produces after a Run. Everything set elsewhere — WFE sliders, NGS fit, NGS/TT/laser geometry, winds, wavelength — carries over, and the timeline always keeps showing the real night.

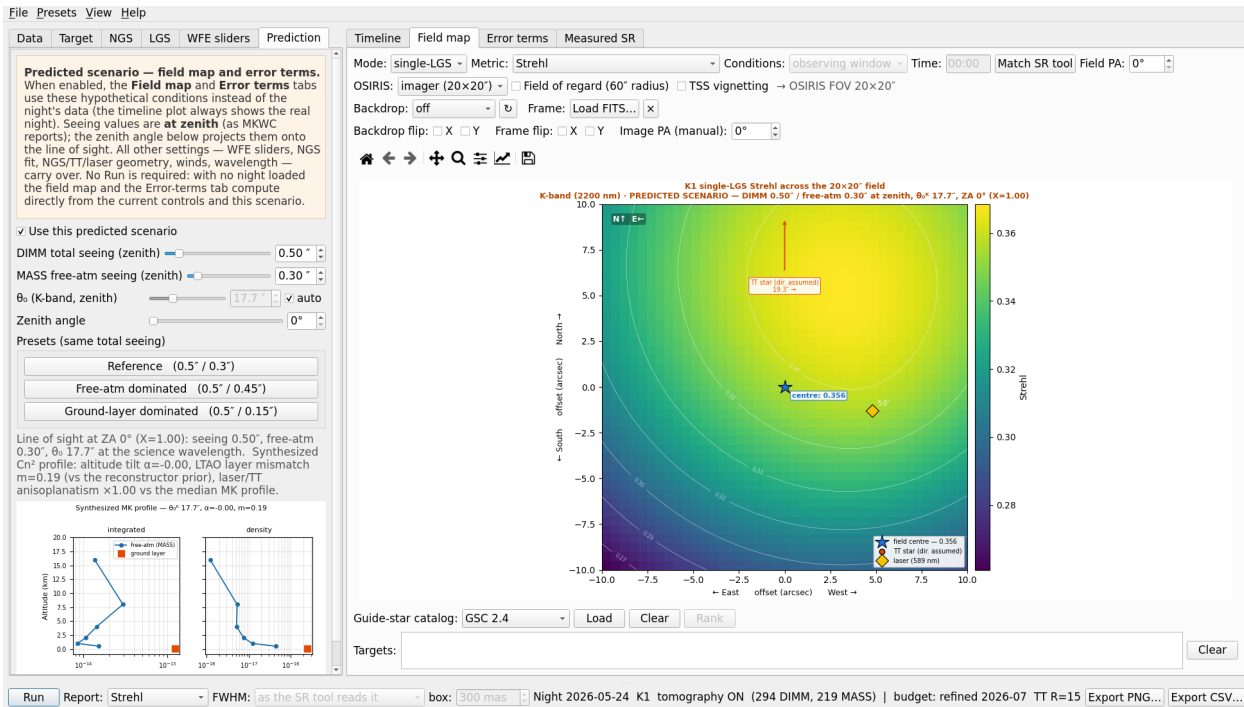


Figure 11 — the Prediction tab driving the field map: PREDICTED SCENARIO title, the zenith seeing / θ_0 / zenith-angle sliders and profile presets, and the twin-panel synthesized C_n^2 profile (integrated and density).

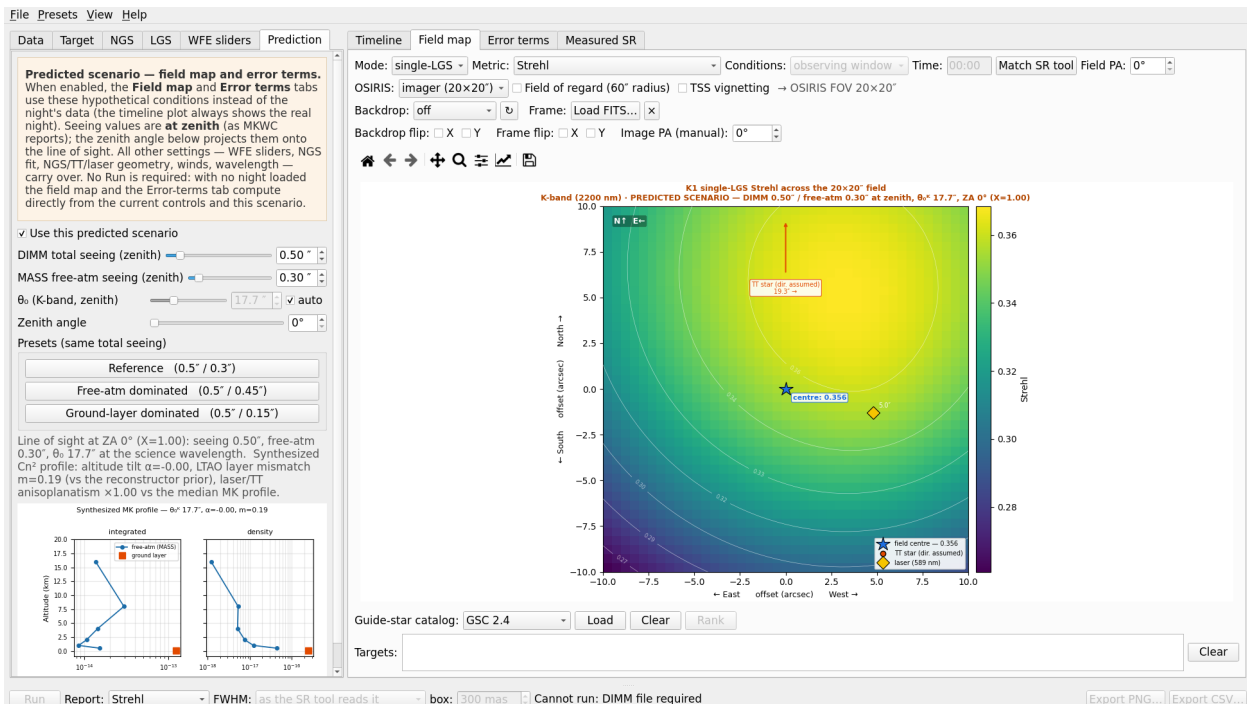


Figure 12 — the Prediction scenario driving the field map in a fresh GUI with no night loaded: the map computes from the current control settings plus the scenario alone (the status bar still reports that a Run is impossible), and matches what the same scenario produces after a Run.

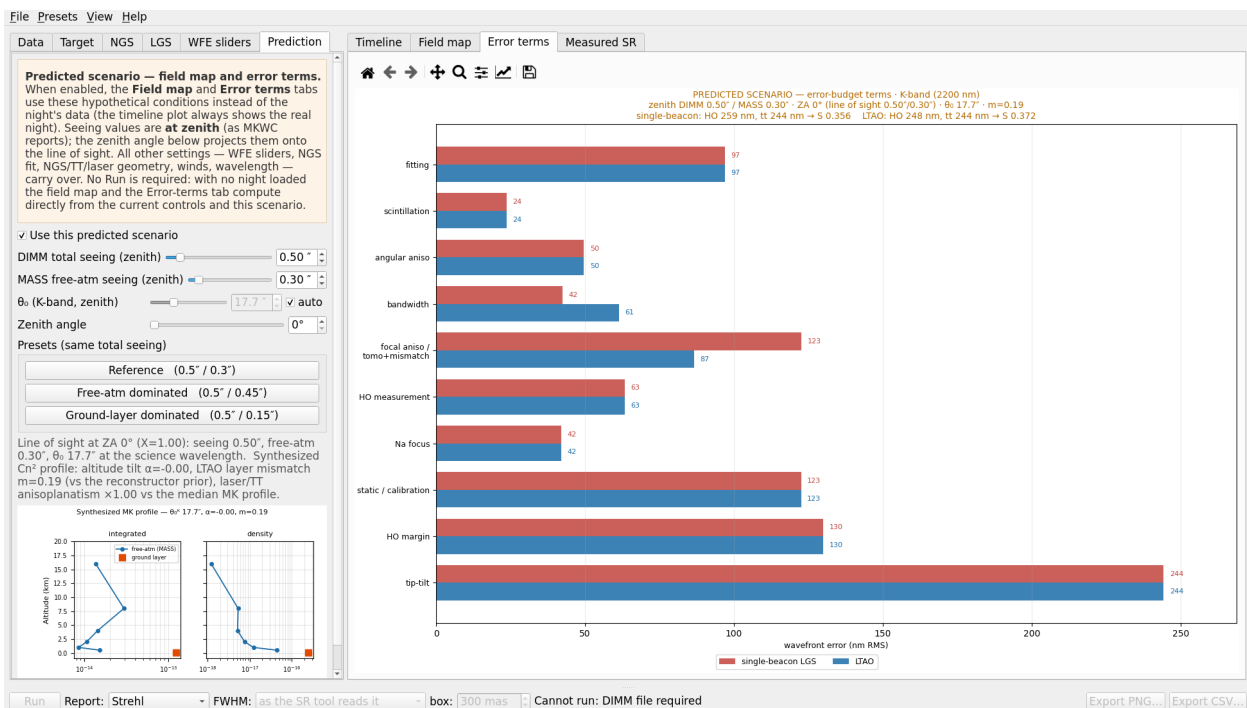


Figure 13 — the Error-terms tab under a predicted scenario: a snapshot term-by-term budget at the scenario's conditions, single-beacon LGS beside LTAO, with the HO/tip-tilt totals and the resulting per-mode Strehl in the title.

Field PA. A Field PA control rotates the entire displayed field — backdrop or frame imagery, every marker, the field-of-regard and FOV outlines, and a compass indicator that keeps pointing at true N/E — by a chosen position angle, North through East. This is purely a display transform: the underlying performance grid and every reported number are computed exactly as before and are unaffected; only the on-screen layout rotates, e.g. to line the view up with a rotated real instrument frame or FOV.

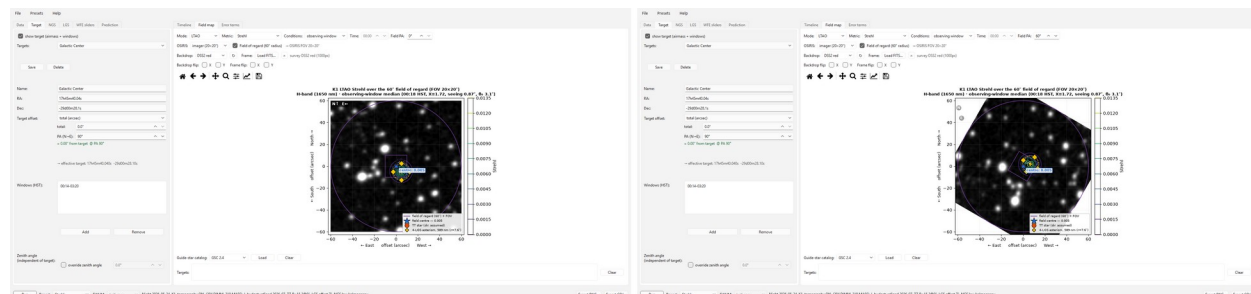


Figure 14 — the same field-of-regard view at Field PA = 0° (left, the default orientation) and Field PA = 60° (right): the DSS2 backdrop, the field-of-regard/FOV outlines and the compass all rotate together; the underlying performance values are unchanged.

Backdrop and frame flip. The survey backdrop and the inscribed local frame can each be independently mirrored in X and/or Y ("Backdrop flip" / "Frame flip" checkboxes), for a loaded FITS whose WCS orientation cannot be trusted. The flip is applied to the already-resampled image at draw time, not the underlying file or its WCS — a display correction, not a re-interpretation of the astrometry.

Image PA — manual override. Separate from the cosmetic Field PA (which rotates the whole displayed field together, backdrop and markers alike), Image PA rotates only the loaded backdrop/frame imagery relative to the catalogue and markers, for the rare case where a file's own WCS orientation is genuinely wrong and needs a by-eye correction against known stars. Because this is a real astrometric override rather than a display convenience, it is labelled "MANUAL OVERRIDE" in its tooltip, and whenever it is non-zero the map itself carries a prominent red "⚠️ IMAGE MANUALLY ROTATED" banner — a corrected image can never be mistaken for a natively correct one.

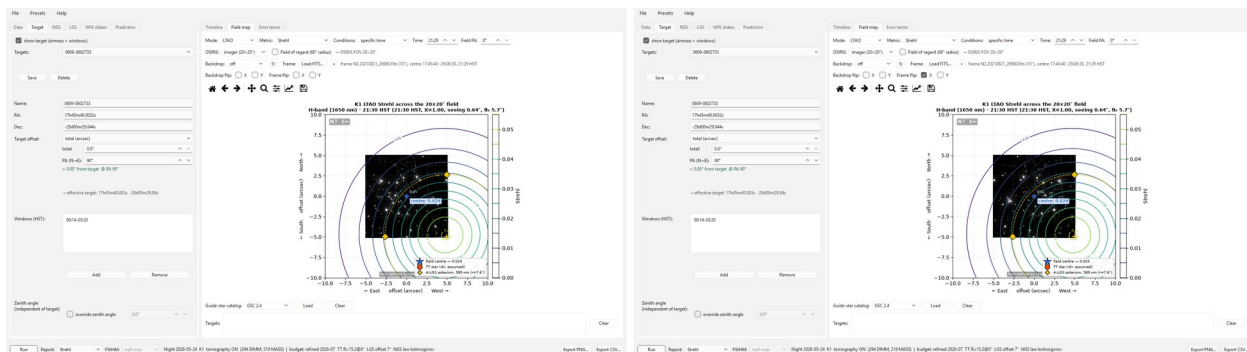


Figure 15 — a loaded local frame unflipped (left) and with Frame flip: X checked (right), showing the mirrored orientation.

Guide-star auto-ranking. A Rank button next to the catalogue controls evaluates every loaded candidate as the guide/TT-tilt star and ranks them by delivered Strehl/FWHM at the science target itself — the decision an observer is actually making, rather than inspecting stars one at a time. The top 3 are badged on the map; the full table (magnitude, offset, RA/Dec, delivered value, and an exclusion reason where relevant) opens separately and is sortable by any column, and double-clicking a row adopts that star exactly as the right-click “Set as TT/NGS star” map action does. No new physics: a ranked value is exactly what the field map itself would show with that star’s position and magnitude dialed into the TT/NGS controls. A ranking clears on catalogue reload, a TT-sensor band switch, or a Mode/Metric change; it does not auto-clear on offset or laser drags, so comparing candidates stays possible while working the map — re-rank after such a change.

Reddening safety on the ranking. A star with only near-infrared photometry has its optical (R) magnitude estimated by a rough nearest-band fallback; in a dusty field that guess can be wrong by many magnitudes; because interstellar dust dims the optical roughly 7× more than K-band, a star bright in K can be optically invisible to an R-band sensor. The tool bounds the optical extinction from the star’s near-IR colour excess (a conservative lower bound, never a silent “fix”) and, for optical sensing of an IR-derived magnitude only, ranks and excludes accordingly, flagging a reddened candidate amber (“⚠ reddened ... verify”) in the table, the inspection line, and the selection line. This is sensor-specific: the identical star stays fully usable for the infrared TRICK sensor, and switching sensors clears the warning.

PNG sky images. The field map can load a PNG carrying an embedded WCS as a backdrop or inscribed frame, alongside FITS — the same placement and orientation handling, dispatched by file extension. This lets a light, shippable sky image travel with the tool where a full FITS is too large; a PNG with no embedded WCS is rejected with a clear message, since a plain screenshot carries no sky position.

Resolving a target and proper motion. The Target tab’s Resolve button queries SIMBAD for a typed name and fills in RA/Dec and, where measured, proper motion, without renaming the target or disturbing a target with no known PM. Proper motion — typeable by hand as well — is applied, not just recorded: the position used everywhere downstream (airmass, the field map, every offset) is propagated from J2000 to the observing date before any target offset is layered on top. This is a real effect at the arcsecond precision this tool otherwise cares about — tens of mas/yr, unremarkable for many real targets, accumulates to over an arcsecond across a few decades. Zero proper motion (the default) is an exact no-op, so nothing about an ordinary target changes, and PM is stored per saved target so switching targets never leaks one target’s motion onto another.

Loading a starlist. Load starlist... on the Target tab reads a standard Keck-format starlist and offers tonight’s targets in a sortable pop-up table, double-click to select. Picking a row zeroes the proper-motion fields, since a starlist carries none and a stale PM from a previous target would otherwise silently mis-point; a non-J2000 equinox is flagged and warned about, since the tool assumes J2000/ICRS throughout and does not precess.

Table and export polish. The ranking and starlist tables sort by clicking any column header (by value, so rank 10 does not sort before 9), and clicking always lands on the correct row regardless of sort order. Export PNG on the Field map or Error terms tab now saves that tab’s own figure rather than always the Timeline’s.

Nighttime mode. A Data-tab checkbox auto-fetches tonight’s MKWC DIMM/MASS/MASSPRO data and re-runs every 5 minutes, matching MKWC’s own publish cadence, so the estimate stays current without the observer re-clicking Run. Enabling it forces fetch-by-date mode on tonight’s UT date and greys out the source-selection controls; a status line always shows the last pull time and the next scheduled one, and a Pull now button forces an immediate re-fetch. The field map’s Conditions selector gains a matching “time of last pull” option, falling back to the whole-night median before any pull has happened. Deliberately not saved in configs — like the guide-star catalogue’s Load

button, enabling it starts background network polling, and a loaded config must never silently do that. No engine change: it reuses the existing cache-staleness auto-refetch, since an in-progress UT night is never “complete.”

Safeties on Nighttime mode. Three safeties keep it from pinging MKWC with nothing to pull. It only enables between sunset and sunrise at Keck (a new engine sun-altitude helper against the standard -0.833° almanac horizon); a daytime enable attempt is refused, with the reason shown in the status line. A session left running past dawn stops itself within one 5-minute cycle, restoring the data-source controls and handing the theme back through the normal disable path. A failed first pull — typically because MKWC has no DIMM file yet so early in the evening — disarms the mode rather than re-hammering the archive every 5 minutes for a file that isn’t there; once a pull has succeeded, later failures are treated as transient and simply retried next cycle.

Dark theme. A checkable View ► Dark theme action restyles all the Qt controls (Fusion style plus a dark palette); unchecking it restores the palette captured at startup. Widgets-only by deliberate choice — the on-screen matplotlib figures and everything Export writes keep their light, print-style rendering, since exported PNGs go into reports and the CLI’s outputs are frozen by the regression harness. Nighttime mode auto-enables dark theme when turned on, unless dark was already the user’s own choice, and switches back when turned off — but only while the auto-switch still owns the theme; touching View ► Dark theme directly, at any point, takes ownership away from Nighttime mode. Dark theme is saved in configs, unlike Nighttime mode itself.

Data-tab summary-stats panel. A new panel reports mean NGS (bright-star) and single-beacon-LGS Strehl for both K1 and K2, mean LTAO Strehl (K1 only — LTAO is K1 hardware), mean DIMM seeing, mean MASS seeing, mean r_0 , and mean θ_0 , over the same period the field map’s Conditions selector already offers (observing window, whole night, a specific time, or — in Nighttime mode — the time of the last auto-pull). A shared engine helper generalizes the exact window/night/time selection the field map’s Cn^2 profile already uses, so the panel can never silently disagree with the map about what “observing window” means. Showing both telescopes needs a second engine call for the telescope not currently selected, which is cheap since it reuses the already-prepared night; this cross-telescope comparison is only physically meaningful with the STRAP tip-tilt sensor (telescope-independent), so with TRICK selected — a K1-only sensor whose dichroic has already fixed the science band for this run — the non-current telescope’s Strehl cells read n/a, with the reason footnoted, and repopulate the instant the sensor goes back to STRAP.

Target definition now also defines the guide star. NGS always assumes the guide star is the science target itself (on-axis) — the tool has no NGS-only exception. LGS defaults to the same on-axis assumption unless a loaded Keck starlist links a real TT-star candidate to that target, the reason a real starlist carries that field: one linked candidate is adopted directly (its absolute RA/Dec and best-available sensing-band magnitude, the same path as picking a catalogue star on the field map); two or more are ranked by delivered Strehl at the target using the same ranking machinery the field map’s Rank button already uses, and the top-ranked one wins. Every “define a new target” path — SIMBAD Resolve, Save, or a starlist pick with no linked candidate — resets to the on-axis default, but re-saving an already-customized target does not wipe out a manually set guide star. Each saved target now remembers its own guide star alongside its own proper motion, so switching between saved targets restores each one’s own pick rather than leaking one target’s guide star onto another.

Guide-star magnitude also follows the target. Picking a starlist entry now also sets the NGS magnitude from the target’s own R-band value — NGS has no starlist exception, the guide star is always the target — and, when there’s no linked TT candidate, sets the TT magnitude from the target’s own magnitude in the sensor’s band too, so a previous target’s guide-star brightness can never linger onto the next. A target with no magnitude at all triggers one asynchronous SIMBAD lookup; a failed lookup, or one with no usable magnitude, warns explicitly rather than leaving a stale value looking deliberate. A separate fix also corrected starlist target= links for space-containing names — the whitespace-delimited format can only encode a space as an underscore (target=IRAS_23077+6707), and matching now treats the two as equivalent.

Starlist picker enhancements. The picker dialog stays reachable without re-parsing the file — a Show starlist button reopens it from the cached entries. Single-clicking a row (double-click still picks it as the science target) opens a detail panel: name, RA/Dec and magnitudes, current Az/EI, Moon separation and Moon illumination, all evaluated at an editable “Evaluate at” date and time-of-day pair — deliberately two independent widgets, a calendar-popup date and the app’s free-typing time field, rather than a combined QDateTimeEdit, which only lets the date portion be freely retyped (the same section-editing limitation the free-typing field was written to fix elsewhere). The control honours the app-wide UTC toggle and labels itself “(HST)” or “(UT)” so the convention in force is never ambiguous, and a new HA (hour angle) column sits beside RA/Dec, refreshed live as the evaluation time changes. Az/EI and Moon separation are colour-coded with the app’s existing theme-adaptive label cues: red for below the horizon or blocked by the telescope’s own pointing limits (the Nasmyth-deck wedge is already modelled, per telescope), amber for vignetted or within 30° of the Moon (red under 15°), green otherwise. Two engine functions back this: a

geocentric Moon separation and illumination pair using the standard Meeus low-precision phase-angle approximation, $k = (1 - \cos \psi)/2$, accurate to ~0.1 %; and an hour angle, $HA = LST - RA$ at Keck’s longitude, which needs no declination or latitude.

Click a row for details (az/el, Moon separation, evaluated at the time below). Double-click to make it the science target (added to tonight's Targets list). Click a column header to sort.

Evaluate at (HST):

2026-07-28

22:30

Now

#		name	RA	Dec	HA	V	R	K	sep (")	role
1	5	syneng100	08:40:44.20	-19:52:24.1	+9.92h	9.9	—	—	—	target (lgs)
2	6	syneng101	11:15:10.72	+59:44:54.8	+7.35h	10.1	—	—	—	target (lgs)
3	7	syneng102	22:04:46.65	-19:55:26.8	-3.48h	10.3	—	—	—	target (lgs)
4	8	syneng103	14:36:44.41	-11:54:00.3	+3.99h	10.0	—	—	—	target (lgs)
5	9	syneng104	17:20:22.68	+12:54:44.9	+1.26h	10.3	—	—	—	target (lgs)
6	10	syneng105	22:08:16.30	+24:53:19.2	-3.54h	10.1	—	—	—	target (lgs)
7	12	SYN-001	19:22:00.40	-23:26:40.2	-0.76h	—	—	—	—	target (lgs)
8	13	SYN-002	02:32:57.23	+23:13:56.4	-7.95h	11.6	—	—	—	target (lgs)
9	14	SYN-003	17:31:57.87	+05:28:28.8	+1.07h	11.8	12.5	9.9	—	target (lgs)
10	15	SYN-004	04:34:40.93	+59:42:29.5	-9.98h	—	—	—	—	target (lgs)
11	16	SYN-005	08:10:32.86	-00:09:19.6	+10.43h	11.9	—	—	—	target (lgs)
12	17	SYN-006	18:51:11.87	+23:21:24.1	-0.25h	14.5	—	—	—	target (lgs)

syneng100 RA 08:40:44.20 Dec -19:52:24.1 (V=9.9)

Az/El at 2026-07-28 22:30 HST: 264.3° / -60.5° — BELOW HORIZON

Moon separation at 2026-07-28 22:30 HST: 141.2" (illumination 100%, 100% = full moon)

TT star → name rows are tip-tilt-star candidates the list explicitly links (target=) to that science target; they can still be picked directly. Starlists carry no proper motion, so picking a row zeroes the PM fields.

Close

Figure 16 — the starlist picker with a row selected: the detail panel with Az/El, Moon separation and illumination, the HA column, and the “Evaluate at” date and time pair labelled (HST). The Az/El line is red here because the star is inside the K1 pointing limit at that time — the colour cue is the point of the panel.

Session-layered edits. Save and Delete on the Target tab now mirror onto the loaded starlist's session layer rather than the file: additions are written to a <name>.session.lst sidecar in the same Keck-starlist format, so they round-trip through the same parser, and deletions of an original-file entry are recorded as names in <name>.session_excluded.txt. The picker always shows the merged view (loaded – excluded + added). The real .lst file is never written to.

Field map: “Redefine field centre here”. A new right-click action alongside the existing “Drop science target here.” Where that one leaves a lightweight, unsaved field-relative marker, this moves the currently-selected target's own RA/Dec to the clicked point — a real re-pointing — converting the plot-frame arcsec offset with the same sign convention the existing “Put TT/NGS star here” actions already establish, and updating the same target-list entry in place rather than stamping out a fresh T1/T2/... on every click. The existing backdrop-shift behaviour (a loaded sky image stays visually fixed while the target moves within it) was already correct; a missing redraw trigger that left the whole map, backdrop included, looking frozen until an unrelated control forced a repaint was fixed at the same time.

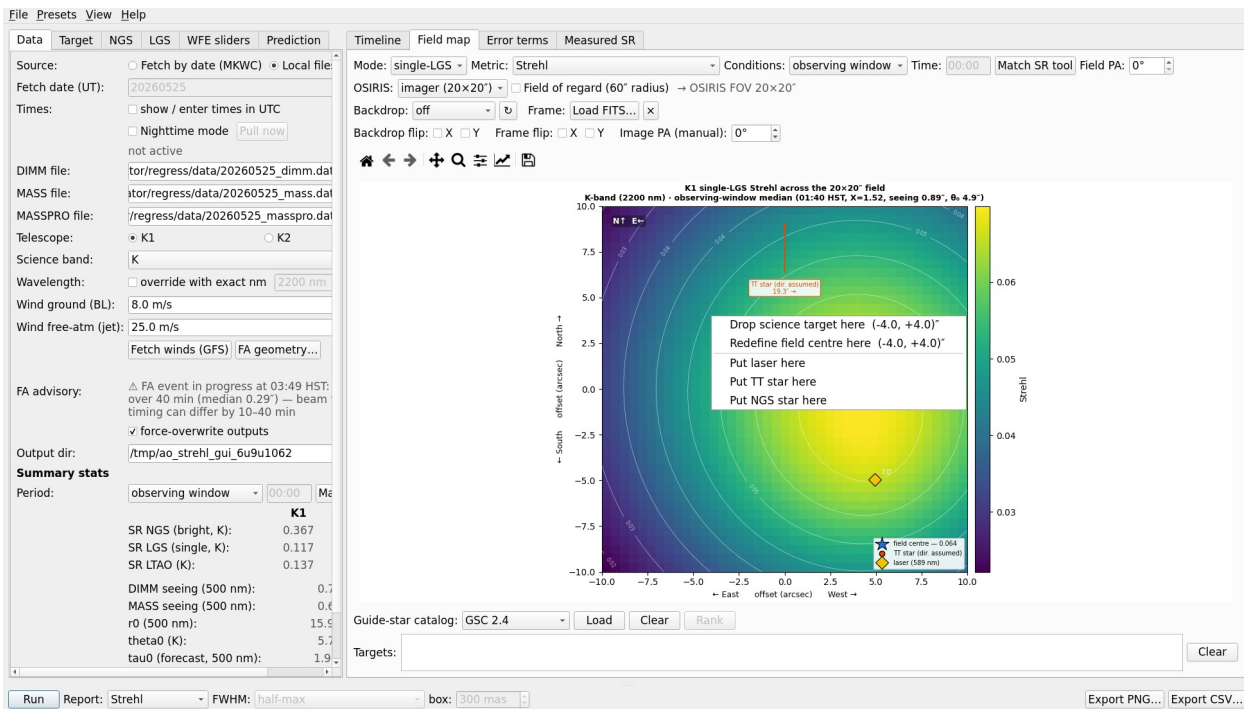


Figure 17 — the field-map right-click menu, showing “Drop science target here” and “Redefine field centre here” as adjacent entries with the same clicked offset — the pair this section warns are easy to confuse. Composite: a Qt menu is its own top-level window, so the menu render is painted onto the window render at the true click position; both halves are real widget renders, not a mock-up.

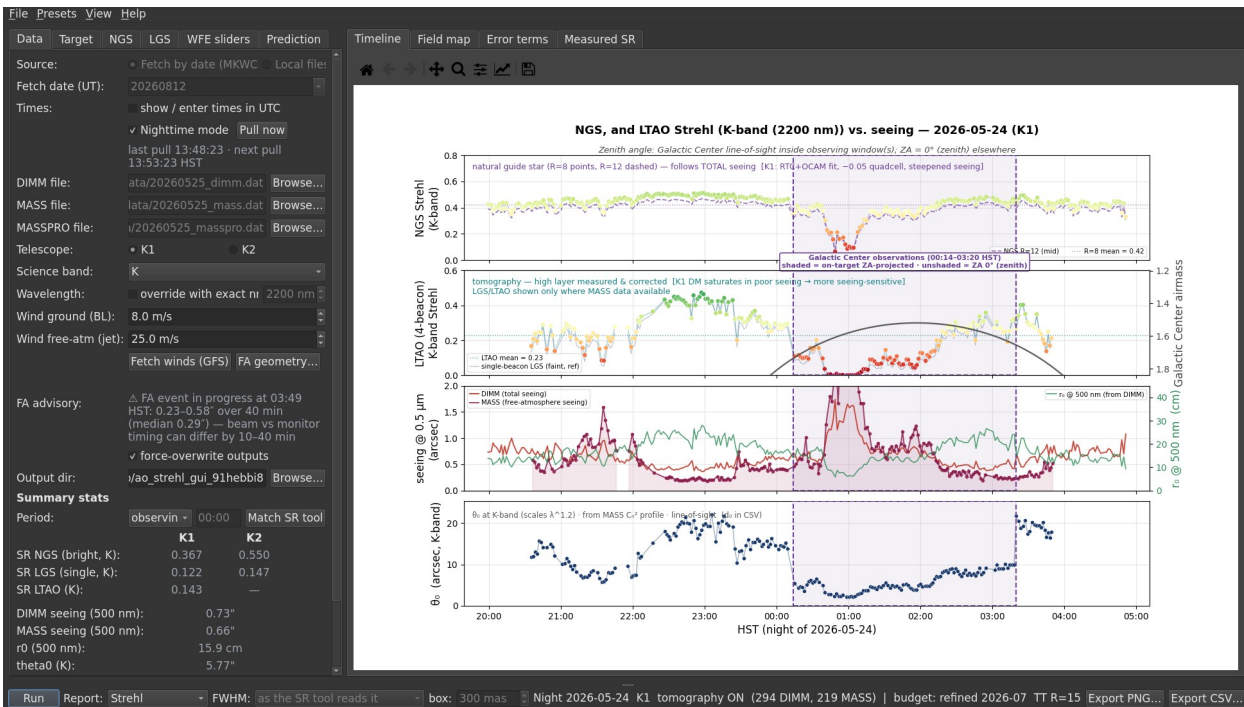


Figure 18 — Nighttime mode enabled: the Data-tab checkbox, the source-selection controls greyed out, the last/next pull status line, and Pull now.

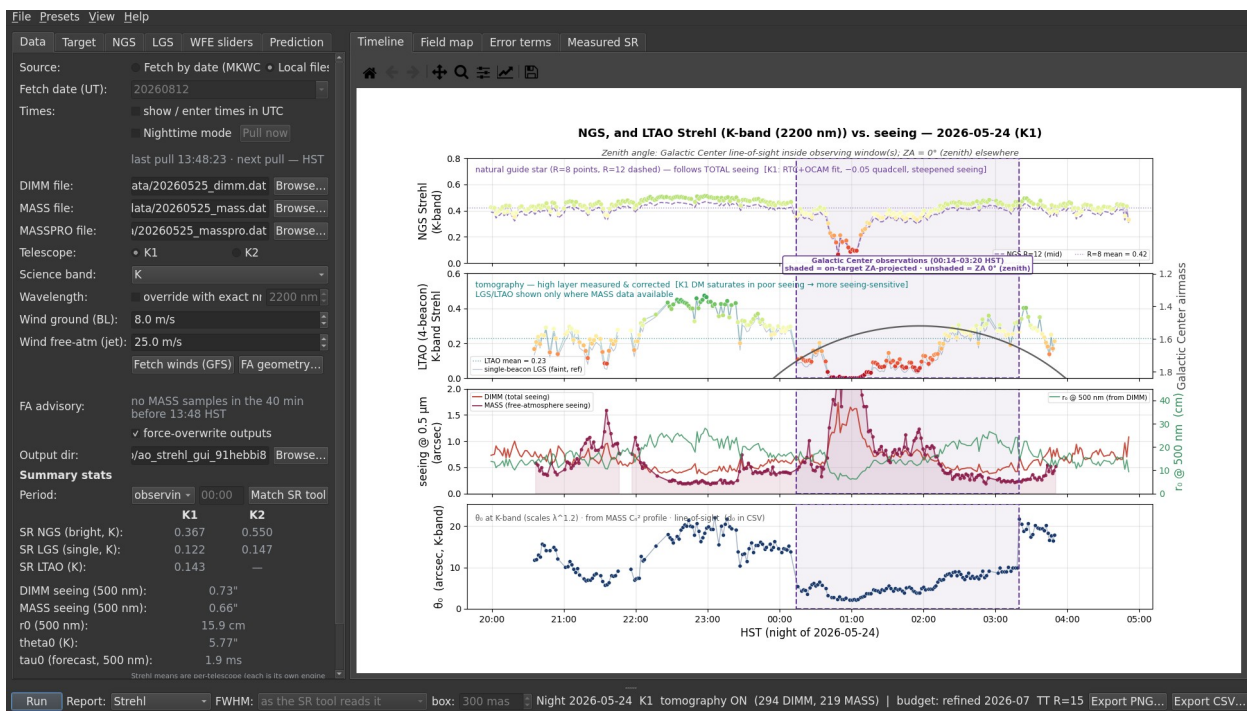


Figure 19 — Dark theme active: the Qt controls restyled dark while the Timeline plot alongside them keeps its normal light, print-style rendering.

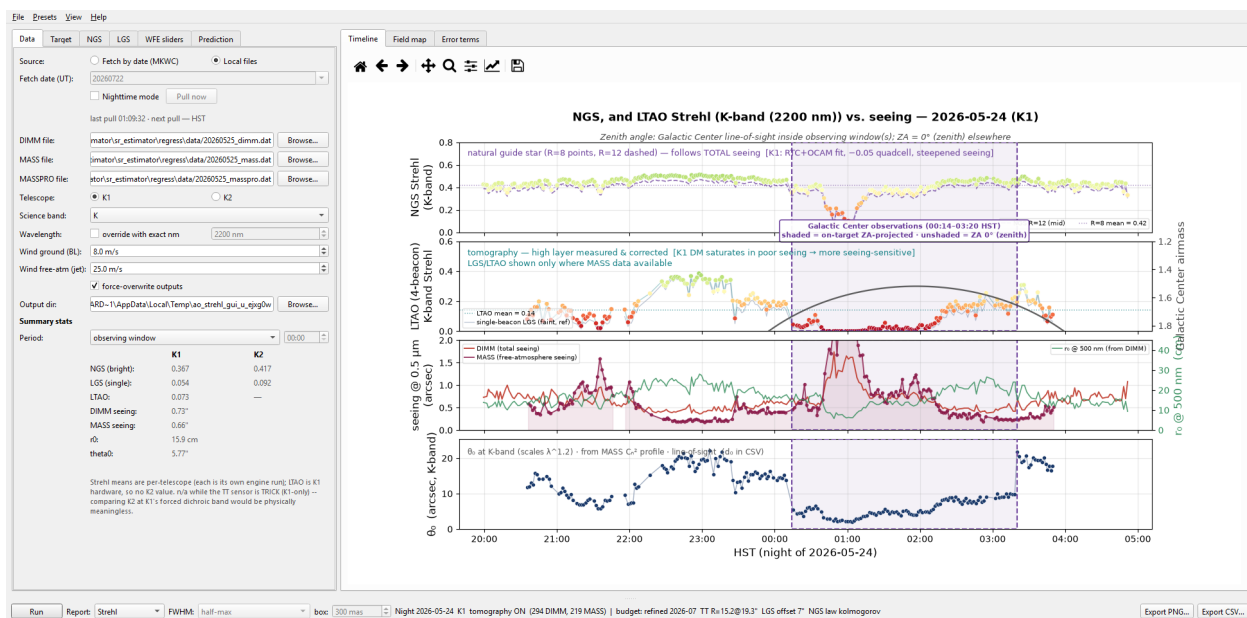


Figure 20 — the Data-tab summary-stats panel with STRAP selected: NGS, single-beacon-LGS, and LTAO Strehl populated for both telescopes (LTAO is K1-only, so K2 has no entry there either way).

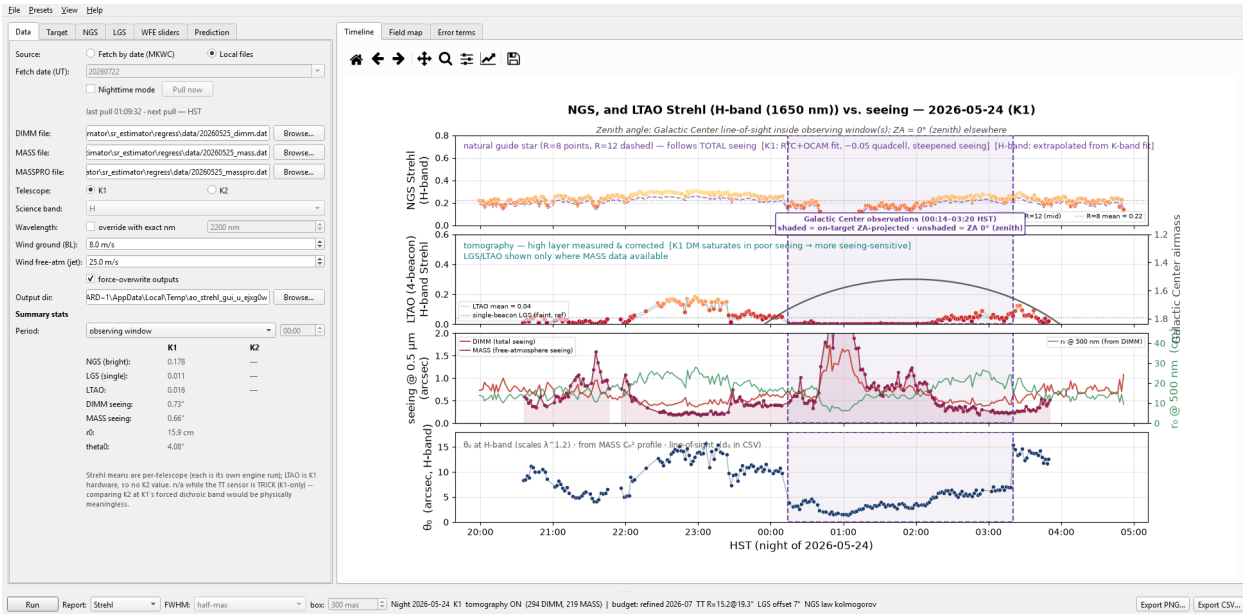


Figure 21 — the same panel with TRICK selected: every K2 cell reads n/a, since a K1-only sensor makes the cross-telescope comparison physically meaningless.

Click a row for details (az/el, Moon separation, evaluated at the time below). Double-click to make it the science target (added to tonight's Targets list). Click a column header to sort.

Evaluate at (HST): 2026-07-28 22:30 Now

#	name	RA	Dec	HA	V	R	K	sep (')	role
1	GuideDemo	15:49:57.7	-03:55:16.0	+2.77h	—	—	—	—	target (lgs)
2	Far_bright	15:50:10.0	-03:55:16.0	+2.77h	—	9.0	—	—	TT star → GuideDemo
3	Close_faint	15:49:58.5	-03:55:20.0	+2.77h	—	16.5	—	—	TT star → GuideDemo
4	Close_bright	15:49:58.7	-03:55:12.0	+2.77h	—	10.0	—	—	TT star → GuideDemo

Click a row to see its details.

"TT star → name" rows are tip-tilt-star candidates the list explicitly links (target=) to that science target; they can still be picked directly. Starlists carry no proper motion, so picking a row zeroes the PM fields.

Close

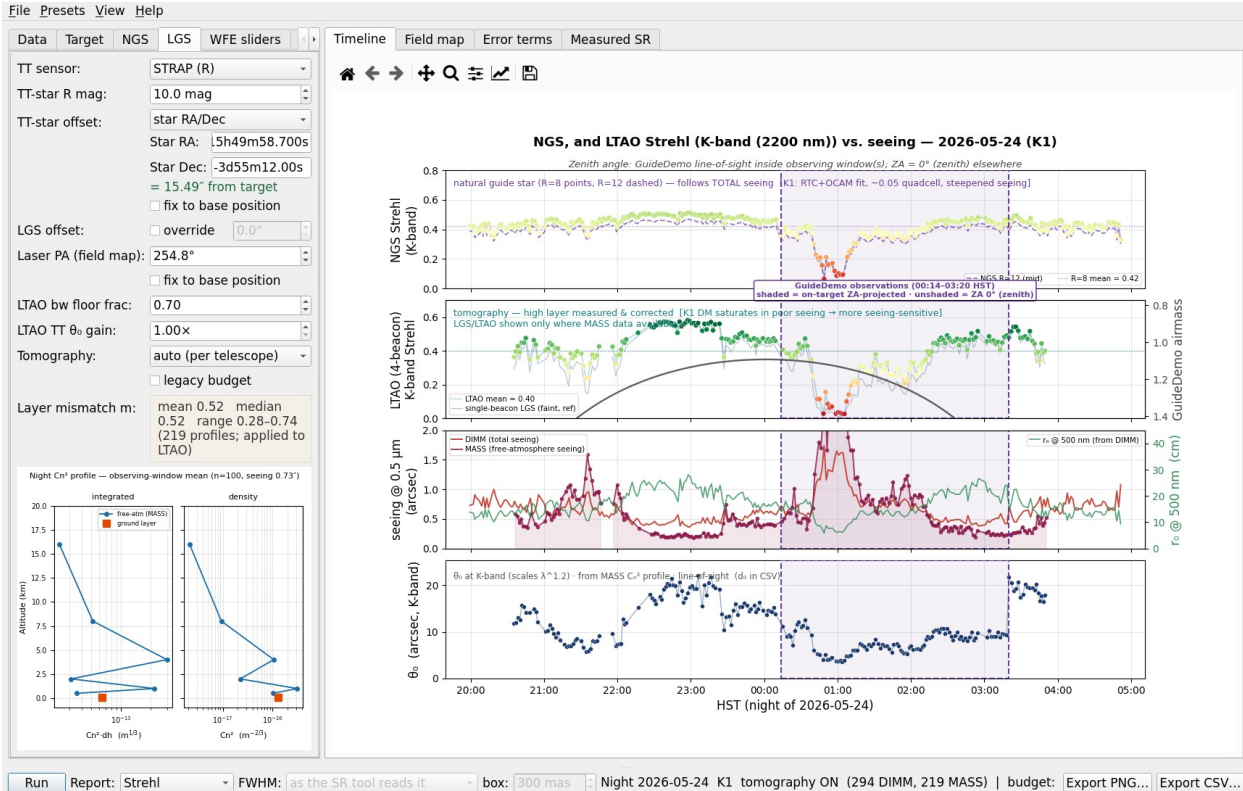


Figure 22 — (a) the Load starlist... picker with three TT-star candidates linked to one target; (b) the LGS tab immediately after picking, with the ranked winner's (Close_bright, $R=10.0$) offset and magnitude filled in automatically.

Real winds from the GFS analysis. A Data-tab Fetch winds (GFS) button replaces the hand-guessed 8/25 m/s wind defaults with the night's own GFS analysis: one bounded HTTP request per night (10 s timeout, no retry loops; a dead service raises a clean error and the estimate keeps whatever winds it had), cached per UT date under the MKWC cache with the same staleness rule. u/v are interpolated in geopotential height to the six MASS bin altitudes and collapsed to the two representative speeds the bandwidth term takes — the free-atmosphere speed by the same 5/3-moment Cn^2 weighting the term itself uses, the ground speed from just above summit level. The button fills the two existing wind fields with full provenance, so the estimate re-runs automatically; the CLI's flags and defaults are untouched. A τ_0 row joins the summary panel — $\tau_0 = 0.314 \cdot r_0/V_{\text{eff}}$ at 500 nm, zenith — labelled “tau0 (forecast)”, deliberately marked because its wind leg is a model product, unlike the measured seeing/ r_0/θ_0 rows. On real nights it matters: 2026-05-25 carried a 41 m/s jet at the 8-km bin (collapsed ≈ 23 m/s) while 2026-07-21 ran quiet at ≈ 9 m/s, against a static 25 m/s guess.

FA timing advisory. A display-only Data-tab advisory line — it deliberately changes nothing about the timeline or any reported estimate — born from the 2026-07-21 free-atmosphere event analysis, which showed a discrete FA event crossing Keck's science beam 22–35 minutes before the MASS/DIMM recorded it: the two instruments pierce a moving layer kilometres apart, so an instantaneous FA reading is representative of the summit, not “what the beam sees right now.” The line states the trailing 40-minute FA window (median and range, with an  FA event in progress” cue when the range is large and the peak stands well above the window median) and, with GFS winds fetched and a target set, the per-layer expected lead of Keck over the monitor — the target's pierce-point offset at each MASS bin altitude projected onto that bin's wind vector, $\pm$ the monitor's own unknown pierce point. The reference time is Nighttime mode's last pull when live, else the newest MASS sample.

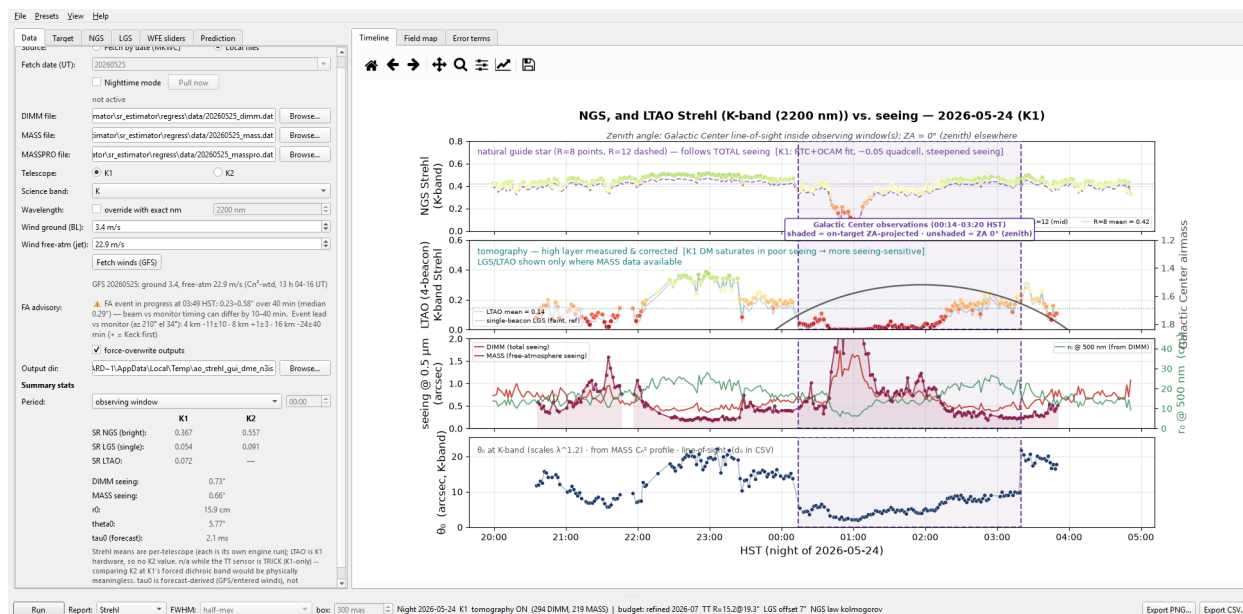


Figure 23 — the FA advisory triggered on the bundled 2026-05-25 night: the  FA event in progress” cue with the trailing-40-minute window stats and the per-layer beam-vs-monitor lead/lag line, with the fetched GFS winds (3.4 / 22.9 m/s) in the wind fields above, beside the Timeline showing the actual FA rise.

FA pierce-point geometry dialog. An FA geometry... button next to Fetch winds opens a non-modal two-panel dialog rendered for the current target, reference time, and GFS winds: a plan view — the monitor's pierce-region circles (radius h for each of the 4/8/16-km MASS bins), the Keck beam's pierce points marching out toward the target azimuth, and each layer's wind vector drawn at its pierce point with the resulting lead/lag annotation — and a side view along the beam azimuth, with the monitor's $\leq 45^\circ$ -from-zenith pierce cone, the beam line at the target elevation, and the per-layer column-separation ranges. Frames auto-scale so a low-elevation target's 16-km pierce point (which can sit more than 20 km out) always fits. Display-only like the advisory itself, and it degrades explicitly: no run $\rightarrow$ message, no target above the horizon $\rightarrow$ monitor region only, no winds $\rightarrow$ no arrows.

MKAM monitor-star model on the geometry dialog. The seeing monitor's pointing is not logged, but it observes one bright single star at a time from a fixed catalogue (~ 130 HR stars, $V \leq 3$, now shipped inside the installed package), re-pointing through the night to stay near zenith. A probabilistic scheduler model weights every catalogue

star by $\exp(-\Delta ZD/5^\circ)$ against the instantaneous nearest-to-zenith star over an observing window — the softness absorbs scheduler hysteresis and operator tie-breaks, and comparable-brightness close doubles (Capella, ζ Tau, η Tau ...) are excluded — yielding the three most probable monitor pointings with probabilities. The dialog draws them live: ranked fading-blue beams with open pierce markers, a probability box giving each star's az/el and its pinned 16-km lead, a summit-zoom inset (± 5.2 km, because near-zenith beams cross within a few km of the origin and are unreadable at the full plan scale), and one line per candidate in the side view at its true zenith angle. The per-layer lead/lag annotation is pinned to the most probable candidate instead of carrying the zenith-monitor ignorance band. On the 2026-01-31 M79 night the model identifies Aldebaran (α Tau) as the monitor star — it transited 3.3° from zenith five minutes before the first frame — collapsing the 16-km lead's ± 43 min pointing-ignorance band to $-1\dots+6$ min across all plausible pointings. Pinned to the frames themselves (20:50 HST) the model gives it $P = 70\%$, rising to 77% at the sequence's first frame. Because the reference time follows the Period selector, note that the default end-of-night instant puts M79 below the horizon and returns an entirely different asterism — the pinning matters. The dialog's reference time follows the Data-tab Period selector whenever that names a single instant. A below-horizon target is now called out in the title with its elevation.

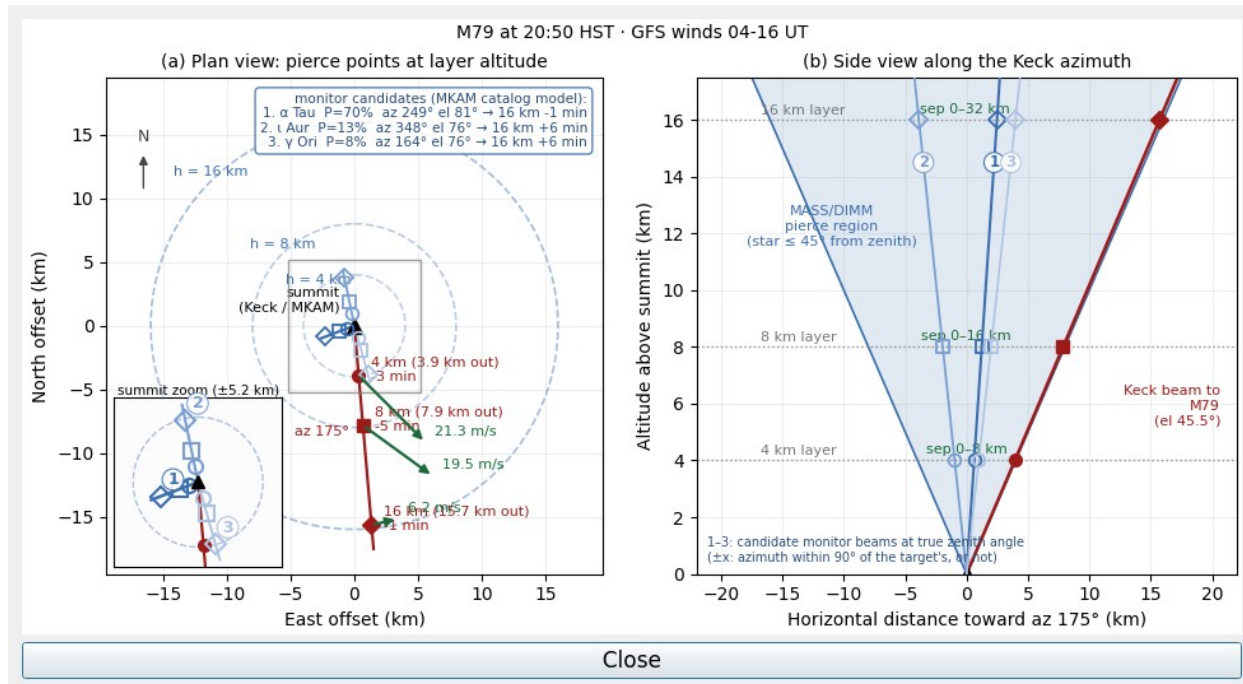


Figure 24 — the FA pierce-point geometry dialog, pinned to the M79 frames of 2026-01-31 (20:50 HST). Plan view: the monitor's pierce-region circles, the beam's per-layer pierce points, each layer's GFS wind vector with its lead/lag annotation, the ranked MKAM monitor candidates with their probabilities, and the summit-zoom inset (left). Side view along the beam azimuth: the monitor's pierce cone, the per-layer column-separation ranges, and one line per candidate at its true zenith angle (right).

Free-typed times and a UTC display mode. The field-map and summary-panel "specific time" fields are now plain free-typed entries — "21:35", "2135", "935", or a bare hour, committed on Enter or focus-out, with an unparseable entry reverting to the last good value — replacing section-based QTimeEdits that never allowed select-all-and-retype. And a Times: checkbox on the Data tab switches every displayed and entered wall time between HST (the tool's native timebase) and UTC: plot axes and the observing-window annotation are relabelled post-render (the plotted data and the CLI's frozen outputs stay HST), the Target tab's windows list converts in place while the engine and saved configs always receive canonical HST, and a typed 13:49 UT resolves to the identical internal instant as 03:49 HST. Persisted in configs.

Measured Strehl from real frames. A fourth plot-side tab, Measured SR, measures delivered Strehl and FWHM directly from NIRC2 and OSIRIS science frames — a Python port of the summit IDL Strehl tools, validated to 0.001 in Strehl against their own golden outputs — including per-star error bars, crowding handling, a measured field map, and an isoplanatic angle fitted from the real data. The port, its validation, and the physics are documented in the new §10; task-level usage lives in the KAON 1556 manual.

10. Measured Strehl from NIRC2 and OSIRIS images

Provenance. The engine now carries a Python port of the summit K2 NIRC2 Strehl tool (IDL, kroot kss/nirc2/strehl/1.0, M. van Dam / A. Bouchez et al. 2003–2024) as three Qt-free modules: nirc2.py (camera plate scales, filter LUTs, KAON 253 pupil-stop dimensions, header conventions including pmrangl = -(ROTPPOSN-EL) +38+90, saturation 32000/DETGAIN), nirc2_psf.py (pupil rasterization and the FFT diffraction-limited PSF with the original's resampling rules, including the PCU fiber-source daytime branch), and image_strehl.py (reduction with superflat/supermask, the astrolib 3×3 sigma filter, DAOPHOT derivative centroid, mean-annulus aperture photometry, sinc-deconvolved 8× Fourier peak upsampling, spline radial-profile FWHM, and the Maréchal WFE conversion).

Validation. Against five IDL-tool golden measurements (Kcont, narrow/largehex, S 0.52–0.55): positions match to 0.05 px, FWHM to 0.4 mas, and Strehl to within 0.001 on every frame. An initial constant +0.004 offset was traced — via IDL-written oracle arrays of the pupil and DL PSF — to POLYFILLV's rasterization convention (vertices truncated to integers, pixel-centre scan lines, closed-interval tie handling), now emulated to 15/262144 pixels; the ported FFT chain agrees with the IDL DL PSF to 1.3×10^{-5} rms/max. The K2 superflat/supermask calibration pair ships inside the package; the IDL sources, raw frames, and golden outputs stay outside the repository, and the golden section of the regression model skips cleanly when they are absent.

OSIRIS. The K1 OSIRIS fork of the tool (k1aoserver; the 2025 widget plus the 2014 find_strehl) is ported into the same engine: osiris.py carries the IFILTER wavelength table, the 9.95 mas post-2016 imager scale, and the pinned open/38° pupil per the tool's own unfinished pupil handling, plus the sfp white-light-source pupil (an 11.14 m circle), the fix_image auto bad-pixel repair (the fork's entire reduction beyond background subtraction — it uses no flat, mask, or saturation ceiling), and the 2048² → central-1024² crop. Validated against four IDL golden measurements (BrGamma): FWHM and positions exact, Strehl within +0.0024 — the same rasterization-scale residual class as NIRC2 before the oracle work. Frames route by header identity (INSTRUME vs CURRINST) in the GUI.

The Measured SR tab. A fourth plot-side tab hosts both instruments' measurement tools, remaking the IDL strehl_widget's layout and flow: numbered-sequence Measure, a double-click frame list, autofind vs click-the-star with a live magnifier, MODEL PSF / MEASURED STAR cutouts, measured-vs-predicted readouts against the loaded estimate (nearest sample by header UTC, series by LSPROP, Maréchal conversion to the frame's wavelength), a crowded-field package (crowding metric, robust and picked sky, unphysical-SR flagging), an instrument-agnostic display stretch, and a measured field map with auto-find, per-star quality gating, and a field self-consistency filter. Full task-level documentation lives in the KAON 1556 user's manual; the engine-side physics is summarized below.

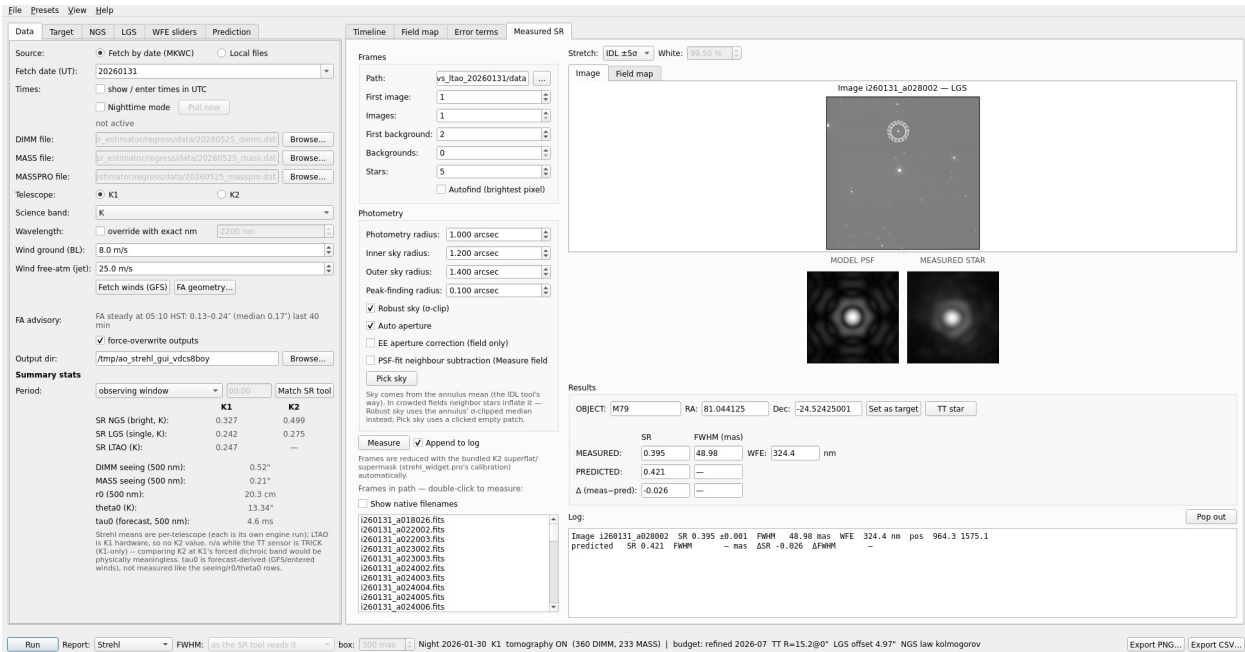


Figure 25 — the Measured SR tab after measuring an M79 OSIRIS frame: Frames / Photometry / Results controls, the frame with its aperture rings, the MODEL PSF and MEASURED STAR cutouts, the TT star button on the OBJECT row, the Append to log checkbox beside Measure, the PSF-fit neighbour subtraction checkbox in the Photometry group, and the aligned measured/predicted log — SR 0.395 ± 0.001 measured against 0.421 predicted. The predicted value carries the measured laser offset (§3.4 (4)) — the status bar reads “LGS offset 4.97”. The FWHM selector in the bottom bar shows the default

convention, “as the SR tool reads it” (§3.7). The predicted FWHM reads “—” here: this frame’s radial-profile fit found no half-max crossing, which is reported as absent rather than as a number (§10).

Photometric errors, apertures, and field statistics. Every measurement carries a 1σ Strehl uncertainty propagated from the frame’s own measured sky noise (the standard aperture-photometry error budget — a lower bound, since photon noise would need the detector gain). An opt-in curve-of-growth auto aperture cuts the photometry radius where growth settles into the noise or turns up again in a sustained way (a neighbour’s wings entering), with the diffraction-limited normalization integral using the same radius so the Strehl ratio stays self-consistent at any aperture; on the goldens S is unchanged, while a 0.6”-separation synthetic pair recovers S 0.90 (vs 0.58 contaminated). An Auto star count keeps accepting stars while the propagated SR uncertainty stays within ± 0.05 . Field statistics report the peak-performance star, the error-weighted field mean and scatter, a weighted best-fit performance gradient, and — with ≥ 4 stars — an isoplanatic angle fitted from the real data. Anisoplanatism gives

$$S(\theta) = S_0 \exp(-(\theta/\theta_0)^{5/3})$$

so $\ln S$ is linear in $\theta^{5/3}$, and a weighted fit about the best-measured star yields θ_0 directly, no model atmosphere involved. With an LGS the tip-tilt star also shapes the field, so this is an effective θ_0 of the delivered correction, not a pure atmospheric one; a 500 nm companion value uses the standard $\lambda^{6/5}$ scaling so it can sit next to the MASS/DIMM θ_0 . Degenerate fields refuse honestly with a stated reason.

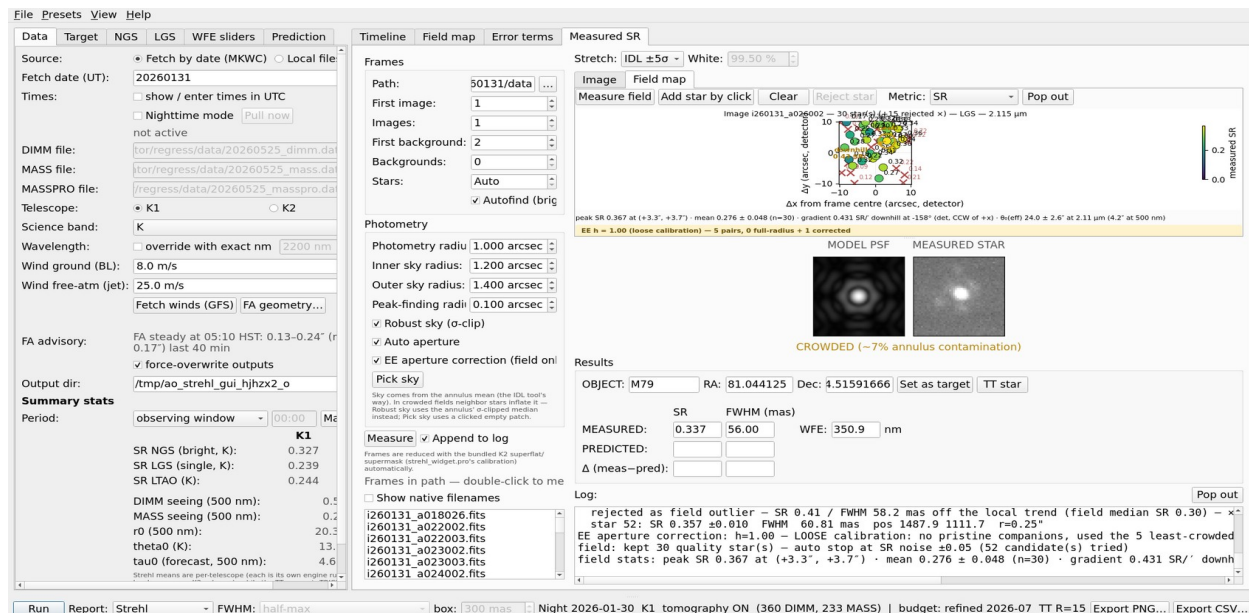


Figure 26 — the measured field map on the same M79 frame, with EE aperture correction enabled: 30 quality stars kept and 15 rejected (auto stop at SR noise ± 0.05 , 52 candidates tried), the Reject star control, the gold peak-star ring and downhill gradient arrow, the field-stats line reporting the data-fitted $\theta_0(\text{eff}) = 24.0 \pm 2.6''$ at $2.11 \mu\text{m}$, and the highlighted EE h badge. Note what the badge says: $h = 1.00$ from a loose calibration on five pairs — this field offered no strictly-clean companion, so the correction is effectively inert here and says so. It is the reporting behaviour that is being illustrated, not the 358-pair validation quoted in the text.

Worked example: M79 single-LGS vs LTAO (2026-01-31 UT). The bundled OSIRIS Kp sequence stepped the telescope 15” North at a time from a 10-mag NGS, once in LTAO and once in single LGS (header AOTSX steps and the WFS background level confirm the split). Each frame runs the full measured-SR chain — auto star count, auto apertures (a globular cluster: crowding everywhere), consistency filter, field statistics. The repeated on-axis anchors expose real conditions drift (LTAO on-axis 0.34 → 0.41 → 0.29 across the hour), so the mode-level fit uses a drift-immune engine primitive: each off-axis frame is normalized by its mode’s time-interpolated on-axis anchor, and $-\ln[S(\theta)/S(0,t)]$ is fitted through the origin against $\theta^{5/3}$, cancelling the atmosphere’s secular change to first order. Result: $\theta_0(\text{eff}) = 72 \pm 2''$ in LTAO vs $60 \pm 13''$ in single LGS (one anchor only — the honest error bar): the tomographic correction measurably holds up better off-axis, and at 45” the single-LGS field has lost ~43% of its on-axis Strehl vs ~38% for LTAO (Figure 27).

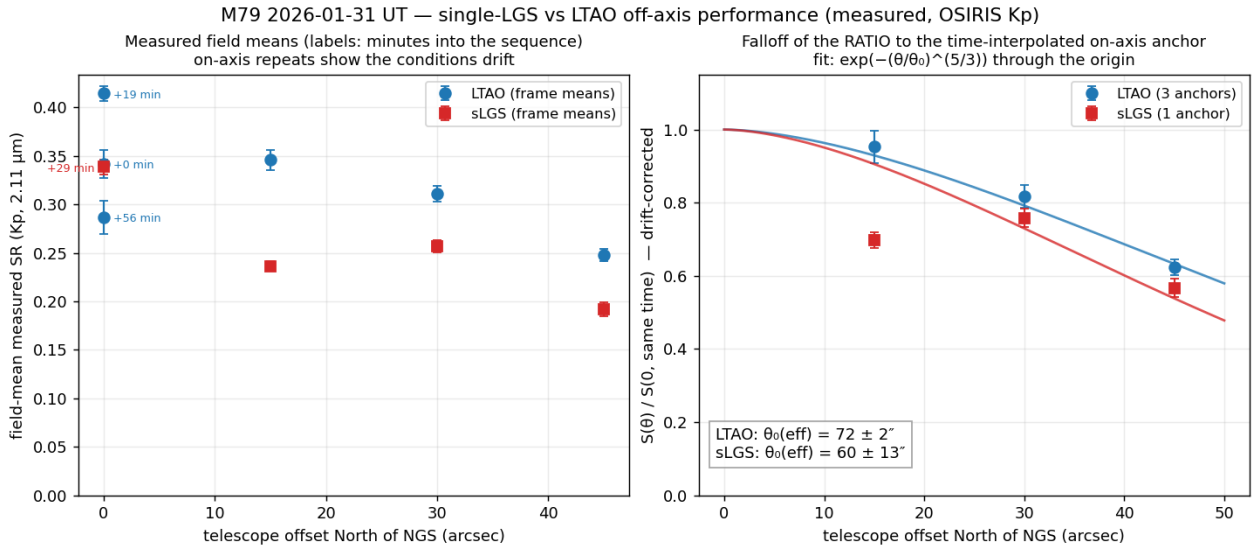


Figure 27 — the M79 worked example (2026-01-31 UT, OSIRIS Kp): measured field-mean Strehl vs telescope offset North of the NGS in LTAO and single-LGS (left; on-axis repeats show the conditions drift), and the drift-corrected ratio falloff with the θ_0 fits (right) — $\theta_0(\text{eff}) = 72 \pm 2''$ LTAO vs $60 \pm 13''$ single-LGS.

Physics basis. Every quantity in the chain is a standard construction: Strehl as the ratio of measured to diffraction-limited peak concentration at matched aperture and wavelength (Maréchal / Roberts et al. 2004 convention); the DL reference from the true Fourier optics of the actual pupil (KAON 253 stop geometry, rotated to the frame's PMRANGL); FWHM from the radial profile; sky from a source-free annulus with outlier-robust estimators; uncertainties from textbook aperture-photometry error propagation; aperture choice from the curve of growth; and the rejection rules are physical constraints ($0 < S \leq 1$, unsaturated detector, positive resolved width, field self-consistency under a smooth anisoplanatic gradient). No empirical fudge factors are introduced anywhere in the chain.

Documented deviations from the IDL originals. A natural cubic spline stands in for IDL SPLINE's tension spline (≤ 0.4 mas effect); the saturation flag reads the raw sub-image maximum rather than a cubic congrid upsample; and fix_image.pro's auto bad-pixel detection is not ported (a mask is required for repair). The summit widget's file-watching/auto-update machinery is deliberately out of scope until an inside-firewall deployment is wanted.

EDGE guard on the auto aperture. A star near the detector edge truncates its own curve of growth — the missing halo area makes the curve flatten early, the optimizer settles at a too-small radius, and peak/flux inflates the Strehl (a real edge star read 0.50 with the auto aperture against a physical 0.24 at the fixed radius). When more than 5 % of the sky annulus's outer disc falls off-array the optimizer is skipped, the fixed radius is used, and the result is flagged: an EDGE tag in the warning row and [edge] in field-log lines. Interior stars and the IDL-golden comparisons are byte-unchanged.

Click-to-inspect and Reject star. Points on the measured field map — embedded and popped-out — are clickable: a click selects the star with a red ring, shows its full measurement in the Results block and its cutout as "FIELD STAR k", and arms a Reject star button. Rejecting removes it and regenerates every field statistic (peak, mean, gradient, θ_0) with a "rejected by user" log line.

TT-star odometer. A TT star button estimates the tip-tilt star's position from the frame's own AO headers. The TT-sensor stage sits on the star and the active pointing origin is the field centre, both in the same f/15 focal-plane frame at 1.375"/mm, so their difference is the delivered star-to-field-centre separation — rotation-free, and ground-truthed on two OSIRIS nights, where the M79 readbacks independently re-measured that night's known TT-stage undershoot of 14.61" per commanded 15". The bench-to-sky direction is not derivable from a single frame (a 60° rotator-configuration degeneracy between the two calibration nights), so the star's identity and magnitude come from a catalogue ring match instead: stars whose radial separation from the frame's RA/Dec matches the odometer's within $\pm 5''$ — the header centre is boresight-referenced and measured 3.8" off on-axis — listed with the best available magnitude, preferring STRAP's R band. The headers carry no STRAP flux keyword usable as a magnitude, so this is catalogue-only, and the band varies by catalogue (GSC's faint stars may carry only the photographic N).

TT-star odometer, method. $\text{sep} = |(AOTSX, AOTSY) - 1000 \cdot (POXPOS, POYPOS)| \times 1.375''/\text{mm}$. On-axis identity was verified to readback precision on two nights, and the per-frame readbacks visibly track the dither pattern, so the

odometer reports the delivered geometry rather than the commanded one. Direction is deliberately not reported: the bench-to-sky rotation differed by exactly 60.0° between the two calibration nights at identical rotator position and image position angle. Identity and magnitude come from the $\pm 5''$ catalogue ring match.

Robust autofind. On the 2026-05-28 TRICK night's M13 frames, autofind's brightest pixel landed on the TT star's saturated charge-bleed plateau, the centroid degenerated, and the measurement failed outright — unchecking autofind and clicking a star worked, which localized the fault. Autofind now falls back to the detection-floor star finder's candidates when the brightest pixel is unmeasurable, takes the first that measures, and logs the substitution. A cleanly-measured saturated star still displays with its SATURATED banner as before: the fallback triggers on measurement failure, not on flags.

Native filenames for archive directories. The Keck Observatory Archive renames every frame on ingest to a date-and-seconds form, which makes the frame list impossible to follow against a night log written in observer numbering. The archive preserves the original observatory filename in each frame's own DATAFILE header card, so a Show native filenames checkbox above the frame list re-labels it from the files themselves — no external metadata table, no renaming on disk. Headers are read on a background worker so a large network directory cannot stall the UI, the on-disk name stays as each item's tooltip, double-click opens the real file and labels the results log with the native name, files lacking the card keep their disk name, and stale scans from an abandoned path are discarded. Off by default, so non-archive directories pay nothing.

EE aperture correction. The crowded-field absolute-Strehl fix. Matched-settings tests showed that the tool's entire historical cross-tool discrepancy is an aperture convention: $r = 0.3''$ photometry read $1.6\text{--}3.3\times$ above the $1''$ -convention value on the same stars with identical FWHM, because small apertures miss halo flux that the diffraction-limited reference does not lose at the same rate. Modelling the PSF as $SR \cdot DL + (1 - SR) \cdot \text{halo}$ gives a one-parameter growth-curve correction family in $h = EE_halo(r)/EE_DL(r)$. Each small-aperture field star also gets a companion full-radius measurement; clean companions calibrate the field's own h , and because h is parametrized by each star's own Strehl rather than by field position, this absorbs anisoplanatic halo variation to first order. The checkbox is gated on Auto aperture, applies to the field map only, and roughly doubles field-measure time. At field completion, stars whose full-radius companion is clean adopt that value verbatim — it is the convention value — crowded stars get the growth-curve correction with the fitted h , and the whole fix runs before the consistency clip so the clip sees final-convention values.

EE aperture correction, method and honesty rule. $SR_true = SR_small \cdot h / (1 - SR_small \cdot (1 - h))$, with $h = EE_halo(r)/EE_DL(r)$ calibrated per field from clean two-aperture companion pairs and never transferred between fields. Validated on 358 M79 pairs (2026-01-31 UT): mean bias $+0.171 \rightarrow -0.005$, with h drifting only $0.42 \rightarrow 0.51$ from on-axis to $30''+$ off-axis. The first Galactic Centre run with the box checked produced nothing and said nothing — every full-radius companion was $\sim 92\%$ annulus-contaminated, so the strictly-clean cut found zero calibration pairs and the method returned quietly. That is fixed: strictly-clean calibrators are tried first, a field offering none (exactly the field type the feature exists for) falls back to the least-crowded quartile with a LOOSE-calibration tag, and the outcome is always logged — calibrated, loosely calibrated, or the explicit “ h uncalibrated, values left in small-aperture convention.” Never silent.

PSF-fit neighbour subtraction (opt-in and off by default). A fourth crowded-field mitigation, addressing the failure the other three cannot: a neighbouring star lying inside the target's photometry aperture rather than in its sky annulus. The three sky-side options — σ -clipped sky, picked sky, and the annulus statistic behind the CROWDED tag — all correct the sky, and a star sitting on top of the target is not a sky problem. The frame's empirical PSF is constructed from the frame itself: isolated, unsaturated, well-exposed stars are stacked on a sub-pixel grid following Anderson & King's effective-PSF definition, so model and data share one pixel convention and no pixel-integration step is required anywhere. That model is fitted simultaneously to the target and to every catalogued neighbour predicted to deposit more than 0.1% of the target's flux inside its aperture — positions solved non-linearly, amplitudes and the local background linearly within each step — and the fitted neighbours only are subtracted. The target is never subtracted, and the aperture measurement that follows is the unmodified default convention: same aperture, same annulus sky, same peak/flux. Only the pixels the neighbours occupied are different. Building the model from the data rather than from theory is the point — it carries the actual static speckle, elongation and instrumental structure of that night, which a diffraction-limited or analytic PSF does not. With the option off, the measurement path is byte-identical to the default (CSV byte-identity harness: 0 mismatches; the five IDL goldens reproduce to 0.00097 SR).

Validated envelope, and two levels of refusal. $|\Delta SR| \leq 0.02$ for Strehl ≤ 0.30 , established on synthetic frames spanning separations $0.10\text{--}1.20''$ and contrasts $0\text{--}5$ mag at three Strehl levels, and confirmed on a full night of 82 OSIRIS frames against the independent encircled-energy method. Above Strehl 0.30 the residual error changes sign and grows (worst measured $+0.07$ at Strehl 0.80 for equal-brightness pairs inside $0.3''$); the cause has been

investigated at length and is not yet identified. Cleaning is not refused above the envelope — the restriction is a statement of confidence, not of correctness — but the star is tagged and the log says the number should be read as an upper bound. The feature also refuses outright in two distinct places, and the log distinguishes them: a donor-level refusal means the frame could not supply the minimum of four usable PSF stars, while a model-level refusal means a model was built and then rejected, because across the brightest stars it predicted more neighbour light than the target's aperture physically contains — which indicates the model's amplitudes are wrong rather than the field merely crowded. The model-level check runs once per frame, before any per-star fitting, so an unusable model costs one log line instead of a refusal on every star. In both cases cleaning does not run and the default measurement stands unaltered.

Which way a cleaned number is likely wrong. Every measurement on which cleaning actually ran carries a statement of the direction of the residual error, not merely its magnitude. Inside the validated envelope the residual is an underestimate — any neighbour light left un-subtracted inflates the aperture flux and therefore depresses peak/flux — which is the conservative direction. Above Strehl 0.30 the sign reverses and the log states explicitly that the value is an overestimate and should be treated as an upper bound. The distinction matters because an unsigned residual error is not usable: an observer who knows a number errs low can bound the true performance from it, whereas one told only that accuracy degrades cannot. On a randomly populated field of moderate density the cleaned result leans high by approximately +0.10 in Strehl, since such targets typically carry several neighbours whose over-subtraction accumulates; this is bounded and reported rather than corrected.

Interaction with Robust sky. On any star for which cleaning succeeds, the Robust sky (σ -clip) option is ignored and the plain annulus mean is used. The two corrections remove the same neighbour light by different means — cleaning removes the neighbours from the image, after which the clipped median would clip an annulus that no longer requires it — and applying both together approximately doubles the residual bias (measured: signed +0.10 with cleaning alone, +0.20 with both). The log records the substitution per star. Robust sky continues to apply normally to any star on which cleaning declined, which is its original purpose. The CROWDED tag itself is driven by the annulus statistic and stays displayed in every case, including after a successful subtraction: it reports that the field is crowded, not that the measurement is uncorrected.

Frame-load resets and display details. Loading a new frame clears the rejected-star $\times$ pool, the map's selection ring, and the EE readout together with the kept-star pool — the same reset the Clear button performs — so a previous frame's markers never reappear on the next image. The EE checkbox is labelled “EE aperture correction (field only)”; the fitted h is shown as a highlighted badge next to the field-map stats rather than only in the scrolling log; the results log has a taller default height, a larger share of the splitter, and its own Pop out button opening a second independently resizable view onto the same document; and the popped-out field map has a “Show image (hollow markers)” checkbox, off by default, which draws the reduced frame as a grayscale background and switches kept-star markers to hollow rings (edge colour still carrying the value) so a measurement can be checked by eye against the real star underneath it.

Structured Image log. Every measurement — single-star or re-inspected from the map — now also appends a structured row alongside the unchanged free-text log, with columns in order: time (UTC), frame number, target name, guide-star magnitude and the source it was resolved from, RA/Dec, measured/predicted/ Δ Strehl, measured/predicted/ Δ FWHM, filter, airmass, detector pixel X/Y, Az/EI, LBWFS seeing, and DIMM/MASS seeing matched by nearest sample. Every numeric column is capped at three decimal places — nothing in this log is meaningful past that. The log's Pop out button now opens a second tab, Image log: a sortable table of these rows with Delete selected and an on-demand Export CSV... button. Export is never automatic. An Append to log checkbox beside the Measure button, checked by default, turns the whole thing off for throwaway re-measurements.

Text log		Image log											
	Time (UTC)	Frame #	Target	Guide mag	Guide mag source	RA	Dec	Measured SR	Predicted SR	Δ SR	Measured FWHM (mas)	Predicted FWHM (mas)	Predicted
1	2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.395	0.569	-0.174	48.982	—	—
2	2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.406	0.569	-0.163	51.939	—	—
3	2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.403	0.569	-0.166	48.612	—	—
4	2026-01-31 06:50:38	i260131_a028002	M79	10	target list entry	81.044125	-24.52425001	0.439	0.569	-0.129	49.721	—	—

Figure 28 — the popped-out log on its Image log tab: real measured rows, the Guide mag source column recording where each magnitude was resolved from, and the Delete selected and Export CSV... controls. The Measured FWHM column carries values while the predicted column reads “—”, which is the absent-value convention described above.

Duplicate guard and resolved guide magnitude. A duplicate-frame guard compares each new row's frame number and detector position against rows already logged, catching both a re-measured directory and a re-clicked point on the field map. On a match the measurement sequence genuinely pauses — the worker gained real pause/resume/abort rather than racing ahead — and asks Append / Overwrite / Don't add, with an “apply to all remaining duplicates in this run” checkbox so a re-measured directory can be settled file-by-file or in one answer; frames already in flight when the pause lands are queued and drained in order afterwards rather than opening stacked dialogs. The guide-star magnitude is now resolved from real evidence rather than read off the TT-magnitude spinbox, which on a frame measured from disk is usually still the estimator default — mislabelling the log and silently meaning the predicted values being compared against were computed for the wrong star. The precedence chain walks this frame's TT-star catalogue ring match, then a starlist target-linked TT star (laser frames only — with no laser there is no separate TT star, so crediting a linked entry would name the wrong one), then the target-list entry's stored magnitude, then on NGS frames the science target itself, falling back to the spinbox only as a last resort and always recording which source was used, so a default can never pass as a measurement. A disagreement of more than 0.3 mag with the magnitude the loaded prediction was computed at is called out explicitly — warn-only, because silently re-running the estimator underneath a measurement the user is reading would be worse. The headers do carry a real STRAP flux keyword, non-zero on every laser frame in the 2026-07-27 dataset, deliberately not converted to a magnitude: no counts-to-magnitude zero point exists anywhere in the repository, and inventing one would manufacture a false number.

Predicted-FWHM band conversion. The measured-vs-predicted readouts convert both predicted quantities to the measured frame's own band. The predicted Strehl is band-converted directly; the predicted FWHM is re-evaluated at the frame's wavelength from the already-band-converted Strehl, at the same matched sample index the Strehl comparison uses — the same function the run itself uses to build the FWHM series, called again at the correct wavelength. The timeline and field-map predictions are single-band and self-consistent already.

Failed FWHM fits are not logged as numbers. The radial-profile FWHM routine returns -1 px as a documented failure sentinel — profile peak off-centre, or no half-max crossing found — which the measurement path treats as such. A non-positive FWHM is recorded as absent — an em dash in the table, an empty cell in the CSV — and the Δ FWHM is dropped with it. The regression harness is byte-identical: this touches the Measured-SR log only, not the estimator. One caveat worth knowing: the on-screen MEASURED FWHM box and the free-text log line still print the sentinel for a failed fit. That behaviour predates this work and would read better as a “no FWHM” tag beside the existing UNPHYSICAL SR one. The sentinel surfaced because autofind on these M79 frames locks onto the R = 10 natural guide star, whose core yields no half-max crossing at all.

One caution for scripted use. The duplicate-frame dialog is modal, which is correct for interactive work but means an automated run with nobody to answer it will hang indefinitely. If you drive the tool from a script, set the duplicate-handling policy up front rather than relying on the prompt.

Why the field map, the summary stats and the SR tool can differ on the on-axis Strehl. A reasonable reader question, and the answer is that they are one model, not three. All three end in the same `lgs_strehl()` call, and given the same instant and the same geometry they agree exactly — verified on 20260525 K1 / HD 141569, where all three return 0.239025. Every visible difference is an input difference, and each is deliberate:

What differs	Why	Size on that night
Period reduction	The summary stats take the mean of the Strehl series over the period; the field map evaluates one representative sample — the one whose line-of-sight seeing is nearest the median. $\text{mean}(S) \neq S(\text{median-seeing sample})$.	+0.022 (whole night)
The SR tool re-resolves the guide star	Its predicted box re-evaluates at the frame's own resolved guide-star magnitude and TT-odometer offset; neither the map nor the stats knows those.	+0.116 (mag 11.0 vs the configured 15.2)
Wavelength	The SR tool works at the frame's EFFWAVE, the other two at the run's science wavelength.	-0.024 (2124 vs 2200 nm)
Which centre offsets are measured from	With a real image defining the field, the map re-measures the guide-star offsets from the image pointing while the budget's offset stays measured from the typed target.	+0.016 per 5" of disagreement

Clicking Match SR tool on both panels removes the first and third. What remains is the second — and that residual is the SR tool being more correct than the other two, since it is using the actual guide star of the actual frame. The laser

marker is not a contributor: the map's on-axis LGS offset is $|0 - \text{laser}|$, which is the LGS-offset magnitude by construction.

11. Appendix: the NGS fit vs the analytic budget (K2)

The NGS Strehl in this tool comes from the on-sky HAKA Gompertz fit, not from the analytic error budget the LGS/LTAO paths use. A natural question is whether the analytic budget — built term by term from the K2 NGS budget sheets and driven across magnitude by the OCAM2K frame-rate / EM-gain look-up table — reproduces that on-sky fit. The exercise both tests the budget and motivates the latency-floor bandwidth model that §3.4 (5) applies to LTAO.

For NGS the on-axis guide star is its own tip-tilt reference, so the focal-anisoplanatism, angular-anisoplanatism, sodium-focus and tip-tilt-anisoplanatism terms all vanish. What remains fixed with magnitude are fitting, scintillation, the static/calibration terms and margin; what varies with magnitude are the high-order bandwidth and measurement and the tilt bandwidth and measurement, all set by the OCAM2K operating point (EM gain compensates flux down to $R \approx 7$, after which the frame rate falls and the wavefront sensor is progressively photon-starved). Each term is seeded from the $R = 5$ sheet and its magnitude scaling calibrated to the $R = 10$ sheet; no term is fitted to the Gompertz curve.

The result: agreement bright, divergence faint

With the sheet's frame-rate-driven bandwidth, the analytic budget matches the on-sky fit at bright-to-medium magnitudes ($R \approx 5$ – 9) but falls off far too fast at the faint end — over-predicting Strehl bright and under-predicting it faint. Crucially, this is a slope (shape) mismatch, not a vertical offset: holding the fixed terms constant and inverting the fit shows the magnitude-dependent error it implies rises only $\times 2.6$ from $R = 5$ to $R = 13$, while the sheet's bandwidth-plus-measurement rises $\times 5.4$ (Figure 29). The two cross near $R = 9$, exactly where the Strehl difference changes sign.

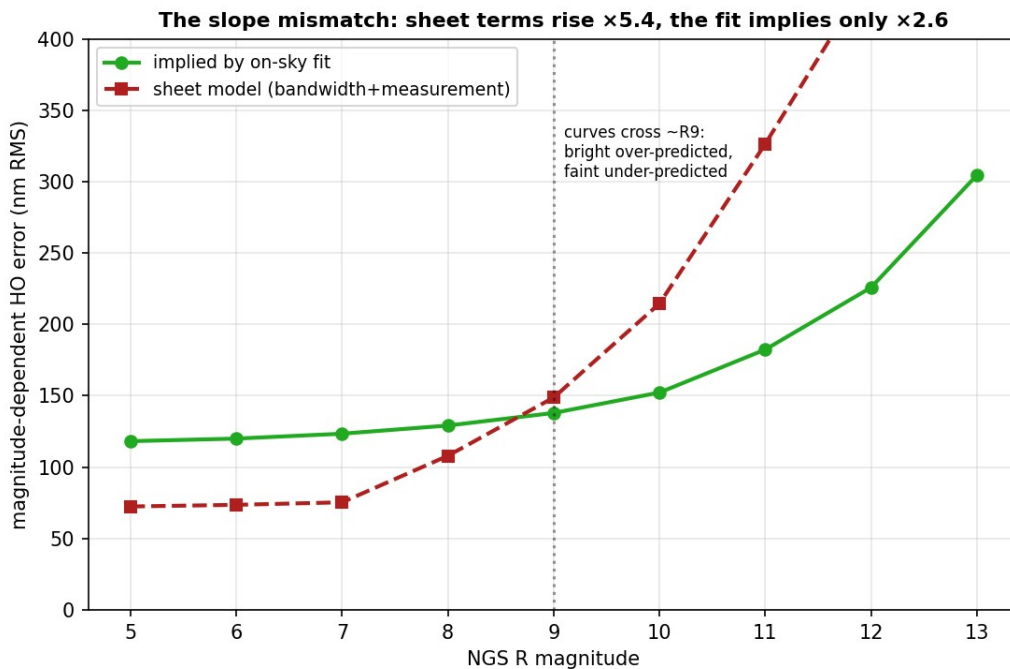


Figure 29 — the magnitude-dependent high-order error implied by the on-sky fit (green) rises gently; the sheet model (red) rises steeply and overshoots beyond $R \approx 9$. The mismatch is in the slope, not the level.

Why a fixed term cannot fix it. Because the discrepancy is in the slope, no magnitude-independent term can repair it. Margin, static aberrations — the entire fixed floor — enter the quadrature identically at every magnitude and only slide the whole Strehl-vs-magnitude curve up or down. Raising the margin to lift the faint end would push the already over-predicted bright end further up. Only the magnitude-dependent terms control the slope.

The fix: a latency-floor bandwidth

The sheet models bandwidth as frame-rate-limited servo lag, which assumes the sample period is the dominant loop delay at every magnitude. In a well-engineered real-time controller the servo lag is instead dominated by a fixed loop

latency — compute plus DM temporal response — until the loop is genuinely photon-starved. Modelling the bandwidth as a fixed latency floor with only a weak frame-rate term on top:

$$\sigma_{bw} = \sqrt{\sigma_{lat}^2 + [k(1500/f_{frame})^{5/6}]^2} \times s_{tot}$$

flattens the slope and tracks the on-sky fit to an RMS of ~0.03 Strehl across R = 5–14, against ~0.17 for the sheet model. The floor the fit prefers is ~92 nm, giving a bandwidth that is nearly flat (92 → 95 nm from R = 5 to R = 10) rather than the sheet's 60 → 149 nm. Figure 30 shows both pictures against the fit and the R5/R10 sheet anchors.

K2 NGS: the budget-sheet bandwidth (frame-rate-driven, 60–149 nm) falls off too steeply vs on-sky.
A fixed loop-latency floor (~92 nm, nearly flat) tracks the HAKA fit to RMS ~0.03 — evidence the sheet bandwidth math over-weights frame rate. Margin (a fixed term) cannot fix a slope mismatch.

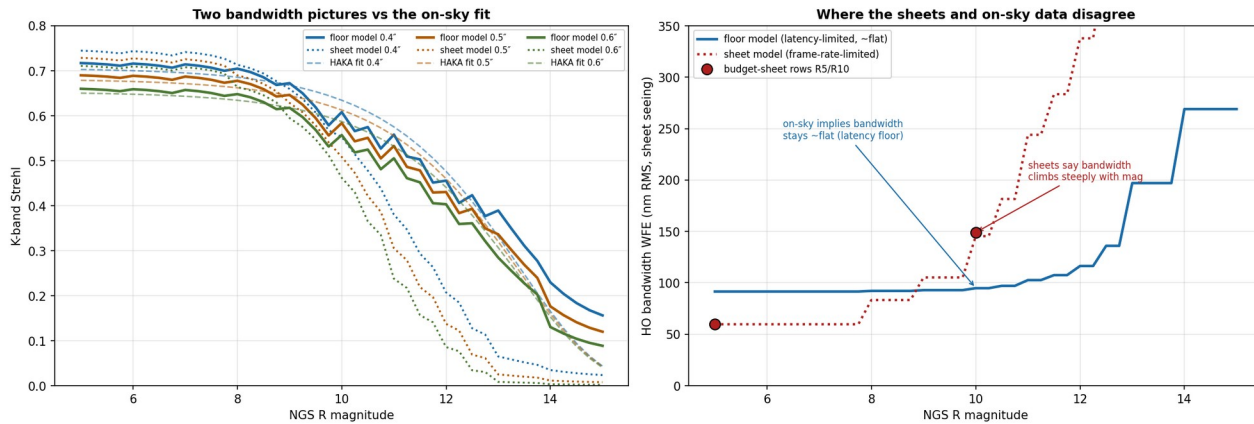


Figure 30 — left: the latency-floor budget (solid) and the sheet budget (dotted) against the on-sky fit (dashed) for three seeing values; right: the two bandwidth pictures, with the R5/R10 sheet rows marked. The sheets say bandwidth climbs steeply with magnitude; the on-sky data imply it stays nearly flat.

Interpretation and connection to LTAO. The budget sheets and the on-sky performance disagree about how bandwidth scales with magnitude, and the on-sky data favour a nearly flat, latency-limited bandwidth — evidence that the sheet bandwidth term over-weights frame rate. This is the same physical insight applied in reverse to LTAO in §3.4 (5): a loop running at a lower frame rate suffers less bandwidth penalty than a pure frame-rate law predicts, because most of the servo lag is fixed latency. A consistency note on the faint end: the raw frame-rate law also runs the bandwidth past the uncorrected-atmosphere ceiling at very low rates, which is unphysical; a loop that has stopped correcting cannot do worse than open loop, so both the bandwidth and measurement terms are soft-capped at the residual atmospheric level.

The same failure mode in the fit's own seeing term. The bandwidth story above — an empirical form calibrated in one regime, extrapolated too steeply into another — repeats inside the Gompertz fit itself. Its Gaussian seeing term $\exp(-A s_K^2)$ was constrained only over the calibration seeing range (K-band FWHM ~0.19–0.38"); when high airmass projects the input 2–3× beyond that range, the quadratic free-falls where the physical Kolmogorov variance ($\propto s^{5/3}$) rolls off far more gently. On 2026-07-06 (HIP 88553, airmass 2.34) the delivered NGS Strehl was ~0.50–0.56 against 0.29–0.42 from the Gaussian form; re-anchoring to the 5/3 law inside the calibration range recovers 0.41–0.49 while changing nothing (<0.005 Strehl) where the fit was actually calibrated (§3.4 (6)). Together the two cases argue the same lesson: within their calibration ranges the empirical forms are excellent, and every extrapolation — in magnitude, seeing, or airmass — should be paced by the physical scaling law rather than the fitted functional form.

This appendix is K2-specific: K1 uses a different OCAM2K LUT and a different fitting error, and would need its own sheets. The NGS path in the tool is unchanged — it still uses the on-sky Gompertz fit — but the analysis validates the fit against first principles where they agree and identifies the bandwidth term as the source of disagreement where they do not.